



DELIVERABLE 3.1

Report on vulnerabilities mapping under the baseline scenario

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Abbreviations

List of abbreviations and acronyms featured throughout the document:

AFIF: Alternative Fuels Infrastructure Facility

AI: Artificial Intelligence

AKIS: Agricultural Knowledge and Innovation Systems

AMI: Advanced Metering Infrastructure

AVE: Alta Velocidad Española (Spain's high-speed rail network)

BEG: Bundesförderung für Effiziente Gebäude (Federal funding for efficient buildings, Germany)

BEHG: Brennstoffemissionshandelsgesetz (Fuel Emissions Trading Act, Germany)

BEV: Battery Electric Vehicles

BH2C: Basque Hydrogen Corridor

BIM: Building Information Modeling

BMWK: Bundesministerium für Wirtschaft und Klimaschutz (Federal Ministry for Economic Affairs and Climate Protection, Germany)

BMWSB: Bundesministerium für Wohnen, Stadtentwicklung und Bauwesen (Federal Ministry of Housing, Urban Development, and Construction, Germany)

CAP: Climate Action Plans

CAP: Common Agricultural Policy

CCS: Carbon Capture and Storage

CEF: Connecting Europe Facility

CH4: Methane

C-ITS: Cooperative Intelligent Transport Systems

CO₂e: Carbon Dioxide Equivalent

CSP: CAP Strategic Plan

DGEG: Direção-Geral de Energia e Geologia (General Directorate of Energy and Geology, Portugal)

EAFRD: European Agricultural Fund for Rural Development

EN-H2: National Hydrogen Strategy

ENPE: Estrategia Nacional contra la Pobreza Energética (National Strategy Against Energy Poverty, Spain)

EP: Energy Poverty

EU: European Union

EV: Electric Vehicles

GDP: Gross Domestic Product

GDPR: General Data Protection Regulation

GEG: Gebäudeenergiegesetz (Building-Energy-Act, Germany)

GHG: Greenhouse gas emissions

GW: Gigawatt

IKOP: Integrated Transport Development Operational Programme (Hungary)

IoT: Internet of Things

IPCC: International Panel on Climate Change

ITS: Hungary's National Intelligent Transport Systems

JTM: Just Transition Mechanism

LTS: Long-Term Decarbonization Strategy 2050

LULUCF: Land Use, Land Use change and Forestry (Ireland)

MACCs: Marginal Abatement Cost Curves

MBO: Model Building Code

MFF: Multiannual Financial Framework

Mtoe: Million tonnes of oil equivalent

MW: Megawatt

NAP: National Access Point

NDCA: National Dialogue on Climate Action

NECP: National Energy and Climate Plans

NRRP: National Recovery and Resilience Plan

NTIDS: National Transport Infrastructure Development Strategy

NZEB: Nearly Zero-Energy Buildings

OIPE: Osservatorio Italiano Sulla Povertà Energetica (National Energy Poverty Observatory, Italy)

PAN: Piano d'Azione Nazionale (National Action Plan, Italy)

PNIEC: Plan Integrado de Energía y Clima (National Integrated Energy and Climate Plan, Spain)

PV: Photovoltaic

RD: Rural Development

RE: Renewable energy

RNC2050: Roadmap for Carbon Neutrality 2050

RTRP: Recovery, Transformation, and Resilience Plan

SME: Small and medium-sized enterprise

TWh: Terawatt-hour

Executive Summary

Purpose and scope

This deliverable is aimed at mapping national baseline scenarios and associated vulnerabilities related to the twin transition process at the national level of six EU member states represented by FITTER-EU. Drawing on an analysis of long-term strategies, policies and action plans implemented at the national level to meet the EU energy transition targets and the Paris Agreement commitments to reduce greenhouse gas emissions and to mitigate climate change, national baseline scenarios are identified in the four sectors of the analysis (energy, transport and mobility, agriculture and food, housing and built environment). Each sector's current situation is analyzed by drawing on a review of national policies and highlighting national context conditions. The specific focus is placed on a review of the impact on inequalities of the transition process in each of the sectors, using indicators of associated vulnerabilities and profiles of structurally vulnerable cohorts.

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1. INTRODUCTION

1.1. OVERVIEW OF FITTER-EU

FITTER-EU aims to contribute to existing research on the origins, dynamics and determinants of inequalities and to enable anticipatory governance to support a fair and inclusive twin transition across Europe. The project innovates through the formulation and development of an ecosystem that includes a Digital Platform. Through a co-creation methodological approach, this ecosystem aims to support policymakers in understanding the structures and systems through which social groups risk being adversely affected by twin transition policies. This understanding is supported through the co-creation of different future scenarios, and the development of mitigation plans, better practice guides, and policy briefs.

1.2. RESEARCH METHOD

Research data for the development of this deliverable was collected from four main sources:

1. **Two online workshops** with international experts in policy making, academic research, social justice, and green and digital transitions (19 September 2024 and 25 February 2025).
2. **Policy review** of national long-term strategies, relevant policies and action plans in each of the six FITTER-EU countries.
3. **Online consultation with experts** in the twin transition process in the four sectors of analysis.
4. **Desk research** of relevant reports and statistical data on the progress of the twin transition process and associated vulnerabilities across the four sectors.

The inquiry was guided by the need to: a) contextualize and systematize the advancement of the twin transition process in each of the sectors and national scenarios, and b) identify potential negative impacts of this transition on existing vulnerabilities and/or drivers of additional inequalities that hinder social justice and fairness.

Main themes addressed during the research process were focused on:

1. Key pillars, targets, commitments and priority action lines of national long-term strategies, policies and action plans to implement the twin transition process.
2. The meaning of vulnerability in the context of the twin transition process: common grounds and sectoral specificities.
3. Critical challenges and emerging opportunities of the twin transition process to ensure social justice and fairness.

1.3. STRUCTURE OF THE DELIVERABLE

The deliverable describes key findings of the conducted analysis of the twin transition process and associated vulnerabilities in the four key sectors in Hungary, Germany, Ireland, Italy, Portugal and Spain. It is structured into six sections. Section 1 introduces the scope of the Fitter-EU project and the methodology used to analyze the baseline situation in four economic sectors across six countries represented by the FITTER-EU consortium. This analysis is framed within the context of national long-term strategies, policies, and action plans related to the key targets of the twin transition process, as well as associated vulnerabilities. Section 2 provides an overview of key targets and priority action lines in the energy sector across six countries—Hungary, Germany, Ireland, Italy, Portugal, and Spain. It focuses on targets related to greenhouse gas emissions reduction, renewable energy, energy efficiency, hydrogen capacity, and energy security. Additionally, this section addresses the issue of energy poverty and the associated vulnerabilities in the transition process. Section 3 examines the mobility and transport sector in Hungary and Spain,

outlining key targets and priority action lines related to renewable energy sources, electric mobility deployment, alternative fuels, and public transport efficiency. It also discusses transport poverty and related vulnerabilities in the transition process. Section 4 explores the housing and built environment sector in Germany and Portugal, with a focus on energy efficiency, housing renovation and insulation, and smart housing. This section also highlights the phenomenon of housing poverty and its implications in the transition process. Section 5 analyzes the agriculture and food sector in Ireland and Italy, emphasizing greenhouse gas emissions reduction, sustainable energy practices, and efficient management of natural resources. It also discusses vulnerabilities associated with the transition process in this sector. Finally, Section 6 presents insights from international experts on the twin transition in the four sectors. It highlights the negative impacts of the transition process on structurally vulnerable groups, as well as identifying social innovation opportunities and action lines to address exacerbated inequalities to ensure that no one is left behind, contributing to a broader discussion on critical challenges of the twin transition process.

2. ENERGY SECTOR

As a backbone of economic and social development, the energy sector is a critical component of the twin transition process. Ensuring access to affordable, reliable, sustainable and modern energy for all is a key priority of the United Nations Sustainable development (United Nations, 2024). Addressing global energy needs in a sustainable and inclusive way to reduce dependence on fossil fuels (oil and gas), to alleviate energy costs and to prevent Energy Poverty (EP) is a driver of the European Growth model and a main challenge of the twin transition process in Europe (European Commission, 2022b).

The Fit for 55 framework and the REPowerEU Plan outline the EU commitment to reduce emissions by at least 55% by 2030 through energy efficiency, security and diversification, compared to 1990 levels. Climate targets of this framework cover all economic sectors with overarching goals aimed at reduction of GHG emissions, boost of natural carbon sinks, update of emissions trading system to cap emissions, investments in the green transition and putting a price on pollution. The priority is also placed on providing social support for citizens and small businesses (European Commission, 2024b).

Energy consumption is estimated to account for over 75% of greenhouse gas (GHG) emissions (World Bank, 2024) and this demands close attention to the performance of the sector's key actors and their impact on the twin transition process. The latest data from the International Energy Agency (2024) highlights that industry is the largest GHG emitting sector with 38%, followed by building emissions representing 26% and the transport sector with 24%. Fossil fuels account for 71% of total primary energy supply in the EU, while only 17% corresponds to renewable energy (RE) sources and 12% to nuclear energy (IRENA, 2024).

2.1. COMMON GOALS AND PRIORITY ACTION LINES OF THE TWIN TRANSITION PROCESS

The European Green Deal (European Commission, 2024d) specifically highlights the commitment of EU member states to reduce GHG emissions by at least 55% in 2030 (as compared to 1990 levels) and achieve zero net GHG emissions by 2050. Industrial decarbonization and an effective transition towards clean energy sources is underpinned by the principles of secure and affordable energy supply; development of a fully integrated, interconnected and digitalised EU energy market; and a priority focus on energy efficiency and RE sources (European Commission, 2022d).

RE is a key component of the twin transition in the energy sector. In 2023, energy from renewable sources (wind power, solar power, hydro power, tidal power, geothermal energy, ambient heat captured by heat

pumps, biofuels and the renewable part of waste) accounted for 24.5% of energy consumed in the EU. This share represents a three-fold increase as compared to 2004. Furthermore, RE sources accounted for 45.3% of total electricity consumption in the EU, with wind, hydro and solar power contributing 38.5%, 28.2% and 20.5% respectively to the total electricity generated from RE. Moreover, in 2023, RE sources accounted for over 26% of the total energy used for heating and cooling, and 10.8% in the transport sector in the EU. The growth of solar power has been most significant in the last 15 years from 7.4 TWh in 2008 to 252.1 TWh in 2023 (Eurostat, 2024e).

By 2030, the EU should meet the target of at least 42.5% and aim for 45% share of RE sources in total energy consumption. In the energy supply side, the green transition prioritises the development of RE infrastructure, energy storage, modernization of power grids, interconnections of gas and electricity, as well as the development of new synthetic fuels. The decarbonization of transport, industry and buildings is heavily dependent on electrification and RE sources, while clean hydrogen and carbon capture and removal play a key role in sectors where electrification is not possible (IRENA, 2024).

The share of renewable hydrogen is established at 42% in total hydrogen consumption by 2030. While clean hydrogen is expected to play a transformative role in the decarbonization process of heavy industry and transport sectors, replacing fossil fuels in the production of iron, steel and chemicals and used as fuel in aviation and shipping (IRENA, 2024), some concerns about the potential environmental and social costs of green hydrogen production have been also raised (Cremonese et al., 2023).

In the mobility and transport sector, the target for the reduction of GHG emissions is established at 14.5% and the share of RE sources in final energy consumption at 29%. Finally, energy efficiency should improve by 11.7% by 2030 to reduce both energy consumption and costs.

2.1.1. MAPPING OF KEY TARGETS

Among six national cases of the FITTER-EU project, there is a shared commitment to achieve climate neutrality by 2050, reducing GHG emissions, and increasing the share of RE source, energy efficiency and decarbonization.

The largest share of energy from renewable sources in gross electricity consumption was recorded in Portugal (63%), followed by Spain (56.93%), Germany (52.2%), Ireland (40.43%), Italy (38.1%) and Hungary (19.5%) (Eurostat, 2024e).

Table 1. Overall share of energy from renewable sources, 2019-2023 (% of gross final energy consumption)

TIME	2019	2020	2021	2022	2023
GEO (Labels)					
European Union	19,887	22,038	21,886	23,087	24,531
Germany	17,266	19,09	19,281	20,913	21,547
Ireland	11,979	16,16	12,996	13,068	15,253
Spain	17,852	21,22	20,550	21,896	24,852
Italy	18,181	20,359	18,883	19,131	19,561
Hungary	12,634	13,85	14,134	15,193	17,361
Portugal	30,623	33,982	33,982	34,675	35,163

Source: Eurostat (2024e)

Context in Germany

The Climate Protection Law of Germany set an achievement of climate neutrality by 2045 and reductions of GHG emissions of at least 65% by 2030 and 88% by 2040 compared to 1990 levels. The achievement of Germany's climate ambitions highly relies on the energy transition, widely known in Germany as "Energiewende", which is seen as the overarching process of fundamentally changing the energy supply in the country: away from nuclear and fossil fuels and towards RE and greater energy efficiency, including the heat and transport sectors. The transition is guided by three fundamental pillars: affordability, reliability of supply, and environmental soundness of the energy supply.

In achieving the goal of climate neutrality by 2025, new energy scenarios have been identified with an emphasis on a major expansion of the electricity grids, the development of a hydrogen grid, the great importance of electrolysis, pipeline-based hydrogen imports, hydrogen storage and hydrogen power plants, as well as large heat pumps and heat storage in heating grids. The System Development Strategy (BMWK - Bundesministerium für Wirtschaft und Klimaschutz, 2023), which provides the framework for the transition to a climate-neutral energy system, established transformation paths for achieving the climate targets efficiently without losing the necessary flexibility to react to changes in environmental conditions through a focus on energy efficiency, the expansion of RE, high and early investments, and intersectoral coordination as cross-sectoral key points in the pathway to climate neutrality.

In the energy demand side, the System Development Strategy focuses on measures for industry, buildings and transport. Meanwhile, in the supply side, the Strategy outlines objectives and measures for:

- Power generation, relying on wind and photovoltaics and on hydrogen in the future.
- Heat supply in heat networks, converted to the utilisation of waste heat and RE, in particular large heat pumps, geothermal energy and solar thermal energy.
- Hydrogen and hydrogen derivatives, using blue hydrogen until green hydrogen is sufficiently available and based mainly on imports (50-70% of the hydrogen requirements).
- Energy imports and trade, aiming to diversify energy imports and reduce Germany's dependence on imports in comparison to current levels, to ensure security of supply
- Flexibility through sector coupling and storage, bearing the potential of electrolyzers, electromobility, heat pumps and storage systems (heat and hydrogen).

Table 2. Key strategic targets of the energy transition in Germany

Component	Strategic goal
Climate neutrality	Climate-neutral by 2045.
	To reduce 65% of GHG emissions by 2030 and 88% by 2040, relative to 1990 levels.
RE	To cover 80 % of gross electricity consumption with RE by 2030. To increase to at least 41% the share of RE in gross final energy consumption in 2030. To completely switch to 100 % RE by 2045.
Coal phase-out and structural change	To complete the coal phase-out by 2038.
Energy efficiency	To reduce by 30 % primary energy consumption by 2030 and by 50% by 2050, relative to 2008 levels.
Clean hydrogen	To expand clean hydrogen capacity to 10 GW by 2030

Source: Own elaboration

Context in Hungary

The transition in Hungary aims at achieving climate neutrality by 2050, increasing the share of RE, enhancing energy efficiency, strengthening energy security, expanding nuclear energy capacity, and integrating into the European energy market. These are supported by the National Clean Development Strategy 2020-2050; the Second National Climate Change Strategy (NCCS-2) 2014-2025; the National Energy Strategy 2030 with an Outlook to 2040 and the National Hydrogen Strategy.

The transition to climate neutrality in Hungary is underpinned by:

- Decarbonisation of energy production: transitioning from fossil fuels to RE sources and nuclear power.
- Energy efficiency improvements: focused on technological innovation and promotion of behavioural changes.
- Carbon sequestration: enhancing natural carbon sinks through afforestation and sustainable land management practices to remove carbon dioxide from the atmosphere.
- Innovation and research: investing in low-carbon technologies and supporting research initiatives.

The energy sector contributes 72% share of total GHG emissions in Hungary, with the transport sector accounting for the largest share of 31%, followed by housing and built environment sectors, agriculture, forestry and fishing. The share of renewable electricity generation accounts for 19% and of natural gas electricity production for 27%. The increase of the RE share in the gross final energy consumption is expected to achieve at least 29% by 2030 through diversification, emission reduction, capitalising on indigenous resources and enhanced energy security (European Commission. Directorate-General for Energy, 2023).

Table 3. Key strategic targets of the energy transition in Hungary

Component	Strategic goal
Climate neutrality	Climate-neutral by 2050.
	To reduce by 50% GHG emissions by 2030, relative to 1990 levels.
RE	To achieve at least 29% share of RE in final energy consumption in 2030, relative to 1990 levels.
Coal phase-out and structural change	Phasing out coal and lignite from domestic electricity production by 2030, transformation of the lignite fired Matra power plant which accounts for 14% of total domestic GHG emissions.
Energy efficiency	To reduce by 20% primary energy consumption, relative to 1990 levels, final energy consumption in 2030 not exceeding 750 PJ.
Clean hydrogen	To reach 240 MW of electrolyser capacity by 2030

Source: Own elaboration

The historical over-reliance of Hungary on import of fossil fuels (natural gas from Russia) poses significant risks, including vulnerability to supply disruptions and geopolitical tensions, as well as exposure to volatile global energy prices. To mitigate these risks, Hungary is investing in a variety of RE technologies such as solar, biomass, and geothermal energy (indigenous geothermal resources among the richest in Europe) with the purpose to stimulate technological innovation and job creation within the RE sector, thus contributing to economic growth and regional development.

Hungary has a substantial solar energy potential, with average annual solar irradiation levels suitable for photovoltaic (PV) power generation. By the end of 2023, installed solar PV capacity reached approximately 5,649 MW, surpassing the initial target of 6,000 MW set for 2030.

The production of green hydrogen production using RE is another priority. One core initiative of the National Hydrogen Strategy is the creation of "hydrogen valleys" in regions like Transdanubia and northeastern Hungary. These areas are intended to integrate hydrogen production, storage, and utilisation into cohesive systems, fostering localised hydrogen economies and advancing clean energy goals. The strategy prioritises green hydrogen production using RE, with an ambitious target of reaching 240 MW of electrolyser capacity by 2030. Other priorities include decarbonizing hard-to-electrify sectors, notably heavy industry and transportation through the use of hydrogen as a substitute for fossil fuels in industrial processes, supporting a shift to low-carbon operations. By 2030, Hungary aims to establish at least 20 hydrogen refuelling stations, creating an infrastructure that facilitates hydrogen-powered mobility. The hydrogen sector is envisioned as a significant catalyst for new employment opportunities and skills development, promoting economic growth and strengthening regional economies.

Furthermore, a range of policies and initiatives across various sectors are aimed at reducing primary energy consumption:

Table 4. Sectoral measures to reduce primary energy consumption in Hungary

Sector	Measures to reduce primary energy consumption
Housing and Built environment	a) Energy-Efficient Renovations of existing residential and public buildings to improve insulation, replace windows and doors, and upgrade heating and cooling systems; b) Nearly Zero-Energy Buildings (NZEB): setting stricter energy performance standards for new buildings to ensure they are highly energy-efficient and incorporate RE sources.
Industry	a) Mandatory energy audits for large enterprises to identify opportunities for energy savings; b) Financial Incentives: Grants and low-interest loans for small and medium-sized enterprises (SMEs) to invest in energy-efficient technologies and processes.
Transport and mobility	a) Promotion of Electric Vehicles (EVs) through subsidies for purchasing and investments in expanding the charging infrastructure; b) Enhancing efficiency of public transport systems to reduce reliance on private vehicles.
Energy supply	a) Upgrading existing power generation facilities to improve efficiency and reduce losses; b) Enhancing the efficiency of district heating networks and integrating RE sources.
Awareness and Education	Educating citizens and businesses about the benefits of energy efficiency and ways to reduce energy consumption.
Training programs	Educational programs to train professionals in energy-efficient technologies and practices.

Source: Own elaboration

Context in Ireland

Climate neutrality by 2050, a significant reduction of GHG emissions and an increase of the RE share in the total energy mix are the main goals of Ireland's Long-term Strategy on Greenhouse Gas Emissions Reduction and Ireland's Climate Action Plans (CAP) of 2021, 2023 and 2024.

The agriculture sector is the largest contributor to GHG emissions at 38.5%, followed by the transport and the energy sectors at 19.4% and 16.6% respectively. Sectoral emissions ceilings were established and approved by the government in 2022. The sectors with the highest levels of emissions reduction targets by 2030 are electricity (75% reduction relative to 2018), transport (50%), commercial buildings (45%), residential buildings (40%), industry (35%) and agriculture (25%) (Government of Ireland, 2023).

CAP 2024 highlights that Ireland has excellent RE resources, already playing a vital role in the domestic fuel mix, with a 36.4% share in 2021 while renewable fuels accounted for 4% of Ireland total final energy consumption in 2021. Wind, solar, marine and bioenergy, alongside new innovative technologies that support the integration of renewables, are key components of the transition in the energy sector in Ireland. By 2030 Ireland's target is to achieve an 80% share of electricity generation supplied by RE sources. The achievement of this target is supported by the commitment to achieve 9 GW of onshore wind, 8 GW of solar PV, at least 5 GW of offshore wind, and 500 MW of community energy by 2030 (Government of Ireland, 2024a, 2024b).

Among the core GHG emissions abatement actions, the focus is placed on shifting demand towards low-emission alternatives while maintaining value added services in society; reducing energy intensity across sectors (without reducing comfort or output) through retrofits, processes or equipment improvement; shifting core energy processes from fossil fuels to electrified substitutes while building out renewables' capacity to secure green power supply; blending in zero-carbon fuels into current fossil applications (H2 and biogas into the natural gas grids and biofuels into road transport); shifting heating in the built environment from localized to district heating networks, particularly in urban areas; and fostering sustainable farming.

Table 5. Key strategic targets of the energy transition in Ireland

Component	Strategic goal
Climate neutrality	Climate-neutral by 2050.
	To reduce by 42% GHG emissions by 2030, relative to 2005 levels.
RE	To achieve a 43% share of RE in final energy consumption in 2030.
Coal phase-out and structural change	To reduce fossil fuel use from 64% of final consumption (2021) to 45% by 2025 and further by 2030. To reduce fossil fuel demand in industry by 10% through energy efficiency measures.
Energy efficiency	To reduce by 12.6% by 2030 final energy consumption, relative to 2022 levels.
Clean hydrogen	To reach 2 GW of renewable hydrogen by 2030

Source: Own elaboration

Table 6. Sectoral measures to reduce primary energy consumption in Ireland

Sector	Measures to reduce primary energy consumption
Housing and Built environment	a) Energy intensity: zero-emission (new); b) Buildings and retrofits of existing buildings; c) Alternative heating: heat pump deployment, roll out of district heating, phase out of fossil fuel heating.
Industry	a) Electrification, industrial heat pumps for low-medium temperatures; b) Zero-carbon heating: zero emissions gases for high temperature heat.
Transport and mobility	a) Promotion of Electric Vehicles (EVs), including build-out of required charging infrastructure; electrification of freight and bus transport, use of clean hydrogen and of biofuels as a transitional measure. b) Enhancing efficiency of public transport systems to reduce reliance on private vehicles.
Electricity	a) Expand renewable power capacity; b) Enhancing grid capacity; c) Flexibility of energy storage.
Agriculture	Enhancing GHG efficient farming and diversification of farm activities
LULUCF (land use, land use change and afforestation)	Increasing afforestation and rewetting of exploited peatland

Source: Own elaboration

Context in Italy

Climate neutrality by 2050 is the overall goal of Italy's long-term strategy and the integrated plan for energy and climate (NECP), to meet the Paris Agreement commitments and the energy union objectives. The priority focus is placed on the growth of RE sources (30% of final consumption); production of renewable gases (biomethane and hydrogen) and other biofuels; the improvement of energy efficiency through building renovations and electrification of final consumption; deployment of electric mobility; and the reduction of GHG emissions. From 1998 to 2018, the contribution of fossil fuels in Italy fell from 95% to under 80% in the total energy mix, with petroleum products and solid fossil fuels gradually being replaced by natural gas. In parallel, renewable energy sources have quadrupled their weight between 1990 and 2018.

Photovoltaics, wind power, and hydrogen are expected to play a transformative role in the decarbonisation process, increasing the share of RE in the national energy mix, enhancing energy security, and reducing GHG emissions. Hydropower (39%), solar sources (21.5%), bioenergy (16%), wind (18%) and geothermal sources (5%) were the main contributors to RE electricity production in 2021 (Italian Ministry of the Environment and Energy Security, 2024).

Among the sectors, transport is the largest contributor to GHG emissions with 30%, followed by industry with 25%.

The latest update of the National Integrated Plan for Energy and Climate sets the following strategic targets for the energy transition in Italy (Italian Ministry of the Environment and Energy Security, 2024):

Table 7. Key strategic targets of the energy transition in Italy

Component	Strategic goal
Climate neutrality	Climate-neutral by 2050.
	To reduce 66 % GHG emissions by 2030, relative to 2005 levels.
RE	To achieve a 39.4% share of RE in final energy consumption in 2030. In the transport sector, the target is set at 34%; at 36 % in the heating and cooling, and at 63% in the final consumption of the electricity sector.
Coal phase-out and structural change	To reduce fossil fuel use by 40% in gross final energy consumption by 2030.
Energy efficiency	To reduce by 32.5% primary energy consumption. The target of primary energy consumption is 123 Mtoe and 102 Mtoe of final energy consumption. Focus on energy efficiency in the civil and transport sectors.
Clean hydrogen	To reach 5 GW of renewable hydrogen by 2030

Source: Own elaboration

The use of clean hydrogen is prioritised for the hard to abate industry, as well as in the transport sector. A significant contribution is expected from bioenergy, by maximizing the development of biogas and its upgrading to biomethane, which can be used not only in thermal end-uses, but also in the production sector. Furthermore, solid fuel consumption should be reduced to a minimum by abandoning the use of coal in the industrial sector and leaving only those solid fuels that can be recovered through the circular economy. The consumption of oil products will be almost entirely directed to the non-energy sector.

Table 8. Sectoral measures to reduce primary energy consumption in Italy

Sector	Measures to reduce primary energy consumption
Housing and Built environment	a) Energy-efficient renovations of existing residential and public buildings to improve insulation, electrification of energy consumption, upgrading heating and cooling systems with heat pumps, automation and control of energy efficiency; b) Deployment of domestic photovoltaic systems.
Industry	a) Reducing solid fuel consumption to a minimum by abandoning the use of coal in the industrial sector, leaving only those solid fuels that can be recovered through the "circular economy"; b) The consumption of oil products will be almost entirely directed to the non-energy sector.
Transport and mobility	a) Promotion of electric mobility; b) Focus on public transport systems to reduce reliance on private vehicles.
Energy supply	a) Upgrading existing power generation facilities to improve efficiency and reduce losses; b) Enhancing the efficiency of district heating networks and integrating RE sources;

	c) Maximizing the development of biogas and its upgrading to biomethane, which can be used in thermal end-uses, but also in the production sector.
Awareness and Education	Educating citizens and businesses about the benefits of energy efficiency and ways to reduce energy consumption.
Training programs	Educational programs to train professionals in energy-efficient technologies and practices.

Source: Own elaboration

Context in Portugal

Portugal's dedication to meeting its commitments under the Paris Agreement and the European Union's Energy Union objectives has been a cornerstone of its energy policy. The country's strategies emphasize the transition to a low-carbon economy while fostering innovation, social inclusivity, and energy independence. Decarbonization of energy systems, enhancing energy efficiency in industrial, residential, and commercial sectors, as well as promoting sustainable mobility, are the key strategic dimensions of the energy transition in Portugal. Specific measures include phasing out coal power plants, increasing the adoption of green hydrogen technologies, and implementing carbon capture and storage (CCS) solutions. A nearly complete shift to renewable energy, aiming for 100% of renewable electricity generation is expected by 2050.

Portugal has prioritized the integration of wind and solar PV systems. A key element of this strategy is hybridizing existing wind farms with solar PV to maximize energy output and grid stability. The country also invests in offshore wind pilot projects to expand its renewable infrastructure. Furthermore, green hydrogen is viewed as a transformative solution for decarbonizing industry and energy storage. The Hydrogen Roadmap outlines the development of 2GW of electrolysis capacity by 2030. Integrating hydrogen production into industrial processes and transport systems is expected to position Portugal as a European hub for hydrogen exports.

Table 9. Key strategic targets of the energy transition in Portugal

Component	Strategic goal
Climate neutrality	Climate-neutral by 2050. To reduce by 55 % GHG emissions by 2030, relative to 2005 levels.
RE	To achieve a 47% share of RE in final energy consumption in 2030.
Coal phase-out and structural change	Achieved in November 2021, by closing the last coal plant Pego.
Energy efficiency	35% improvement in energy efficiency
Clean hydrogen	To reach 2 GW of renewable hydrogen by 2030

Source: Own elaboration

Portugal faces several challenges in implementing its energy strategies, including securing investment for large-scale renewable energy projects, modernizing its grid infrastructure, and ensuring equitable access to clean energy. However, the country's favourable geographic conditions for solar and wind power, combined with its political commitment, provide significant opportunities for success. Policies like the NECP and RNC2050 set a clear direction for reducing carbon intensity while fostering innovation in energy technologies.

Portugal's energy strategy is underpinned by ambitious goals to ensure alignment with global climate imperatives, regional objectives such as the European Green Deal, and national socio-economic priorities. These goals are structured to achieve a balance between decarbonization, energy security, technological innovation, and social equity.

The cornerstone of its energy transition strategy is decarbonization. By committing to reduce GHG emissions by 55% by 2030, Portugal seeks to align its energy systems with the Paris Agreement. The planned phase-out of coal-fired power plants by 2030 is a critical milestone, complemented by a substantial increase in renewable energy capacity. Investments in wind, solar, and hydropower are pivotal to this transition, with the aim of achieving carbon neutrality by 2050.

Energy security remains a top priority, given its reliance on imported energy in the past. The government aims to reduce energy dependency by diversifying energy sources and increasing renewable energy production. Energy storage systems, particularly battery and hydrogen storage, are being developed to ensure grid stability and manage intermittency in renewable energy supply. This strategy not only enhances resilience but also positions the country as a leader in green hydrogen production.

Portugal's energy transition strategy emphasizes fostering economic competitiveness by supporting innovation in renewable technologies and energy efficiency. Public-private partnerships and financial incentives have been introduced to encourage investment in the green economy. SMEs play a significant role in this transition, with targeted programs to improve their energy efficiency and reduce operational costs. By leveraging cutting-edge technologies such as AI and IoT, Portugal is driving down the costs of renewable energy integration.

Portugal's energy strategy is structured around several priority action lines designed to achieve its ambitious climate and energy goals. These action lines encompass expanding renewable energy capacity, enhancing energy efficiency, modernizing energy infrastructure, promoting hydrogen as a key energy vector, and fostering innovative technologies for a sustainable future. Each action line aligns with the overarching objectives of the National Energy and Climate Plan (NECP) and the Paris Agreement, aiming to ensure energy security, economic growth, and environmental sustainability.

Expanding renewable energy capacity remains a cornerstone of Portugal's energy strategy. The country has committed to achieving 47% renewable energy in its gross final energy consumption by 2030. This involves significant investments in wind, solar, and hydroelectric power. Notably, hybrid systems combining wind and solar technologies have been prioritized to maximize energy output and reduce variability. The integration of these renewable sources into the national grid is supported by substantial EU funding, which aims to add 12 GW of capacity by 2030.

Key initiatives under this action line include offshore wind development, leveraging Portugal's extensive coastline to develop offshore wind farms; solar PV expansion; and small-scale renewables to promote decentralized renewable energy systems, such as rooftop solar panels and community wind projects.

Improving energy efficiency across sectors is another critical priority. The government aims to reduce energy consumption in buildings, transportation, and industry by implementing rigorous efficiency standards and retrofitting programs. For instance, the "Casa Eficiente 2020" program provides financial incentives for energy-efficient home renovations, targeting a 30% reduction in energy consumption in residential buildings.

Table 10. Sectoral measures to reduce primary energy consumption in Portugal

Sector	Measures to reduce primary energy consumption
Housing and Built environment	Upgrading insulation, windows, and HVAC (heating, ventilation and air conditioning) systems in residential and commercial buildings.
Industry	a) Introducing energy efficiency standards for industrial equipment and processes; b) Transition from fossil fuels to renewable energy sources. c) Promotion of green technologies and innovation to reduce energy costs.
Transport and mobility	d) Promotion of electric mobility; e) Focus on public transport systems to reduce reliance on private vehicles.
Energy supply	f) Advanced Metering Infrastructure (AMI): Rolling out smart meters to households and businesses. g) Energy Storage Integration: Developing large-scale battery storage systems to balance supply and demand. h) Digitalization: Employing IoT and AI technologies to predict and manage energy consumption patterns.
Awareness and Education	Educating citizens and businesses about the benefits of energy efficiency and ways to reduce energy consumption.
Training programs	Educational programs to train professionals in energy-efficient technologies and practices.

Source: Own elaboration

The modernization of Portugal's energy infrastructure is essential to accommodate the increasing share of renewables and ensure grid stability. Smart grids are being deployed to enhance energy distribution, improve demand-response capabilities, and integrate advanced metering systems. These grids enable real-time monitoring and management of energy flows, reducing losses and optimizing resource utilization. Furthermore, hydrogen is emerging as a pivotal energy carrier in Portugal's transition to a low-carbon economy. The National Hydrogen Strategy (EN-H2) outlines plans to produce green hydrogen using renewable electricity, which can be used for energy storage, industrial processes, and transportation. By 2030, Portugal aims to deploy 2 GW of green hydrogen capacity and establish hydrogen clusters in industrial hubs. Key activities under this action line include scaling up electrolyser installations for green hydrogen production; utilizing hydrogen in steelmaking, chemical production, and other energy-intensive industries; and positioning Portugal as a key exporter of green hydrogen within Europe.

Context in Spain

Aligned with the European Green Deal, the REPowerEU and the Paris Agreement, Spain is committed to climate neutrality by 2050. The Long-Term Decarbonization Strategy 2050 is a roadmap to move forward with mid-term objectives in 2030 and 2040 (Ministerio para la Transición Ecológica y el Reto Demográfico, 2020).

The focus on reduction of GHG emissions is intertwined with priorities of energy efficiency, reduction of energy consumption and introduction of carbon-neutral technologies. The Energy and Climate National Integrated Plan 2023-2030 is a national strategic framework that outlines energy and climate policies and targets in accordance with national and European regulations. The Plan has been recently updated from the initial version, outlining the framework for 2020-2030, to maximize the impact of energy transition and to present new revised commitments (Ministerio para la Transición Ecológica y el Reto Demográfico, 2024). The reduction in GHG emissions for 2030 has been increased to 32%, and the target for final consumption of RE is fixed at 48% (with 81% of the electricity generation), and of energy efficiency at 43%. National energy dependence is expected to drop from 73% in 2019 to 50% in 2030. In parallel, the electricity demand has grown 34%, and the electrification of the economy reaches 35%. Spain is the first large European country in which renewable electricity generation has covered more than 50% of demand.

The average energy expenditure of households is expected to fall from 7.8% of their income in 2019 to 5.7% in 2030, with a higher incidence in low-income households. The installation of 76 GW of PVs sources (with 19 GW of self-consumption), 62 GW of wind sources, 22.5 GW of storage capacity and 12 GW of electrolyzers to obtain renewable hydrogen are expected by 2030 (Ministerio para la Transición Ecológica y el Reto Demográfico, 2024).

Table 11. Key strategic targets of the energy transition in Spain

Component	Strategic goal
Climate neutrality	Climate-neutral by 2050.
	To reduce by 32 % GHG emissions by 2030, relative to 1990 levels.
RE	To achieve a 48% share of RE in final energy consumption in 2030. 81% of RE in electrification by 2030.
Coal phase-out and structural change	To complete coal phasing-out in 2025. Five remaining coal plants to be closed.
Energy efficiency	43% improvement in energy efficiency, 50% share of indigenous energy production in total energy mix by 2030, total installed power of 214 GW in the electricity sector by 2030.
Clean hydrogen	To reach 12 GW of renewable hydrogen by 2030.

Source: Own elaboration

To achieve the ambitious targets of electricity generation from RE sources by 2030, Spain prioritises the promotion of large generation projects, the deployment of self-consumption installations, and measures to boost flexibility of energy storage and energy demand management. The country has high potential for RE sources due to Mediterranean and Atlantic winds, high levels of sunshine, extensive forests and remarkable hydraulic resources, which are complemented by an important business, technological, innovation and knowledge fabric.

In 2023, renewable energy surpassed 50% of total electricity generation for the first time in history, making Spain the first among major European economies to achieve this milestone. Solar PV experienced the highest growth, increasing by 193% from 8,747 MW to 25,549 MW, excluding self-consumption, which is largely integrated on the demand side. This growth has generated local employment and reduced energy costs for SMEs, households, and public administrations. Furthermore, wind power capacity increased by 20% over this period, rising from 25,678 MW in 2019 to 30,810 MW in 2023. In 2022, renewable energy sources created over 130,000 jobs, a 54% increase compared to 2018.

The promotion of RE self-consumption is also a high priority, enabled by the availability of RE resources in Spain; the allocation of 2,000 million euros by the Recovery, Transformation and Resilience Plan to incentivise energy self-consumption; the modularity of the facilities; the reduction of costs; and the new regulation introduced since 2018, which abolishes taxes and charges for self-produced energy and allows economic compensation for surpluses injected into the energy grid. RE self-consumption is expected to achieve 19 GW by 2030, which could cover 11% of energy demand.

Energy efficiency is a key driver of the twin transition process in Spain, expected to meet the target of 43% of total energy demand in 2030. On the path to full decarbonisation, efficiency strategic goals aim to significantly reduce fossil fuel consumption, while in parallel reducing global energy needs, that could result in a lower need for renewable deployment and associated infrastructures.

Spain's "Hydrogen Roadmap: A Commitment to Renewable Hydrogen" is a national strategy aimed at developing a competitive renewable hydrogen sector as a key pillar of the country's energy transition and decarbonization goals (Ministerio para la Transición Ecológica y el Reto Demográfico, 2020). Approved in October 2020, the roadmap aligns with EU hydrogen strategies and Spain's Integrated National Energy and Climate Plan (PNIEC 2021-2030) to promote a clean, secure, and competitive energy system. Spain's geographic advantages (high solar and wind potential) and commitment to renewable energy make it well-positioned to become a major renewable hydrogen hub for Europe. The roadmap aims to reduce energy dependence, enhance industrial competitiveness, and contribute to the EU's carbon neutrality goals by 2050. Key Objectives of Spain's Hydrogen Roadmap include the expansion of clean hydrogen production from solar, wind and other green sources; the deployment of 12 GW of installed electrolyser capacity; prioritizing hydrogen use in hard-to-abate sectors, developing hydrogen transport, storage and refuelling infrastructures, fostering hydrogen valleys (industrial hubs integrating hydrogen production, distribution, storage and consumption within a specific region), boosting R&D and innovation, as well as economic and employment opportunities.

As of 2024, the following hydrogen valley initiatives, supported by public and private investment, operate in Spain (AEH2-Asociación Española de Hidrógeno, 2024; Casas, 2025):

- Basque Hydrogen Corridor (BH2C): one of Spain's most advanced hydrogen valleys that focuses on industrial decarbonization, transport applications, and hydrogen infrastructure development, led by Petronor (Repsol), with over \$1.5 billion investment and participation from 80+ companies;
- Catalonia Hydrogen Valley, which focuses on mobility (hydrogen-powered buses, trucks, and trains) and port operations in Barcelona and Tarragona, and aims to develop a regional hydrogen economy by integrating production, storage, and usage in key industries;
- Green Hydrogen Hub in Andalusia, led by Cepsa and aimed at creating one of Europe's largest hydrogen production sites. This hub focuses on exporting hydrogen to Northern Europe and decarbonizing heavy industry and shipping;
- Aragon Hydrogen Valley, built around Huesca's hydrogen production center, with strong support from public institutions and companies like Iberdrola and focused on industrial applications and hydrogen-powered transport solutions;
- Murcia and Valencia Hydrogen Initiatives, which are projects aimed at supporting local industries, mobility solutions, and infrastructure development, with stakeholders from energy companies, universities, and regional governments.

Finally, ensuring the diversification of the national energy mix, reducing external energy dependence, guaranteeing security of supply and access to energy sources, as well as promoting the use of indigenous energy resources are expected to provide energy national security. While, in 2019, external energy dependence of Spain was 73% due to a high reliance on fossil fuels (coal, oil and gas) in the national energy mix, the target for 2030 is 50%. On the other hand, the current interconnection of Spain's electricity grid with Europe is low, with less than 5% of the generation capacity, being the only country in continental Europe below the 10% target of the EU. The target of interconnection of the Spanish electricity grid is set at 15% by 2030. New interconnections are planned with Portugal and France, expected to increase the exchange capacity.

Table 12. Sectoral measures to reduce primary energy consumption in Spain

Sector	Measures to reduce primary energy consumption
Housing and Built environment	a) Upgrading insulation, cooling and heating systems in residential and commercial buildings; b) Increasing the deployment of RE self-consumption installations; c) Promoting green and sustainable architecture and enabling easy adaptation of the building stock (both new and existing buildings) to the best existing standards of energy and water consumption.
Industry	a) Coal phasing-out and reduction of fossil fuel consumption; b) Energy efficiency improvements; c) Increasing the share of RE sources in transport, heating, cooling and electrification systems
Transport and mobility	a) Modal shift from combustion vehicles to collective public and shared transport, and non-emission modes of electric mobility; b) Reducing use of private vehicles; c) Delimitation of low-emission zones in cities of more than 50,000 inhabitants; d) Promotion of smart urban mobility solutions. e) Focus on sustainable and advanced biofuels.
Energy supply	a) Decarbonization of the energy sector; b) Electrification of the energy final consumption; c) Support to establishment of energy communities for self-generation, self-consumption and distribution of RE.
LULUCF	Increasing CO2 sequestration and reducing GHG emissions.
Awareness and Education	Educating citizens and businesses about the benefits of energy efficiency and ways to reduce energy consumption.
Training programs	Educational programs to train professionals in energy-efficient technologies, key competencies of digitalization and energy efficiency and circular economy practices.

Source: Own elaboration

2.2. IMPACT OF THE TWN TRANSITION ON STRUCTURALLY VULNERABLE COHORTS

While the transition is expected to generate substantial social and economic benefits such as improved economic resilience, research and innovation leadership, employment, and reliable and affordable access to renewable energy sources, a commitment to globally shared prosperity means that the benefits and burdens of the transition process should be equally distributed and no one can be left behind, in particular, those who remain dependent on fossil fuels, or lack access to energy (IRENA, 2024).

2.2.1. ENERGY POVERTY: KEY INDICATORS AND DETERMINANTS

Energy Poverty (EP) is one of the critical challenges of the twin transition process in the EU. 8%-16% of the EU population face EP. Low income, a high proportion of household expenditure spent on energy and low energy performance of buildings and appliances are considered three critical determinants of EP. (European Commission: Energy Poverty Advisory Hub, 2023).

The multidimensional and complex nature of the EP phenomenon is contextualized by the EU Energy Poverty Advisory Hub (European Commission: Energy Poverty Advisory Hub, 2023) that highlights the need to consider dimensions of Climate, Facilities and Housing, Mobility and Socioeconomic Aspects and their corresponding sub-topics as key indicators of EP.

Climate conditions impact household energy needs and thermal comfort, influencing EP vulnerability. Colder climates increase heating demands, while hotter climates raise cooling costs, exacerbating financial strain on energy-poor households. Moreover, climate change is intensifying extreme weather events, making this issue increasingly relevant.

The **Facilities and Housing** dimension relates to the building stock quality, energy efficiency, and fuel use, which directly impact EP as poorly insulated or inefficient housing increases energy consumption, making thermal comfort unaffordable for structurally vulnerable groups. Improving energy efficiency can significantly reduce EP.

The **Mobility** dimension explores the link between transport poverty and EP, as energy-poor households often struggle with access to affordable, reliable transportation. Limited mobility options can lead to higher transport costs, reinforcing financial struggles and worsening EP. This dimension highlights a **double energy vulnerability** phenomenon, where households face both high transport and energy costs, deepening economic hardship.

Socioeconomic Aspects cover factors such as income, employment, energy prices, housing costs, and health impacts related to EP. Economic hardship and high energy prices create a vicious cycle, where EP worsens financial instability, leading to further vulnerability. Some socioeconomic factors are direct causes of EP, while others contribute to or result from broader vulnerability settings (European Commission: Energy Poverty Advisory Hub, 2023). Furthermore, the gender dimension of EP has been highlighted by Murauskaite-Bull et al. (2023) who emphasize that lower levels of disposable income of women and their overrepresentation in single parent households lead to their frequent exposure to EP, with consecutive adverse health risks and social exclusion.

Table 13 outlines the key primary macro dimensions of energy poverty (EP), along with their corresponding sub-topics and specific indicators, which highlight the complexity and multidimensional nature of the EP phenomenon and must be considered for a holistic assessment of energy sector inequalities.

Table 13. Energy Poverty indicators

Topic	Subtopic	Indicator
Climate		Cooling degree days
		Heating degree days
Facilities/ housing	Building Stock	Dwellings with energy label A
		Final consumption expenditure of households ¹
		Pop. Liv. Dwelling with presence of leak, damp and rot
		Pop. Liv. Dwelling equipped with heating
		Pop. Liv. Dwelling equipped with air conditioning
		Pop. considering their dwelling as too dark
	Energy Consumption and Equipment	Final consumption expenditure of households ²
		Final energy consumption in households by energy use
		Final energy consumption in households by type of fuel
Mobility		Final consumption expenditure of households ³
		Pop. who cannot afford a regular use of public transport
Socioeconomic aspects	Socio Economic and Living Conditions	Arrears on utility bills
		At risk of poverty or social exclusion
		Disposable annual household income
		Inability to keep home adequately warm
		Final consumption expenditure of households ⁴
		Housing cost overburden rate
		Pop. Liv. Dwelling comfortably cool during summer time
		Pop. Liv. Dwelling comfortably warm during winter time
	Energy Expenditure and Energy Markets	Energy expenses by income quintile
		Energy Prices
		High share of energy expenditure in income (2M)
		Low absolute energy expenditure (M/2)
	Health	Causes of death
Excess winter mortality/deaths		
Final consumption expenditure of households ⁵		
Pop. Reporting a chronic disease		

Source: European Commission: Energy Poverty Advisory Hub (2023)

Economic hardship and unemployment are key indicators of vulnerability that also contribute to energy poverty (EP). Additionally, gender and age dimensions further shape vulnerability, as illustrated in Tables 14 through 19. While over one in five people in the EU face the risk of poverty or social exclusion, women are more affected than men (21.8% compared to 20.0%). The situation is particularly challenging in Spain and Italy for women aged 18–64, where the rates are 27.4% and 24.6% respectively, with little change over the past five years (Tables 14 and 15).

Table 14. Female share of population at risk of poverty and social exclusion aged 18 to 64, 2019-2023 (%)

TIME	2019	2020	2021	2022	2023
GEO (Labels)					
European Union	22,0	22,1	22,4	22,0	21,8
Germany	18,2	20,3	21,1	21,3	21,4
Ireland	21,3	19,8	19,5	18,9	19,0
Spain	28,6	28,7	29,7	27,3	27,4
Italy	27,2	27,3	27,4	26,6	24,6
Hungary	20,5	18,6	18,7	19,1	18,1
Portugal	22,1	19,6	22,3	19,8	19,8

Source: Eurostat (2024g)

Table 15. Male share of population at risk of poverty and social exclusion aged 18 to 64, 2019-2023 (%)

TIME	2019	2020	2021	2022	2023
GEO (Labels)					
European Union	20,2	20,4	20,9	20,2	20,0
Germany	16,2	19,0	19,6	19,9	20,2
Ireland	18,8	17,1	17,0	15,5	15,6
Spain	25,8	25,9	27,3	24,3	24,8
Italy	24,7	24,6	26,0	23,5	21,7
Hungary	19,3	18,3	17,7	17,7	17,7
Portugal	20,2	18,2	21,0	19,8	19,0

Source: Eurostat (2024g)

Among the population aged 65 and over in the EU, women experience a significantly higher risk of poverty and social exclusion than men (22.1% compared to 16.6%), as shown in Tables 16 and 17. Germany, Portugal, and Spain exceed the EU average rates for both women and men in this age group, while Hungary shows a rising impact for older women over the last five years.

Table 16. Female share of population at risk of poverty and social exclusion aged 65 or over, 2019-2023 (%)

TIME	2019	2020	2021	2022	2023
GEO (Labels)					
European Union	22,0	22,6	22,1	22,8	22,1
Germany	20,8	22,6	22,1	21,8	22,8
Ireland	13,2	18,4	21,9	28,7	21,0
Spain	18,3	23,4	23,1	23,4	23,2
Italy	22,0	21,7	20,0	22,8	21,1
Hungary	19,0	21,1	21,7	20,4	24,7
Portugal	22,4	23,9	26,9	22,7	22,4

Source: Eurostat (2024g)

Table 17. Male share of population at risk of poverty and social exclusion aged 65 or over, 2019-2023 (%)

TIME	2019	2020	2021	2022	2023
GEO (Labels)					
European Union	16,2	16,8	16,2	16,7	16,6
Germany	16,8	19,0	18,5	17,4	17,9
Ireland	10,3	13,6	15,7	20,8	16,3
Spain	17,9	19,3	17,2	18,6	18,1
Italy	15,7	16,3	15,6	16,5	16,9
Hungary	12,9	19,1	16,6	15,3	15,9
Portugal	17,5	18,0	20,5	17,5	17,2

Source: Eurostat (2024g)

Employment status also plays a critical role in vulnerability. Spain, Italy, and Portugal report the highest unemployment rates for both women and men among the six countries (Tables 18 and 19). Although these rates have slightly declined over the past five years, unemployment in Spain remains nearly double the EU average.

Table 18. Male share of unemployed people aged 20-64 in the labour force (%)

TIME	2019	2020	2021	2022	2023
GEO (Labels)					
European Union	6,4	6,8	6,6	5,7	5,6
Ireland	4,9	5,5	6,0	4,1	4,1
Spain	12,1	13,6	12,9	11,0	10,3
Germany	3,3	3,9	4,0	3,3	3,2
Italy	9,0	8,5	8,5	7,0	6,7
Hungary	3,2	4,0	3,7	3,6	3,9
Portugal	5,9	6,8	6,5	5,7	6,0

Source: Eurostat (2024h)

Table 19. Female share of unemployed people aged 20-64 in the labour force (%)

TIME	2019	2020	2021	2022	2023
GEO (Labels)					
European Union	7,0	7,4	7,2	6,3	6,2
Ireland	4,3	5,5	5,5	4,2	3,7
Spain	15,7	17,2	16,6	14,7	13,6
Germany	2,5	3,1	3,1	2,8	2,8
Italy	10,9	10,3	10,5	9,2	8,7
Hungary	3,2	4,1	4,1	3,4	4,0
Portugal	7,1	7,1	6,9	6,4	6,7

Source: Eurostat (2024h)

In 2023, 10.6% of European citizens were unable to keep their homes warm, with the highest rate of 20.8% in both Portugal and Spain, followed by Italy (9.5%), Germany (8.2%), and 7.2% in both Ireland and Hungary (Eurostat, 2024f). This data is consistent with the share of people in these countries at risk of poverty or social exclusion.

Table 20. Share of population unable to keep home adequately warm, 2019-2023 (%)

TIME	2019	2020	2021	2022	2023
GEO (Labels)					
European Union	6,9	7,5	6,9	9,3	10,6
Germany	2,5	7,0	3,3	6,7	8,2
Ireland	5,1	3,6	3,4	6,8	7,2
Spain	7,5	10,9	14,2	17,1	20,8
Italy	11,1	8,3	8,1	8,8	9,5
Hungary	5,4	4,2	5,4	4,7	7,2
Portugal	18,9	17,5	16,4	17,5	20,8

Source: Eurostat (2024f)

Context in Germany

The German energy policy recognizes the impact of the energy transition on people in vulnerable positions in the following ways:

- 1) The affordability goal of the transition, which entails efforts to ensure that energy is accessible and affordable for all, including people in low-income households and socially excluded groups. However, the Federal Audit Office (Bundesrechnungshof) indicates that high electricity prices might have regressive distributional impacts on German households. Therefore, the federal government should define what it means by an affordable electricity supply. Measures related to affordability and consumer protection mentioned in the NECP include:
 - An Energy Arbitration Service to settle disputes regarding grid connection, energy supply and energy metering;
 - The use of the “Marktwächter Energie”, an early warning system, to monitor the energy market from the consumer’s perspective and identify undesirable developments at an early stage;
 - Protection against supply disruptions;
 - Housing costs support for low-income households with a CO₂ component;
- 2) The support to regions affected by the coal phase-out, particularly to workers in coal mining regions in North Rhine-Westphalia, Brandenburg, Saxony, and Saxony-Anhalt. Under the Structural Development Act (Strukturstärkungsgesetz), the federal government has allocated around €40 billion to support coal regions. This includes also the recognition of regional disparities as some regions are expected to experience more significant structural change compared to economically diversified regions.
- 3) The reskilling and upskilling programmes to help workers transition to jobs in the growing renewable energy sector.

However, some aspects related to the impact of the transition on structurally vulnerable cohorts have little recognition on policy and strategy documents, despite their increasing recognition in the public debate. First, the risk of increasing EP or the barriers faced by households living in EP to enjoy the benefits of the transition have not been sufficiently recognized and integrated in Germany’s national plans, including the NECP. Second, despite the emergence of gender as an important topic for a just energy transition, there is little integration of the gender and intersectional dimensions in German energy policy.

In addition, as regards working towards a fair and just transition, the Federal Government has committed to carrying out a participatory process in the planning and development of new energy projects and, in general, to foster dialogue with all stakeholders in order to ensure a high level of transparency and public support for the energy transition.

Context in Hungary

Hungary's energy policy acknowledges the potential adverse effects of the twin transition on vulnerable groups, highlighting the phenomenon of EP of households that have difficulties in securing their basic energy needs (European Commission. Directorate-General for Energy, 2023).

A significant focus is on regions heavily reliant on specific industries, such as the Mátra Power Plant area. The NECP highlights the plant's dual role as a critical electricity provider and a major employer, recognizing that its transformation or closure could disproportionately impact local communities. To mitigate such

effects, the plan outlines measures aimed at ensuring a just transition, balancing environmental objectives with social equity.

Furthermore, Hungary's energy strategy emphasizes the need for comprehensive impact assessments of planned policies. This includes evaluating macroeconomic, health, environmental, employment, education, skills, and social implications, with a particular focus on just transition aspects. Such evaluations are intended to identify and address potential negative consequences on vulnerable populations, ensuring that the benefits of the twin transition are equitably distributed (European Commission. Directorate-General for Energy, 2023).

Specific national policies and action plans to address EP of the most vulnerable groups in Hungary are systematized by the EU Horizon 2020 Research and Innovation Programme project POWERPOOR. Actions range from funding and grants for energy efficiency, deep renovations of housing, to social care programmes aimed at most disadvantaged municipalities, and promotion of energy communities for self-consumption of RE.

Context in Ireland

Relevant regulatory frameworks in Ireland, such as the Climate Action Plan or the National Adaptation Framework, recognize the impact of climate actions on vulnerable groups, while the Long-Term Strategy on Greenhouse Gas Emissions Reduction refers to vulnerable groups specifically in the context of just transition. According to the Strategy, vulnerable households need to be supported through the transition, and the policies will be guided by the Just Transition Framework (first articulated in the Climate Action Plan 2021), with a “gender-responsive” approach that also recognizes the needs of people with disabilities. The Climate Action Plan 2024 (CAP 2024), also proposes vulnerability assessments across society, economy, the environment and health, to “ensure a just transition across all sectors” and it reaffirms the just transition principles guiding Irish policy making and implementation, and these are as follows:

- An integrated, structured and evidence-based approach to identify and plan our response to just transition requirements
- People are equipped with the right skills to be able to participate in and benefit from the future climate neutral economy
- The costs are shared so that the impact is equitable and existing inequalities are not exacerbated
- Social dialogue to ensure impacted citizens and communities are empowered and core to the transition process.

The transformation is also acknowledged to be long-term, resulting in shifts in Ireland's economy and society. For these two reasons, the Government aims to focus on potential impacts, risks and opportunities, further identifying and implementing required policy responses (incl. interventions and targeted supports), as well as undertaking a proactive engagement with citizens, especially communities, sectors or regions which are challenged by either acute or long-term impacts of the transition. This includes the National Dialogue on Climate Action, already in place. Finally, to support the government in their commitment to long-term transition management, the government established a Just Transition Commission, which will act as an independent and evidence-driven advisory body.

In addition to the above, it should be noted that Civil Society Organisations also identify vulnerable groups in relation to the transition, with more specific categories discussed, to include: older citizens 60+, members of the Traveler Community, those in private rental sector who are at risk of being displaced and vulnerable in relation to housing energy efficiency, as well as those currently employed in the electricity sector.

The Energy Poverty Action Plan (Government of Ireland, 2022) refers to a recent study by the Economic and Social Research Institute (ESRI 2022), which estimated that 29% of Irish households are in energy poverty. The Action Plan sets out the range of measures implemented across Government during winter 2022-23 to support people with energy costs, as well as the long-term actions taken to ensure those most at risk of energy poverty can adequately heat and power their homes. It is developed, implemented and overseen by cross-departmental and inter-agency Steering Group, chaired by the DECC. At the core of the Steering Group work is facilitating a structured, whole-of-government engagement with key stakeholders. The near-term actions were implemented throughout 2022 and 2023 across three main areas: Income supports (total 2.9bn); Targeted social protection (1.2bn social protection lump sum) and consumer protection (through a package of strengthened obligations on suppliers and network operators mandated by CRU). In addition, further measures are implemented to strengthen the safety net for most vulnerable people.

Context in Italy

The National Energy and Climate Plan, adopted in 2019, already identified the fight against EP as an important energy policy choice, bearing in mind that the transition path towards a sustainable energy system requires special attention to the most vulnerable end users.

In order to ensure the institutional coordination of the analysis and the fight against EP, and in accordance with the provisions of the INECP 2019, the National Energy Poverty Observatory (OIPE) was established. This is an inter-institutional body promoted and managed by the Ministry of the Environment and Energy Security, with tasks including the monitoring of the EP phenomenon and the development of a strategy to combat it.

EP in Italy is defined as the difficulty to purchase a minimum amount of energy goods and services, or access to energy services exceeding a normal value. 2.2 million families (8.5%) in Italy were estimated in EP in 2021 (OIPE, 2025).

As part of the previous Development and Cohesion Plan (instrument for the implementation of the FSC) 2014-2020, a concrete measure to combat EP is reported, namely, the creation of a revolving fund for the construction of photovoltaic installations for self-consumption in favour of households in a state of economic distress. The measure, which aims to combat EP, reduce energy costs and promote the development of photovoltaic self-consumption, is addressed to Southern regions and demonstrates a national commitment to enable a just and fair transition process. It is estimated that cohesion funds could be used to install around 100-180 MW of photovoltaic systems to supply 30,000-60,000 households, with the possibility of installing more systems to redirect the soil.

Context in Spain

The right to access energy is a key premise of the National Integrated Plan for Energy and Climate 2023-2030. The Plan highlights the potential of energy rehabilitation of buildings and deployment of self-consumption systems to mitigate EP and vulnerability (Ministerio para la Transición Ecológica y el Reto Demográfico, 2024). It establishes substantial financial aid to support communities and workers transitioning from fossil fuel industries and investments in green skills training programs to ensure that workers have access to new employment opportunities in the renewable sector. Additionally, the Plan focuses on protecting consumers by promoting energy efficiency in homes and ensuring equitable access to clean energy, particularly in economically disadvantaged areas.

Furthermore, Spain's Just Transition Strategy (Ministerio para la Transición Ecológica y el Reto Demográfico: Instituto para la Transición Justa, 2020) ensures that decarbonization process is fair and

inclusive, with a strong focus on supporting workers and communities affected by the phasing out of fossil fuels. The strategy outlines economic diversification efforts, promoting new industries in renewable energy, green technology, and sustainable agriculture sectors. It emphasizes the creation of quality jobs and provides extensive retraining programs to equip workers with skills for the green economy. Strategic timelines align with the phase-out of coal and other carbon-intensive industries, ensuring a gradual transition that minimizes economic disruption. The plan includes financial support for municipalities to develop local projects that stimulate economic growth and improve infrastructure. By fostering partnerships between government, industry, and communities, the strategy aims to create resilient local economies. For consumers, the Just Transition Strategy promotes access to affordable clean energy and supports initiatives to reduce energy costs, particularly for low-income households. The plan also incorporates social safeguards, such as income support and community development funds, to mitigate potential negative impacts of the energy transition. Regular assessments and stakeholder engagement ensure that the strategy remains responsive to the needs of all sectors, prioritizing fairness and equity throughout Spain's energy transformation.

The National Strategy Against Energy Poverty 2021-2024 (ENPE) (Ministerio para la Transición Ecológica y el Reto Demográfico, 2019) focuses on reducing and ultimately eradicating EP in Spain. Launched in 2019, it adopts a comprehensive approach, with key targets set for 2024. Measures include expanding the Electricity Social Bonus (which provides financial support to vulnerable households) and implementing the Minimum Vital Supply to ensure that basic energy needs are met. ENPE addresses energy inequities by protecting low-income consumers from rising energy prices and enabling access to affordable, clean energy. The strategy also includes initiatives to promote energy efficiency in housing, reducing consumption and costs. By 2024, ENPE aimed to decrease the number of households experiencing EP significantly, ensuring that the benefits of the energy transition are shared equitably. The plan supports a just transition by integrating social protections for the most affected groups, including older populations and low-income families. It also emphasizes the importance of community-based solutions, such as shared energy projects, to empower local populations.

Context in Portugal

One of the primary impacts of the energy transition in Portugal is on EP. High energy costs and inadequate housing conditions have historically burdened low-income households. Portugal's strategy includes retrofitting 30% of homes to improve energy efficiency and providing subsidies to make clean energy more affordable. These measures aim to lower energy bills and enhance living conditions for vulnerable populations. The "Casa Eficiente 2020" program is a notable example, offering financial support for home energy upgrades, thereby reducing EP rates.

Ensuring social inclusion is a critical component of Portugal's energy strategy, as marginalized groups, including rural communities and underrepresented demographics, often face barriers to accessing the benefits of renewable energy. A just and fair energy transition emphasizes inclusivity, equity, and sustainability in addressing the socio-economic and environmental challenges associated with the shift towards renewable energy. Portugal has shown significant commitment to ensuring that the energy transition benefits all sectors of society, including structurally vulnerable cohorts, while meeting its national and international obligations.

Portugal's commitment to a just energy transition aligns with the European Union's Green Deal principles and the United Nations Sustainable Development Goals (SDGs). The country's primary objectives in this context include:

- **Economic Diversification:** Supporting regions and communities dependent on traditional energy industries to diversify their economies.
- **Social Equity:** Ensuring equitable access to energy resources and services, addressing disparities in EP.
- **Workforce Reskilling:** Providing training and education programs to equip workers with skills for emerging renewable energy industries.
- **Public Participation:** Involving stakeholders, including marginalized communities, in decision-making processes.
- **Environmental Restoration:** Investing in projects that mitigate the environmental impacts of decommissioned energy infrastructure.

Portugal has adopted several policy instruments and frameworks to ensure the fair implementation of its energy transition strategy. The National Energy and Climate Plan (NECP 2021-2030) forms a cornerstone of the country's energy transition, focusing on energy efficiency, renewable energy deployment, and significant reductions in greenhouse gas (GHG) emissions. The plan includes targeted measures to alleviate EP through subsidies for low-income households and comprehensive housing retrofits, while also promoting job creation in renewable energy sectors via public and private investments. Complementing the NECP, the Roadmap for Carbon Neutrality 2050 sets forth an ambitious vision of a decarbonized economy characterized by inclusive growth. This roadmap prioritizes investments in sustainable energy and infrastructure projects, particularly in underdeveloped regions, and incorporates policies to retrain workers transitioning from fossil fuel industries.

2.2.2. ENERGY POVERTY: ASSOCIATED VULNERABILITIES

The EP phenomenon is related to a series of demographic determinants, including age, gender, income, employment status, level of education and place of residence. These factors influence households' vulnerability to high energy costs and inadequate access to energy services.

Furthermore, housing conditions and the energy efficiency levels of buildings are important indicators of a potential EP. The intersectionality of determinants that shape EP is particularly relevant as low-income households, the elderly, single parents, and rural populations are usually the most affected.

Income Levels

- Low-income households are the most affected by EP, as they allocate a higher percentage of their income to energy costs.
- Rising energy prices and economic instability (e.g., inflation, financial crises) further increase their vulnerability.
- The cost-of-living crisis in Europe, exacerbated by the COVID-19 pandemic and geopolitical conflicts (e.g., Russia-Ukraine war), has disproportionately impacted lower-income groups.

Employment Status and Job Security

- Unemployment or precarious employment increases the risk of EP, as these groups struggle with unstable income and high energy costs.
- Retired individuals on low pensions often face difficulty maintaining adequate heating or cooling in their homes.
- EP is often linked to seasonal work or low-wage jobs, particularly in sectors with high job insecurity.

Housing Conditions and Energy Efficiency

- Poorly insulated homes or outdated heating systems increase energy consumption, leading to high energy bills.

- Social housing and rental properties are more likely to have low energy efficiency, as tenants often lack control over home improvements.
- In Eastern and Southern Europe, many older buildings lack proper insulation, contributing to higher rates of EP.

Education and Awareness

- Households with lower educational attainment often have limited access to energy-saving information and financial assistance programs.
- Awareness of energy efficiency measures and government support programs can help reduce EP, but information gaps persist in many regions.

Age

- Elderly individuals are particularly vulnerable due to fixed incomes, health issues, and reduced mobility, making it harder to adapt to energy shortages.
- Children in low-income households suffer from EP, impacting their education, health, and well-being.

Disability and health

- People with disabilities and people with chronic illnesses require stable indoor temperatures for their health, making them more sensitive to EP.

Gender

- Single-parent households, particularly those led by women, are at a higher risk due to lower income levels and caregiving responsibilities.
- Women, especially elderly women living alone, experience higher EP rates due to longer life expectancy and lower pensions.

Regional Disparities and the Urban-Rural Divide

- Rural areas often face higher EP rates due to lower incomes and fewer job opportunities, limited access to energy-efficient infrastructure and higher reliance on fossil fuels like coal or wood for heating.
- Urban areas also experience EP, particularly in low-income neighbourhoods with poorly maintained housing.

2.2.3. ECONOMIC AND LABOUR MARKET EXCLUSION TRIGGERS

While transitions to RE sources, energy efficiency and decarbonization drive sustainability and foster new business innovation and employment opportunities, they also pose risks of economic and labour market exclusion for structurally vulnerable groups.

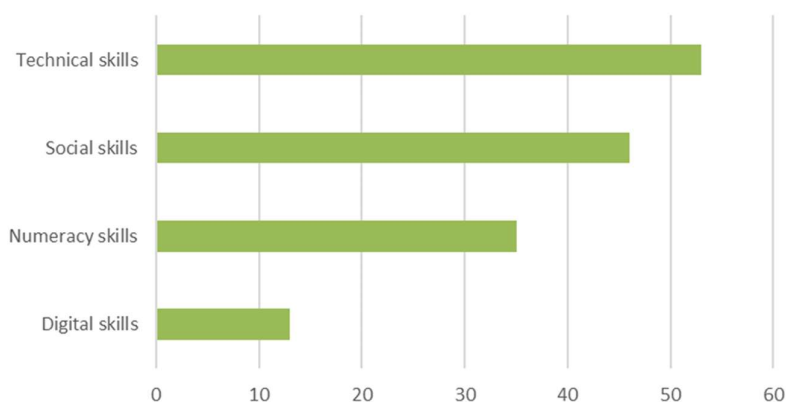
In 2023, employment in the EU's renewable energy sector reached 1.82 million with solar PV, solid biofuels and wind power as the most dynamic sub-sectors (IRENA, 2024).

Table 21. Estimated direct and indirect jobs in RE sectors, EU, 2023 (thousands)

INDUSTRY	2023
Solar photovoltaic	720
Liquid biofuels	150
Hydropower	71
Windpower	282
Solid biomass	333
Solar heating and cooling	22
Biogas	49
Geothermal energy	7
Concentrated solar power	5
TOTAL	1812

Source: IRENA (2024)

While this data highlights the strong potential of the clean energy sector to generate employment, labour shortages remain a key challenge, with around 25% of companies producing electrical equipment in the EU struggling to find workers. Job vacancy rates in energy supply have continued to highlight the need for skilled professionals (European Commission: Joint Research Centre et al., 2023). A general shortage of employees with technical backgrounds in science, technology, engineering and mathematics persists in the RE sector in the EU. The rapid pace of technological innovation can also produce skills obsolescence among highly skilled employees in ICT, managerial and engineering positions. It is estimated that 22% of plant operators, 20% of electrical trades' workers and 20% of science and engineering technicians face risk of skills obsolescence without continuous on-the-job training. Traditional energy workers (e.g., coal miners, oil refinery workers) often lack the technical qualifications needed for emerging green jobs. Older workers may struggle more with retraining, thus leading to structural unemployment and economic displacement. Furthermore, digitalization of the energy sector (e.g., smart grids, AI-driven energy management, blockchain for energy trading) can widen the digital divide. Increased automation in energy production and grid management may replace low- and mid-skilled jobs, leading to further exclusion (European Commission, 2023c). Furthermore, women account only for 32 % of the workforce in the renewable energy sector (Murauskaite-Bull et al., 2023).

Table 22. Skill gaps in the energy supply sector (% of workforce)

Source: European Commission, Joint Research Centre (2023c)

IRENA (2024) identifies skilling as one of the pillars of the just twin transition process in key areas of clean energy, and with a high priority for workers in fossil-fuel industries to prevent their unemployment risks and transition them to RE related sectors. The decline of traditional energy jobs and sectoral disruptions in the phasing out of coal, oil, and gas industries disproportionately affects workers in fossil fuel-dependent regions, leading to job losses and economic decline in areas reliant on mining and heavy industry.

2.2.4. COMMITMENT TO ENABLE A JUST AND FAIR TRANSITION PROCESS

Germany government's strategy includes large-scale funding mechanisms and targeted regional initiatives to support affected workers, businesses, and communities. In addition to the funding from the EU Just Transition Fund to support coal-dependent regions, Germany has allocated additional national co-financing, bringing total investments in affected areas to over €2.5 billion (European Commission, 2022e). The priority focus is placed on reskilling and upskilling workers in coal regions, business incentives to attract new industries and infrastructure modernization for sustainable economic growth. Germany's Coal Phase-Out Compensation & Structural Strengthening Act (Kohleausstiegsgesetz) provides €40 billion in federal funding for coal regions (Lusatia, Rhineland, Saxony-Anhalt and Thuringia) through 2038 and direct compensation to coal companies and workers to mitigate job losses to diversify the local economy beyond coal (Bundesregierung, n.d.). Investments are channeled into research and innovation centers in affected regions, new industries such as green hydrogen, battery manufacturing, and renewable energy production, as well as expansion of public transport and digital infrastructure to enhance regional competitiveness.

In **Hungary**, the NECP includes specific measures to support communities and workers currently reliant on fossil fuel industries, especially in regions heavily dependent on coal. Reskilling programs are integral to the plan, offering job training and educational opportunities for workers to transition into renewable energy sectors. Additionally, the NECP encourages the creation of green energy clusters in former coal-mining areas, helping to revitalise these communities through new job opportunities and sustainable energy production. From a consumer standpoint, the NECP aims to stabilise energy prices by fostering diverse energy sources, including renewables and local generation, protecting low-income households from EP.

In **Ireland**, the NECP emphasises that the green economy sectors will become a source of significant high quality employment growth. These include: retrofitting sector, the circular economy, clean mobility, green and blue infrastructure, sustainable agriculture and the bioeconomy. While it is recognized that the transformation will have geographic, economic and capability-based implications for Ireland's workforce, it has also been noted that the ongoing transformation of the economy through information and communication technologies will help to facilitate the transition to a climate neutral economy.

Italy recognizes that achieving climate neutrality and digital modernization must go hand in hand with protecting workers, vulnerable communities, and regional economies. The country was allocated €1 billion from the EU Just transition Fund to support green transition in Taranto, Apulia, and in Sulcis Iglesiente, Sardinia to foster economic diversification and job creation in green sectors, including RE (European Commission - Press release, 2022). In the Taranto province, the priority is placed on the transformation of the largest steel mill of Europe through the development of new business models, clean hydrogen and retraining of 4,300 workers for green jobs and a circular economy. Leveraging the potential of women in new green jobs and circular economy as well as supporting the most vulnerable is also emphasised. Furthermore, allocated funding will support the development of green hydrogen, wind turbines and geothermal installations to ensure affordable RE sources. Italy is investing in vocational training, STEM education, and upskilling programs to prepare workers for green and digital jobs. The National Recovery and Resilience Plan (NRRP) allocates funds for bridging the digital divide, especially in rural and underdeveloped areas. In the region of Sulcis Iglesiente, which hosts Italy's last coal mine, the focus is on the diversification of the local green economy, agriculture, tourism and marine economy sectors. 2,250 workers are trained in new entrepreneurship skills. The creation of RE communities to reduce EP and to enhance energy efficiency is also planned in these regions. In parallel, special incentives are provided to small and medium-sized enterprises for the adoption of sustainable and digital technologies.

The Recovery, Transformation, and Resilience Plan (RTRP) is **Spain's** strategic framework for post-pandemic economic recovery, aligning with European Green Deal principles. It allocates significant funding toward renewable energy, energy efficiency, and green hydrogen development, with key projects scheduled for completion by 2026. The plan also emphasizes modernizing the energy network to support increased renewable energy integration and resilience. A just transition underpins RTRP's objectives, focusing on economic revitalization and social inclusion. The plan prioritizes retraining programs for workers in carbon-intensive sectors, facilitating a shift to sustainable jobs. It also provides targeted investments in regions disproportionately affected by the energy transition, promoting economic diversification and infrastructure development. For consumers, RTRP includes measures to alleviate energy costs and improve living conditions, such as retrofitting homes for energy efficiency. The plan ensures that vulnerable populations are not left behind, with specific initiatives aimed at reducing EP. By fostering collaboration between public and private sectors, RTRP aims to create a resilient and inclusive energy landscape that benefits all, while meeting Spain's climate commitments.

In **Portugal**, the RE and hydrogen sectors present significant opportunities for job creation. Transitioning from fossil fuels to renewables is expected to generate over 15,000 jobs in Portugal, particularly in the wind, solar, and hydrogen industries. Training programs and reskilling initiatives are essential to equip workers from traditional energy sectors with the skills needed for these emerging industries. This not only supports economic growth but also ensures that workers are not left behind in the transition.

Supported by the European Union, the Just Transition Mechanism (JTM) provides critical financial and technical assistance to regions most impacted by the transition to sustainable energy. The JTM focuses on fostering economic diversification, creating jobs, and ensuring social inclusion, thereby addressing challenges faced by vulnerable communities. Stakeholder engagement is another key element of Portugal's energy policies, emphasizing participatory governance through mechanisms like public consultations and workshops. Regional and local governments play a crucial role in representing their communities' interests, ensuring that the energy transition is both inclusive and equitable.

Capacity-building programs, such as "Futuro Verde" and "Requalifica," are integral to Portugal's strategy, providing workforce reskilling opportunities, particularly for individuals displaced from traditional energy sectors. These programs are developed in partnership with educational institutions and industries to ensure alignment with labor market demands. Financial support mechanisms, including subsidies and grants, further enhance the transition by enabling low-income households to access energy-efficient housing and renewable energy installations. Additionally, funding is allocated to economically disadvantaged regions to stimulate local economies and foster sustainable growth.

Portugal's policies seamlessly integrate social and environmental goals, addressing socio-economic disparities while promoting environmental restoration. Renewable energy projects are strategically invested in regions with high unemployment and EP rates. Meanwhile, decommissioned coal plants and fossil fuel infrastructure are being repurposed to support sustainable energy initiatives, ensuring a comprehensive and equitable transition for all sectors of society.

2.3. CHALLENGES AND OPPORTUNITIES OF JUST TWIN TRANSITION PROCESS

The just twin transition in the energy sector requires a careful balance between decarbonization goals and the economic, social, and technological inclusion of vulnerable populations. A multidimensional approach, considering factors such as energy poverty, the digital divide, regional disparities, employment vulnerability, and the skills gap, is essential to advancing green and digital innovation while ensuring that no one is left behind. This is crucial for meeting the EU's climate neutrality target by 2050.

Key Challenges of the Just Twin Transition:

1. **Energy Poverty and Socioeconomic Vulnerability.** Access to energy remains a major challenge for disadvantaged groups due to low incomes, rising energy costs, and employment instability, particularly in regions affected by the phase-out of fossil fuel industries. Those at risk of poverty often struggle to afford adequate heating, cooling, and electricity, making energy poverty a pressing issue in the transition process.

2. **The Digital Divide and Technological Barriers.** The digitalization of the energy sector poses new inclusion challenges. On the consumer side, access to smart meters, renewable self-consumption technologies, and energy efficiency tools is increasingly digitized, creating barriers for individuals with limited digital skills or access to digital tools. On the infrastructure side, the transition to smart grids, AI-driven energy management, and digital energy solutions requires advanced digital infrastructure, which is not yet uniformly available across the EU. Additionally, there is a workforce skills gap, necessitating large-scale reskilling and upskilling programs to support workers transitioning into green and digital industries. Moreover, the rise of cybersecurity risks in digital energy systems further underscores the need for highly skilled experts in cybersecurity and data protection.

3. Regional inequalities also pose a significant challenge in the transition. The modernization of energy grids and renewable energy integration require large-scale investments, which weaker economies may struggle to attract. The geographic distribution of renewable resources is uneven—while some countries benefit from abundant wind and solar energy, others lack the conditions for large-scale deployment. Regions facing the closure of coal plants and fossil fuel industries suffer from job losses and economic downturns, highlighting the need for targeted just transition policies.

Opportunities for an Inclusive Energy Transition:

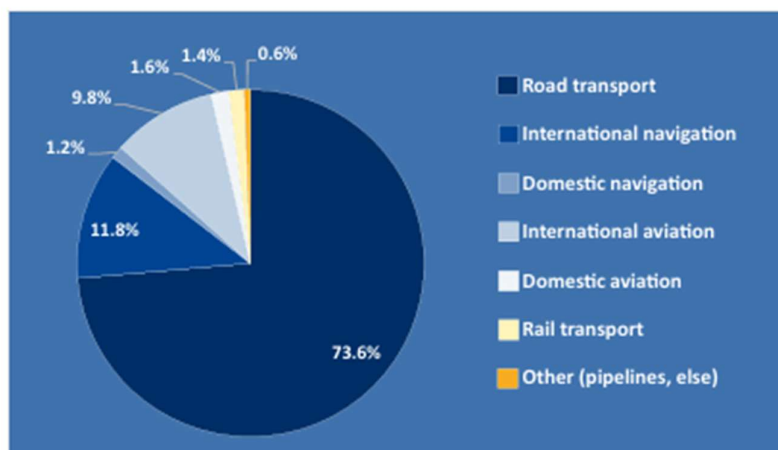
1. Job Creation and Economic Revitalization. The expansion of renewables supports new industries, such as battery manufacturing, solar panel production, wind turbine components, and smart grid management. Energy storage, digital energy solutions, and circular economy practices, such as battery recycling and repurposing old infrastructure, offer diverse employment opportunities. Reskilling programs and technical training in advanced digital and renewable technologies will be key to bridging employment gaps and fostering social stability.

2. Strengthening Energy Security and Price Stability. Renewable energy enhances energy independence, reducing reliance on imported fossil fuels and shielding consumers from energy price crises. Local energy self-consumption projects empower communities and promote public participation in the transition. Decentralized energy systems and community-led renewable projects can help democratize access to energy resources, making the transition more inclusive.

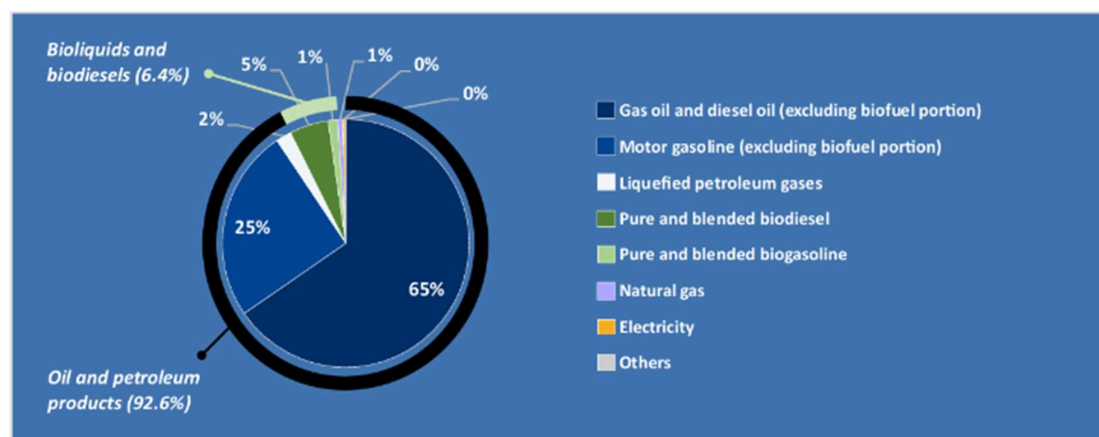
3. Advancing Social and Regional Cohesion. Just Transition Plans and EU funding mechanisms, such as the Just Transition Fund, help mitigate regional disparities by directing financial support to coal-dependent and economically struggling areas. Policies promoting green and digital skills training ensure that the workforce remains adaptable to the changing energy landscape. Citizen-led energy cooperatives and community solar projects empower local populations, fostering greater energy equity.

3. MOBILITY AND TRANSPORT SECTOR

The mobility and transport sector is a key economic and social pillar connecting people and places, enabling accessibility to critical social services, and fostering economic development and innovation. The sector contributes 5% to the EU GDP with 1.3 million public and private companies that employ 10.2 million people. Since 1995, the volume of transported goods has increased by 41%, and passenger transport grew by 33% (European Commission: Directorate-General for Mobility and Transport, 2024). The sector is highly dependent on oil-derived fuels with 92.7% of total energy consumed in 2022, with road transport accounting for 73.6% of total energy consumption.

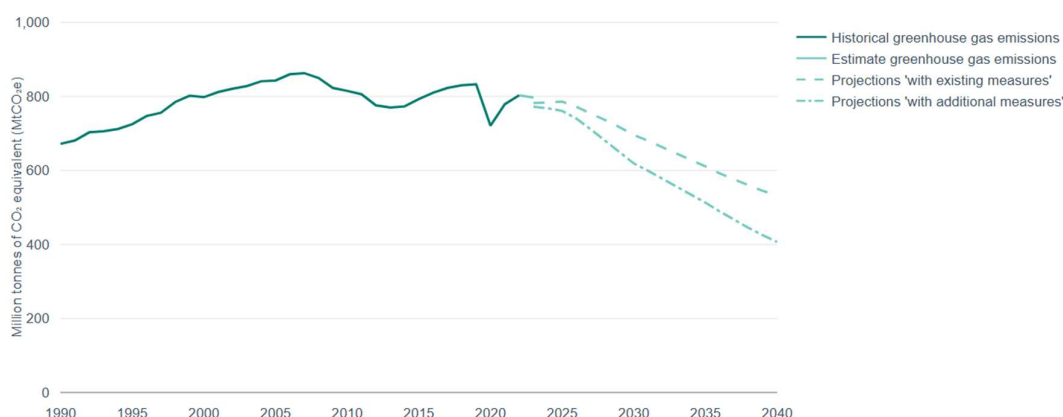
Figure 1. Consumption of energy by transport modes, 2022

Source: European Commission: Directorate-General for Mobility and Transport (2024)

Figure 2: Energy consumption in road transport by type of fuel in the EU, 2022

Source: European Commission: Directorate-General for Mobility and Transport (2024)

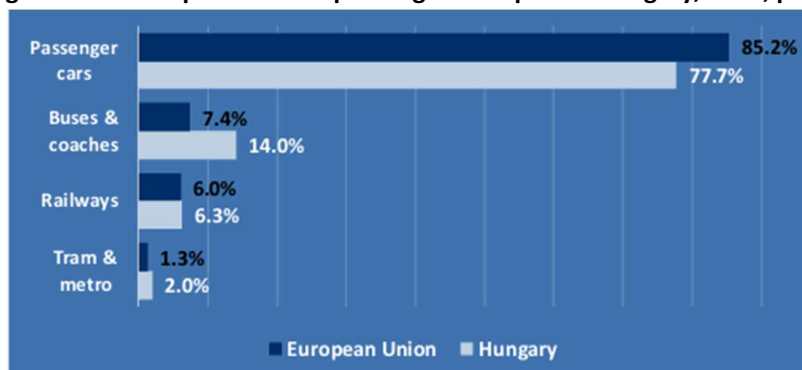
While the share of energy from RE has increased to 10.1% in 2023, the sector is a heavy emitter of CO₂ and nitrous oxide N₂O contributing 25% to the EU total GHG emissions. Road transport is the most important GHG emitter with 73.2% of all EU transport sector emissions in 2022. Therefore, a specific focus of EU energy efficiency and green transition measures is aimed at road transport (European Environment Agency, 2024c).

Figure 3. Greenhouse gas emissions from transport in Europe

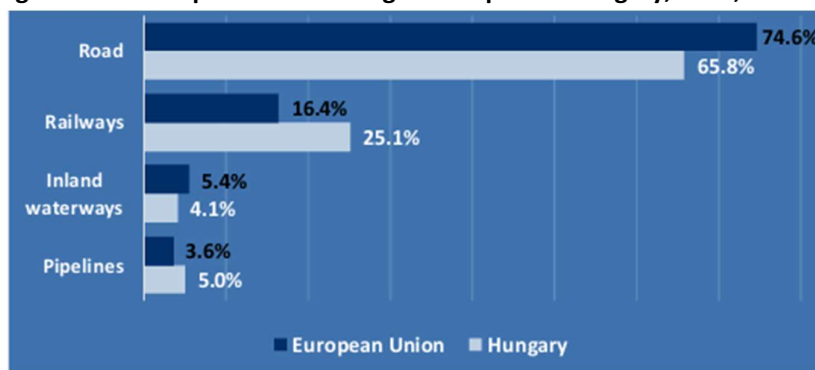
Source: European Environment Agency (2024)

An overview of the mobility and transport sector in Hungary

Hungary's transport and mobility sector accounts for 5.3% of the national GDP and employs around 259,000 people, around 6.5% of Hungary's total labour force (KSH -Hungarian Central Statistics Office, 2024). The Hungarian Recovery and Resilience Plan supports the green transition in the mobility and transport sector through EUR 1.414 billion investment in sustainable transport solutions aimed at enhancing public transport options, improvement of the rail network and renovating the fleet of buses with electric vehicles. Road and railway transport are the main transport modes in Hungary with twice as many number of buses and coaches as the EU average (European Commission: Directorate-General for Mobility and Transport, 2024).

Figure 4. Modal split of inland passenger transport in Hungary, 2021, passenger-km in %

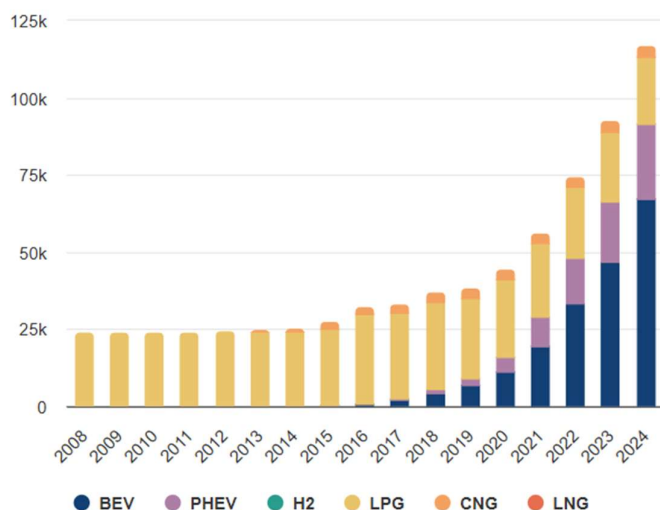
Source: European Commission: Directorate-General for Mobility and Transport (2024).

Figure 5. Modal split of inland freight transport in Hungary, 2021, tonne-km in %

Source: European Commission: Directorate-General for Mobility and Transport (2024)

In 2023, Hungary had 4,743,787 registered passenger cars and light commercial vehicles. Only 1.95% (116,572) of this fleet corresponded to low- and zero-emission vehicles (BEV, PHEV, H2, LPG, CNG, LNG) (European Commission: European Alternative Fuels Observatory, 2025a).

The passenger car market grew 13% in 2024 with 121,607 new registered vehicles, with a 47.8 % increase in electric vehicles registration, accounting for 7% of the total fleet (MTI-Hungary Today, 2025).

Figure 6. Alternative fueled passenger and light commercial vehicles in Hungary

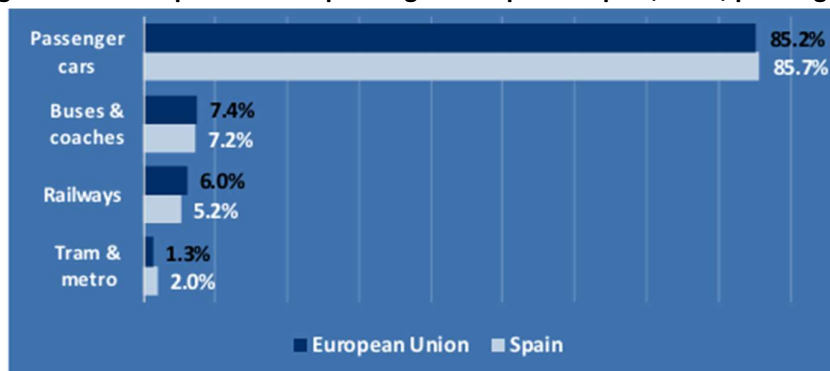
Source: European Commission: European Alternative Fuels Observatory (2025a)

An overview of the mobility and transport sector in Spain

Spain's transport and mobility sector is a vital component of its economy with its network of air, land, rail and maritime infrastructures. The sector contributes 4.4% to Spain's GDP, has an annual turnover of 149,000 million euros with around 218,000 companies providing 1 million jobs. The share of road transport is 90% in total passenger and freight inland transport.

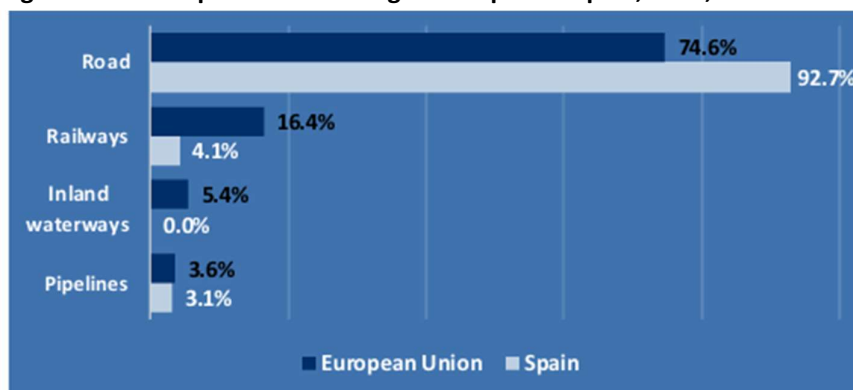
Spain is leader in EU and second worldwide for its high-speed rail network, with over 3,900 km. Its 10 ports and 10 airports ranked among the busiest in the EU, and it has an extensive motorway network with over 17,550 km (Ministerio de Economía, Comercio y Empresa, n.d.).

Figure 7. Modal split of inland passenger transport in Spain, 2021, passenger-km in %



Source: European Commission: Directorate-General for Mobility and Transport (2024)

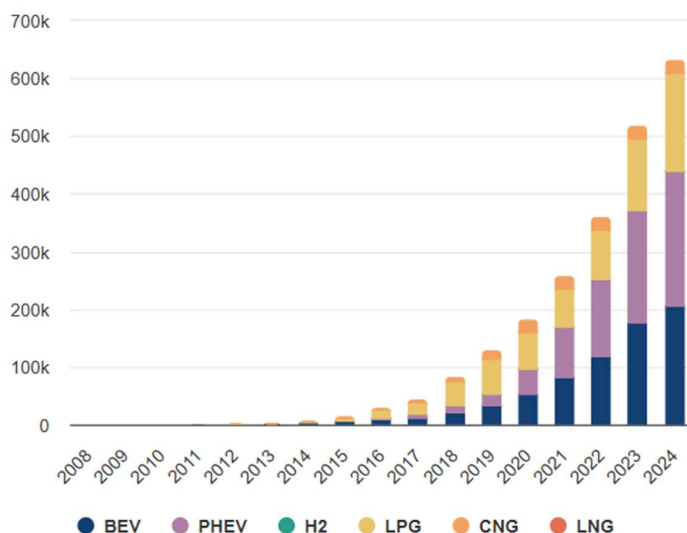
Figure 8. Modal split of inland freight transport in Spain, 2021, tonne-km in %



Source: European Commission: Directorate-General for Mobility and Transport (2024)

The Recovery, Transformation and Resilience Plan in Spain supports the green transition of the mobility and transport sector through EUR13.2 billion investments in sustainable mobility aimed at improving rail infrastructure, creation of low-emission zones in urban areas, green public buses, urban public transport and deployment of electric charging stations (European Commission: Directorate-General for Mobility and Transport, 2024).

In 2023, Spain had 30,041,334 registered passenger cars and light commercial vehicles. Only 1.72% (631,712) of this fleet corresponded to low- and zero-emission vehicles (BEV, PHEV, H2, LPG, CNG, LNG) (European Commission: European Alternative Fuels Observatory, 2025). The passenger car market grew by 7.1% in 2024, reaching sales of over 1 million vehicles, with 2% increase in BEV+PHEV, reaching 115,939 units and 11.4% of the total market. Registration of light commercial vehicles increased by 13.6% reaching 165,847 units. The sales of industrial vehicles, buses, coaches, and minibuses close the year with 12.5% growth, reaching 36,509 units (ANFAC, 2025).

Figure 9. Alternative fuelled passenger and light commercial vehicles in Spain

Source: European Commission: European Alternative Fuels Observatory (2025b)

3.1. COMMON GOALS AND PRIORITY ACTION LINES OF THE TWIN TRANSITION PROCESS

Fostering green and digital transformation of the transport and mobility sector is an overarching goal of the EU, as outlined in its Sustainable and Smart Mobility Strategy containing 82 specific initiatives that aim at boosting the sector's efficiency, resilience and smartness. By 2050, the Strategy aims to achieve climate-neutrality of almost all road vehicles; to double rail freight traffic, and to have implemented a multimodal Trans-European Network for sustainable and smart transport with high-speed connectivity. The targets for 2030 aim at: at least 30 million zero-emission cars; 100 climate-neutral European cities; 50% increase in high-speed rail traffic; carbon-neutral scheduled collective travel for under 500 km; deployment of automated mobility and launch of zero-emission marine vessels. By 2035, zero-emission large aircraft should be ready in the market (European Commission, 2021c). Sustainability, smartness and resilience drive the green and digital transition in the sector. Specific additional targets for 2030 are focused on:

1. Deployment of 3 million public charging points for electric mobility.
2. Development of zero-emission airports and ports.
3. Providing healthy and sustainable options for interurban and urban mobility by increasing high-speed rail traffic and cycling infrastructures.
4. Introducing carbon pricing strategies to reflect environmental costs of mobility for users.
5. Developing connected and automated multimodal mobility.
6. Fostering innovation in data use and artificial intelligence for smarter mobility options.
7. Increasing public-private investment in structural changes aimed at sectors' efficiency and modernization.
8. Ensuring affordability, availability and accessibility of mobility and employment opportunities for all.
9. Significantly increasing the sector's safety and security.

Context in Hungary

Hungary's strategic approach in the mobility and transport sector is interconnected with broader national priorities in sustainable development, economic resilience, and environmental protection. By integrating transport objectives with energy and environmental policies, the government aims to create a cleaner, more efficient, and resilient transport infrastructure. Recognizing the importance of domestic hydrogen production to support the planned transport infrastructure, Hungary is investing in renewable hydrogen production facilities. The National Hydrogen Strategy focuses on green hydrogen to achieve zero-emission solutions from production to consumption. The government plans to support pilot projects in hydrogen production through electrolysis, especially utilising excess renewable energy to ensure that hydrogen remains a sustainable option for the long term. That means the production of 36,000 tons of hydrogen annually by 2030, with 16,000 tons from green and other carbon-free sources and 20,000 tons from low-carbon hydrogen, including the development of 240 MW of electrolysis.

Hydrogen-powered vehicles produce zero tailpipe emissions, which significantly reduces harmful pollutants like nitrogen oxides in urban areas. By focusing on sectors where traditional fuel alternatives are less feasible, such as long-haul freight, hydrogen-powered heavy-duty vehicles are expected to play a pivotal role in reducing air pollution hotspots around highways, industrial zones, and city outskirts. This shift also aligns with Hungary's goals for improving urban air quality and enhancing public health by reducing reliance on diesel-powered freight, which is a major contributor to air pollution in many regions. To promote the uptake of hydrogen-powered vehicles, Hungary is implementing a comprehensive policy framework that includes tax benefits, purchase subsidies, and lower registration fees for hydrogen-fuelled heavy-duty vehicles.

The National Transport Infrastructure Development Strategy (NTIDS) is Hungary's comprehensive long-term plan for enhancing its transport infrastructure up to the year 2050, approved in 2014. The NTIDS prioritises the development of transport infrastructure that supports economic growth and boosts competitiveness. By improving connections within Hungary and across borders, the strategy seeks to make infrastructure developments aimed to reduce transit times, improve logistics, and provide more efficient links between urban and rural areas. Environmental sustainability is a core pillar of the NTIDS, which promotes eco-friendly transport modes and technologies to mitigate the impact of transportation on the environment. Initiatives focus on reducing greenhouse gas emissions, encouraging the adoption of electric vehicles, expanding the rail network, and supporting urban transit solutions that lower reliance on private car travel.

The strategy also prioritises the development of a transport system that provides equitable access to services, education, employment, and other opportunities across the country. It aims to reduce disparities in access and improve quality of life through better infrastructure. An improved public transport service, road expansions, and the construction of new routes will provide a good environment to foster inclusive growth, ensuring that all citizens benefit from Hungary's economic advancements.

Hungary's National Intelligent Transport Systems (ITS) strategy is a structured framework designed to improve the efficiency, safety, and sustainability of its transportation infrastructure. The approach is part of a broader European effort to enhance the interoperability and seamlessness of intelligent transport systems across member states. ITS involves the use of advanced information and communication technologies to monitor, manage, and optimise transportation operations, reduce congestion, improve safety, and provide real-time information to users. Under the Integrated Transport Development Operational Programme (IKOP) Hungary allocated significant resources, around EUR 3.92 billion including national co-financing, for projects aimed at developing and implementing ITS solutions between 2014-2020. The program primarily targeted improvements in road safety, traffic management, public

transportation, and multimodal transport logistics, creating a more connected and resilient national transport network. This funding supported infrastructure upgrades, technology deployment, and the creation of supportive platforms like the National Access Point (NAP) for data sharing. The NAP enables the effective collection, sharing, and utilisation of transport-related data among stakeholders, such as government authorities, transport operators, and road users.

Furthermore, the country has invested in developing advanced traffic management and information systems to monitor and control road traffic in real-time. This includes the implementation of intelligent traffic signals, traffic detection sensors, and electronic signage to inform drivers of current conditions and provide guidance.

Currently, Hungary continues to enhance its ITS infrastructure with a focus on Cooperative Intelligent Transport Systems (C-ITS) through the C-ROADS project (C-Roads, n.d.). C-ITS are an advanced branch of ITS that enables communication between vehicles, infrastructure, and other transport elements. C-ROADS is part of an EU-wide program to establish and implement C-ITS infrastructure, which supports real-time communication between vehicles and road infrastructure (e.g., traffic lights and road signs). This technology allows for the exchange of information about road conditions, traffic incidents, and hazards, which enhances safety and helps to prevent accidents.

Smart mobility is a central aspect of Hungary's ITS strategy, involving the use of digital tools to create "digital twins" of real-world transport situations. These projects allow for better planning and management of transportation systems by predicting traffic patterns and identifying potential bottlenecks.

Hungary's cycling infrastructure expansion plan, outlined in the National Cycling Strategy 2030, is a key element of the country's broader sustainable transport goals. By 2030, Hungary aims to increase its cycling network by 20%, reaching a total of approximately 15,000 kilometres of cycling paths nationwide (Active Hungary, 2023).

National Battery Industry Strategy 2030 outlines a transformative roadmap aimed at developing a sustainable battery production sector that should contribute to both economic growth and environmental objectives. The Strategy considers battery technology as a must-have of Hungary's approach to create sustainable transportation and declares its vital role in the shift from combustion engines to EVs. By 2030, Hungary aspires to be a leader in the European battery value chain. Supporting the production of EVs through local battery manufacturing is expected to significantly lower transport-related emissions. With a domestic supply of batteries for EVs and energy storage, Hungary seeks to mitigate energy insecurity and economic vulnerability to fluctuating global fuel prices. Significant infrastructure developments are in place already (battery factories in Göd, Debrecen). Also, the country has ongoing investments in promoting research and development in battery technology. As it is a constantly evolving industry, the strategy envisions an increase in work opportunities, thus provisioning training programs to equip the workforce with expertise in battery and EV technology. The strategy also prioritises social policies that ensure an equitable transition. E.g. retraining programs for workers to facilitate entry into emerging sectors, and financial support for small and medium-sized enterprises adapting to new technologies (ITM, n.d.).

Context in Spain

The mobility and transport sector plays a significant role in national energy demand, with 43% share in 2018. At the same time, GHG emissions of this sector, at 30.8% in 2022, are of highest concern with over a 50% increase since 1990. The Long-Term Decarbonization Strategy 2050 (LTS) and the Energy and Climate National Integrated Plan 2023-2030 outline the key targets of the twin transition in the mobility and transport sectors (Ministerio para la Transición Ecológica y el Reto Demográfico, 2020; 2024). In addition, the Safe, Sustainable and Connected Mobility Strategy 2030 outlines priority action lines and key measures

to ensure the effective transition process of the sector by 2030 (Ministerio de Transportes y Movilidad Sostenible, 2021). The Strategy 2030 is a key pillar in Spain's long-term commitment to reducing transport-related emissions and promoting eco-friendly, efficient transportation systems. Among the key strategic dimensions outlined by the Strategy, the priority is placed on mobility for all; new investment policies; safe, low emission and smart mobility; smart intermodal logistics chains; European and international routes, as well social and labor aspects of the transitions and structural changes needed on the public administration level.

The strategy emphasizes urban and metropolitan areas by prioritizing active mobility (walking and cycling), improving public transport infrastructure, and introducing low-emission zones. Strategic targets include reducing the use of private vehicles in cities by increasing the attractiveness and reliability of public transport, enhancing the quality of urban environments, and creating jobs related to infrastructure development. The strategy also addresses equity by making public transportation more affordable and inclusive, particularly benefiting marginalized groups. Investments in new mobility services, like shared electric scooters and bikes, are supported to provide convenient alternatives. Local governments receive support to implement measures that align with national goals, ensuring widespread and fair benefits across regions

In addition, the Shock Plan for Sustainable, Safe, and Connected Mobility in Urban and Metropolitan Environments is a core component of Spain's Recovery and Resilience Plan (Gobierno de España, Plan de Recuperación, Transformación y Resiliencia, n.d.). This Shock Plan is aimed at transforming urban mobility to meet climate and safety goals. Key targets include reducing traffic congestion, enhancing public transit networks, and promoting active transportation like cycling and walking. By 2025, the plan aims to create extensive low-emission zones in major cities, improve the safety of urban infrastructure, and integrate smart mobility technologies. Moreover, a major focus of this plan is enabling a fair transition for both manufactures and consumers. For transportation companies and service providers, economic incentives and financial aid are available to adopt cleaner technologies and upgrade fleets to electric or low-emission vehicles. The plan also promotes public-private partnerships to foster innovation in urban mobility solutions. Consumers benefit from enhanced safety measures, more reliable and accessible public transport, and cleaner urban air quality. The Shock Plan also addresses social equity by ensuring that sustainable mobility options are affordable and available across all neighbourhoods, particularly in marginalized areas.

Several factors for success have been identified in the LTS for Spain:

- An opportunity for the Spanish vehicle manufacturing sector, including components and infrastructure for alternative fuel charging, to adapt and lead in meeting emerging national and global demands.
- Potential for renewable fuel production facilities to drive reindustrialization in areas needing economic revitalization and job growth, supporting rural development initiatives.
- Advancing the circular economy concept.
- Retrofitting combustion vehicles, particularly in the freight transport, with hydrogen, gas, or electric power—requiring advancements in European-level certification for these conversions.
- Producing fuels from waste resources without jeopardizing material recovery targets.
- Strengthening the recycling industry and developing second-life uses for batteries.
- Accessing European structural funds and support from the new Multiannual Financial Framework (MFF) to expand infrastructure for alternative fuels and invest in research for low- and zero-emission transportation technologies.
- Introducing aggregators to manage electric vehicles' storage capacity.

- Developing renewable hydrogen production at competitive costs using renewable energy.
- A major shift towards smart and connected mobility systems by 2050, fostering collaboration among diverse sectors like energy, transportation, telecommunications, and insurance. Shared mobility will increase vehicle utilization, making investments in vehicle acquisition more efficient.
- Expanding opportunities for research and innovation to drive the development of new mobility solutions.

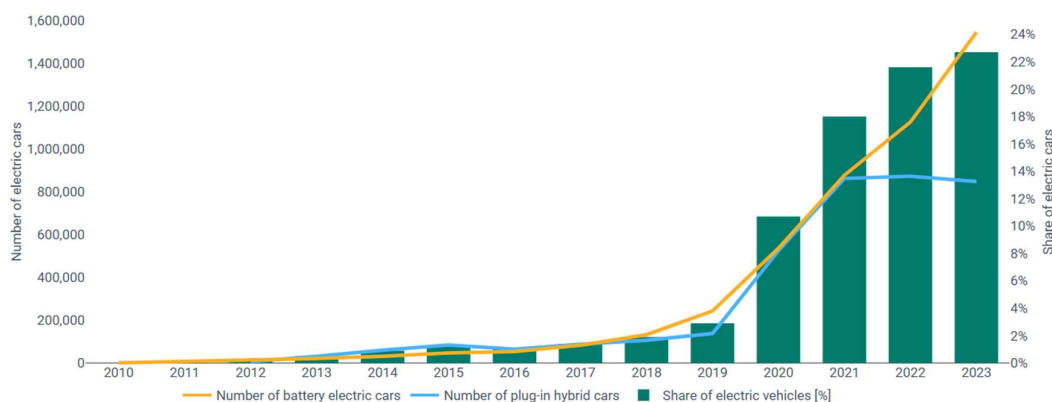
3.1.1. MAPPING OF KEY TARGETS

As one of the main global contributors to GHG emissions with 24% (IRENA, 2024), the mobility and transport sector faces critical challenges of the twin transition process in the EU. To meet the target of 90% reduction in GHG emissions by 2050 and reach 14% share of energy used from renewable sources by 2030 established by the Sustainable and Smart Mobility Strategy, the sector should significantly reduce its high dependence on fossil fuels. A priority is placed on phasing-out new polluting vehicles by 2035; the replacement of combustion vehicles by electric vehicles; a significant boost to the deployment of EV charging stations and hydrogen refuelling infrastructure, as well as the use of alternative fuels.

Electric mobility

In 2023, 22.7% of new car registrations in the EU countries corresponded to electric vehicles, with battery electric vehicles (BEV) accounting for 15% and plug-in hybrid electric vehicles for 8% of the total of new car registrations. The registration of BEV increased by 37% compared to the previous year, totaling 2,4 million. 54% of all BEV registrations in 2023 corresponded to Germany, France and Norway, however in Poland and Croatia electric vehicles registrations were lower than 5% (European Environment Agency, 2024).

Figure 10: New registration of electric cars, EU, 2023



Source: European Environment Agency (2024)

Among the different types of road vehicles, electric vehicles accounted for almost 15% among new registered passenger vehicles and collective transport vehicles in 2023 in the EU. The following tables outline the share of different types of electric mobility across the six national case studies of FITTER-EU.

Table 23. Share of registered battery-only passenger vehicles, 2019-2023
(% of total registration of vehicles)

TIME	2019	2020	2021	2022	2023
GEO (Labels)					
European Union	1,889	5,32	8,958	12,069	14,525
Spain	0,758	1,964	2,655	3,727	5,63
Germany	1,754	6,654	13,575	17,747	18,428
Italy	0,544	2,152	4,554	3,665	4,161
Hungary	1,159	2,376	3,532	4,216	5,383
Portugal	3,075	5,443	9,237	11,643	18,206
Ireland	2,906	4,41	8,469	14,748	18,466

Source: Eurostat (2024b)

Table 24. Share of registered battery-only buses, motor coaches, and trolley buses, 2019-2023 (% of total registration of vehicles)

TIME	2019	2020	2021	2022	2023
GEO (Labels)					
European Union	3,937	6,209	9,235	11,805	15,108
Spain	2,093	1,841	6,608	5,79	15,777
Germany	2,796	5,727	9,113	12,922	14,727
Italy	1,553	0,81	6,453	2,954	8,711
Hungary	0	3,142	4,327	5,729	7,1
Portugal	4,775	5,387	5,68	4,241	38,755
Ireland	0	1,123	0,584	4,796	12,663

Source: Eurostat (2024b)

Table 25. Share of registered battery-only road tractors, 2019-2023
(% of total registration of vehicles)

TIME	2019	2020	2021	2022	2023
GEO (Labels)					
European Union	0,03	0,03	0,033	0,091	0,456
Spain	0,247	0,094	0,076	0,154	0,457
Germany	0,01	0,019	0,013	0,159	0,704
Italy	0,008	0,01	0,021	0,035	0,099
Hungary	0	0,021	0,114	0,034	0,074
Portugal	0	0	0	0	0
Ireland	0	0	0	0	0,429

Source: Eurostat (2024b)

Table 26. Share of registered battery-only goods road motor vehicles <= 3.5 tonnes, 2019-2023 (% of total registration of vehicles)

TIME	2019	2020	2021	2022	2023
GEO (Labels)					
European Union	1,37	1,958	2,918	5,248	7,326
Spain	1,417	2,039	2,565	4,332	5,153
Germany	2,225	3,271	4,798	7,938	7,994
Italy	0,594	0,718	2,019	2,861	3,303
Hungary	0,287	0,596	1,203	3,214	3,255
Portugal	0,619	0,989	1,313	4,48	8,98
Ireland	1,531	3,63	2,463	2,293	3,529

Source: Eurostat (2024b)

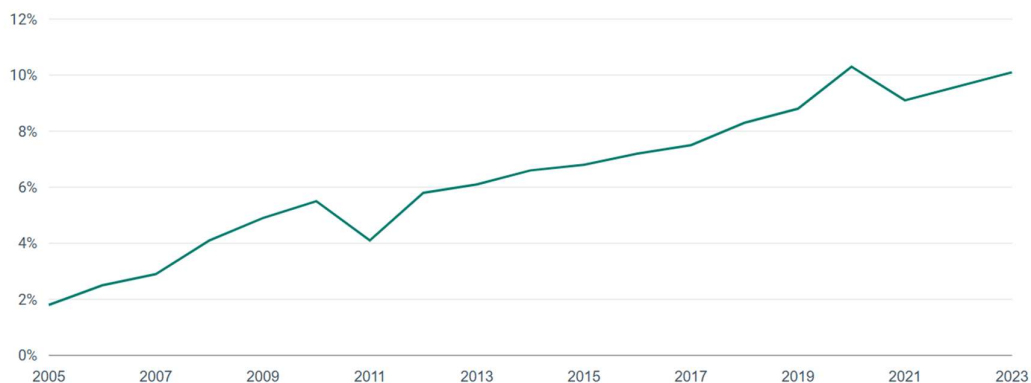
Table 27. Share of registered battery-only lorries > 3.5 tonnes, 2019-2023 (% of total registration of vehicles)

TIME	2019	2020	2021	2022	2023
GEO (Labels)					
European Union	0,437	0,809	0,984	1,557	3,184
Spain	0,044	0,115	0,277	2,222	2,43
Germany	1,199	2,185	2,473	2,111	4,326
Italy	0,099	0,182	0,064	0,162	0,58
Hungary	0	0	1,034	0,223	2,298
Portugal	0	0	0,168	0,176	0,879
Ireland	0	0,405	0,542	0,624	0,983

Source: Eurostat (2024b)

Use of Renewable Energy

The share of energy from RE sources in the transport and mobility sector has increased to 10.1% in 2023. The sources range from renewable electricity to biofuels, hydrogen and synthetic fuels of renewable origin. Among the EU countries, Sweden and Finland lead the use of RE sources with 29.5% and 19.4% respectively. Netherlands, Malta, Belgium, Denmark, Austria, Italy, Germany, and Spain reached the share of at least 10% (European Environment Agency, 2024c).

Figure 10: Share of energy from RE sources in the EU transport sector

Source: European Environment Agency (2024c)

Sustainable Public transport and climate-neutral cities

Efficiency, sustainability, availability and accessibility are the key EU priorities for ensuring daily mobility of people and addressing new mobility patterns related to the increasing prominence of teleworking, digital commerce, and shared and collaborative mobility solutions. The 2030 EU targets include the goal to achieve 100 climate-neutral and smart cities in the EU, that should become innovative labs and a reference of sustainable urban mobility, efficient public transport systems and active solutions of walking and cycling for other EU cities. All 27 EU countries feature candidate cities for this smart urban challenge. The role of Artificial Intelligence, big data analytics and digital innovation is expected to be critical in fostering efficient and safe smart mobility solutions in EU cities. Online booking and ticketing, multimodal planning of trips, access to shared and collaborative mobility services, automated mobility or traffic congestion prevention are some of the smart mobility solutions expected to meet the challenge of the climate-neutral EU by 2050.

Context in Hungary

Hungary aims to decrease reliance on fossil fuels by promoting alternative energy sources and enhancing energy efficiency in the transport sector. Transport remains a major contributor to Hungary's GHG emissions. In 2022, the transport sector accounted for approximately 25.3% of Hungary's total GHG emissions, up from 21.9% in 2021. Road transport is the largest source of these emissions. To address this, Hungary's strategy focuses on decarbonising the sector, in line with the EU's *"Fit for 55"* package, which mandates a collective reduction of GHG emissions by at least 55% by 2030 compared to 1990 levels.

The revised version of the National Energy and Climate Plan outlines key targets in the transport sector for Hungary (European Commission, 2024e). The use of energy from renewable sources should reach at least 29% of gross energy consumption in the transport sector by 2030; a reduction in greenhouse gas intensity of at least 14.5 % in the transport sector should be achieved by 2030. To meet this target, Hungary aims to increase the share of first-generation biofuels produced from food and feed crops to almost 4 % by 2030, while the share of waste and second-generation (or advanced) biofuels and biogas will be increased to at least 4.5 % and renewable transport fuel of non-biological origin to a minimum of 1% in final energy consumption in transport. The remaining part needed to reach the 29 % target will be achieved through a significant increase in the use of electricity and hydrogen for transport in fuel cells.

The process includes investments in electric vehicle infrastructure, meaning the installation of 12.000 EV charging points by 2030 to support the EV stock in the country. Also, there are incentives in place for public and private entities to adopt these changes, with the ambitious goal to double the annual number of EV registrations to reach the target of 200.000 EVs by 2030.

The strategy includes having higher fuel efficiency standards for vehicles to reduce fuel consumptions and GHG emissions. The government has two targets to reach to support this action by 2030: a) a 20% improvement in fuel efficiency and b) increasing the share of low-emission vehicles in public transport to at least 50%.

Hydrogen is key for decarbonising heavy-duty and long-haul transport sectors. Given its energy density and low emissions, hydrogen presents a viable alternative for industries and sectors that are challenging to electrify fully. By 2030, Hungary aims to establish at least 50 hydrogen refuelling stations, strategically located across major transportation corridors and urban centres to support both domestic and cross-border travel for hydrogen-fuelled vehicles. Initial stations will be placed in high-traffic zones like Budapest and along critical highways, with an incremental roll-out plan to ensure connectivity throughout the country. Infrastructure development will focus on accessibility for both commercial and passenger vehicles, ensuring seamless, wide-reaching hydrogen availability.

Another goal is to deploy approximately 4,800 hydrogen-powered vehicles by 2030, with a significant emphasis on heavy-duty vehicles with a minimum of 500. To achieve this goal, industries that rely heavily on logistics, such as freight, construction, and agriculture will be supported through the provision of initial incentives and subsidies to logistics companies and fleet operators for the purchase of hydrogen trucks, with plans to scale this support as adoption rises. Collaborations with vehicle manufacturers and international logistics providers aim to promote the shift toward hydrogen-powered fleets, with the intention of positioning Hungary as a leader in sustainable heavy-duty transportation within Central Europe (Innovációs és Technológiai Minisztérium, 2021).

Hungary's electric mobility plan outlines specific goals to increase electric vehicle adoption and the necessary supporting infrastructure. By 2030, Hungary aims for EVs to make up at least 50% of new car sales, with significant strides toward full electrification of public transport and a vast network of EV charging stations across the country (Ministry of Agriculture of Hungary, 2017).

The plan includes subsidies for EV purchases, tax incentives, and low-interest financing options to make EV ownership more accessible to consumers. The government has committed to expanding the EV charging network, especially in urban areas, main highways, and public transportation hubs, to ensure convenience and address "range anxiety" issues for EV users. For manufacturers, the plan encourages investment in EV manufacturing and the development of related supply chains, creating job opportunities in battery production, maintenance, and charging infrastructure installation sectors. The action plan also includes retraining programs for workers transitioning from traditional automotive roles to electric-focused ones, helping ensure that economic benefits are widely shared. Partnerships with local and international companies are encouraged to foster technology transfer and build a skilled workforce in electric mobility technologies. The long-term objective of this plan is to make EVs a viable option for a wide range of consumers, including those in low-income and rural areas, supporting a fair transition for all.

Table 28. Key strategic targets for the mobility and transport sector in Hungary by 2030

Component	Strategic goal
RE sources	At least 29% of final energy consumption in the transport sector.
	To reduce by at least 14.5% GHG emissions.
Electric mobility	Installation of 12,000 EV charging points and expected number of 200,000 registered electric vehicles (50% of new car sales), significant progress to the full electrification of public transport.
Alternative fuels	To increase to 4% the share of first-generation biofuels (produced from food and feed crops); to at least 4.5% of the second-generation (advanced) biofuels and biogas; and to a minimum of 1% of fuel of non-biological origin in final energy consumption in transport.
Energy efficiency	20% improvement in fuel efficiency; At least 50% share of low-emission vehicles in public transport.
Hydrogen	At least 50 hydrogen refueling stations and deployment of 4,800 hydrogen-powered vehicles, at least 500 of them should be heavy-duty vehicles.

Source: Own elaboration

Context in Spain

The main pillars of decarbonization in the mobility and transport sector in Spain are modal shift, the deployment of electric mobility, and the use of advanced biofuels. Modal shift and electric mobility are also key for achieving the energy efficiency targets.

By 2030, Spain aims for renewable energy to constitute 81% of electricity production, supported by 60,000 new public EV charging points. Targets include enhancing efficiency of energy use, achieving a 40% improvement over 1990 levels, and significantly reducing GHG emissions (Ministerio para la Transición Ecológica y el Reto Demográfico, 2024). The National Integrated Energy and Climate Plan 2023-2030 and the Safe, Sustainable and Connected Mobility Strategy 2030 outline key strategic dimensions of the transition.

The reduction of GHG emissions is aimed at 16.3% due to a progressive shift from combustion vehicles to collective public transport, shared mobility, and non-emitting modes of transport such as electric mobility. The shift to electric mobility signifies a broader change towards energy efficiency per kilometre travelled. Penetration of electric, hybrid, and plug-in hybrid vehicles in motorcycles, cars, and light vans is expected to grow rapidly, with significant adoption projected by 2030.

Promoting shared mobility in urban and interurban areas through carpooling and car-hiring is also expected to help decouple vehicle fleet growth from population growth and reduce urban congestion, through increased vehicle occupancy and considering the mixed effects on public transport usage. Furthermore, the implementation of low-emission zones in cities with more than 50,000 inhabitants since 2023 aims to contribute to the development of sustainable transport and cities. Spanish municipalities with more than 50,000 inhabitants and island territories must have adopted sustainable urban mobility plans before 2023. Promoting active mobility such as cycling, travelling by foot and public transport options are also important pillars of sustainable mobility.

Renewable hydrogen is expected to play an important role in the decarbonisation process of the mobility and transport sector in Spain. By 2030, Spain should have a fleet of at least 150-200 renewable hydrogen fuel cell buses across the national territory, with a significant presence in urban bus fleets in cities with more than 100,000 inhabitants; a fleet of at least 5,000-7,500 hydrogen fuel cell light and heavy vehicles for freight transport, and a network of at least 100-150 publicly accessible hydrogen refuelling stations. These stations must be located in easily accessible areas, evenly distributed across the country, with a maximum distance of 250 km between each station and the nearest one (Ministerio para la Transición Ecológica y el Reto Demográfico, 2020).

Table 29. Key strategic targets for the mobility and transport sector in Spain by 2030

Component	Strategic goal
RE sources	At least 29% of final energy consumption in the transport sector.
	To reduce by 16.3% GHG emissions.
Electric mobility	Installation of 60.000 EV charging points and expected number of 5,5 million registered electric vehicles.
Alternative fuels	To increase to 5.5% the share of the second-generation (advanced) biofuels and biogas; and 1% share of non-biological origin renewable fuels in final energy consumption in transport.
Energy efficiency	40% improvement;
Hydrogen	100-150 hydrogen refuelling stations, deployment of 5,000-7,500 hydrogen-powered vehicles, and 150 buses.

Source: Own elaboration

While the country still has important gaps in EV charging infrastructure, especially in rural areas, its leadership in solar and wind power provides a competitive edge in electrifying transport. Moreover, the rise of green hydrogen can make Spain a hub for clean fuel technologies, benefiting freight transport and aviation. The NextGenerationEU funds for green and digital transitions are an important financial support for Spain for high-speed rail, electric buses, and multimodal transport hubs. The strategic project of the Recovery Plan for the development and manufacturing of electric and network-connected vehicles has already secured funding of more than 3.776 billion euros, with over 900 innovation and industrial projects. Currently, there are already around 50,000 charging points. In February 2025, Spain secured EUR 72.7 million of EU funds to finance the deployment of alternative fuel supply infrastructure under the 2024 call of the Connecting Europe Facility (CEF) - Alternative Fuels Infrastructure Facility (AFIF). The funding will support the installation of 589 electric charging points, 26 green hydrogen refuelling stations, and decarbonize the ports of Barcelona, Valencia and Gijón (La Moncloa.Gobierno de España, 2025).

3.2. IMPACT OF THE TWIN TRANSITION PROCESS ON STRUCTURALLY VULNERABLE COHORTS

The twin transition process is redefining the mobility and transport sector in the EU. While the expected outcomes of increased energy efficiency, sustainability and accessibility are significant for meeting the targets of climate-neutrality, critical challenges for vulnerable groups, including low-income individuals, elderly populations, people with disabilities, and rural communities should not be overlooked. Costs related to the ownership and use of low-emission vehicles and alternative fuels may not be affordable by low-income individuals. Policies restricting high-emission vehicles in urban areas (e.g. low-emission zones) may limit mobility options for those who cannot afford cleaner alternatives. EV charging stations and integrated public transport modes are often prioritized for urban areas, leaving rural residents with fewer mobility options.

Furthermore, advanced digital technologies increasingly introduced in transport for ticketing and route planning options may create barriers for people who lack basic digital skills or do not have access to digital devices. Elderly people, low-income groups and people with cognitive disabilities are most affected by the digital divide.

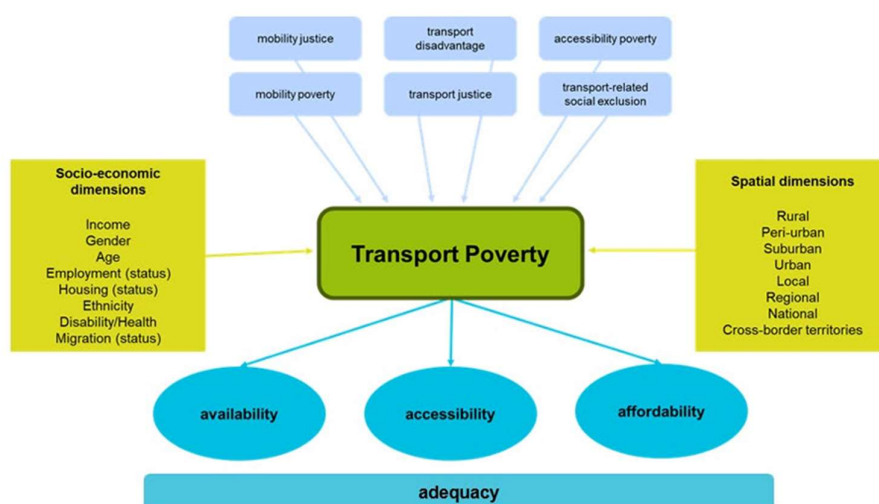
3.2.1. TRANSPORT POVERTY: KEY INDICATORS AND DETERMINANTS

Transport poverty is a critical social phenomenon in the EU, related to the lack of availability or accessibility to transport services enabling the access to essential social inclusion enablers such as education, health care, work or welfare services, or the inability to pay for these transport services (European Parliament, n.d.). The share of personal transport expenditure is the third most important in final household expenditure on consumption, at around 11.5-13.5%, following expenditures on housing and food and beverages. Low-income populations, older people and people with disabilities are most affected by transport poverty. For instance, 21% of households at risk of poverty are not able to afford transport services (European Commission, 2024f).

The lack of availability of either public or private transport options or very low frequency of service; the lack of barrier-free accessibility options to transport services due to disabilities or reduced mobility; or low affordability of transport services due to related costs are the three key dimensions of transport poverty, together with the lack of safety of available transport options or inadequate transport routes as shown in Table 30.

Income, gender, age, employment (status), housing (status), ethnicity, disability/health, and migration (status) are considered relevant cross-cutting socio-economic dimensions of transport poverty. Furthermore, the urban-rural divide needs to be considered for the assessment of transport poverty (European Commission, 2024f). The gender dimension is also particularly relevant in this context, as women are more likely to have complex and multi-purpose trip patterns when taking care of children or other family members. On the other hand, women are more likely to have low-paid or part-time jobs and experience more security concerns when using public transport (Murauskaite-Bull et al., 2023). Furthermore, rural or peripheral areas can exacerbate the impact of transport poverty due to the lack of transport infrastructure and lack of direct connections. The multidimensional nature of transport poverty phenomenon is reflected in its conceptual definition (European Commission, 2024f).

Figure 11. Conceptualisation of Transport Poverty



Source: Oeko-Institut and University of Manchester (UofM), European Commission (2024f).

Table 30. Specific EU-level indicators for each of the transport poverty dimensions are the following:

Transport poverty dimension	EU-level indicators
Availability	- Materially and socially deprived individuals who own a car; - Public transport stop 'too far away'; 'Very difficult' access to public transport; - Access to public transport is too difficult for persons with reduced mobility.
Accessibility	One-way commute to work of more than 30 minutes.
Affordability	- Enforced lack of a car; - Public transport is 'too expensive'; - Expenditure on transport exceeds 6% of total expenditures of a household; - Share of total expenditures spent on transport by a household exceeds twice the national median (2M).

Source: European Commission (2024f)

3.2.2. TRANSPORT POVERTY: ASSOCIATED VULNERABILITIES

Transport poverty is related to a series of demographic determinants, including age, gender, income, employment status, and ethnicity. These factors significantly influence transport mobility choices and can play a critical role in shaping transport accessibility and equity.

Age

- **Elderly People:** Older adults may face mobility issues due to physical disabilities, limited driving capabilities, lack of digital skills or inadequate public transport options tailored to their needs. Adoption of digital ticketing or smart mobility solutions may be challenging.
- **Youth and Students:** Young people, particularly those in rural areas, may struggle with limited or unaffordable transport options, restricting access to education and social participation.

Gender

- **Safety Concerns:** Women are more likely to experience safety-related issues while using public transport, affecting their willingness to travel at certain times or locations.
- **Trip Patterns:** Women often engage in more complex travel patterns, balancing employment, childcare, and household responsibilities, which may not be adequately supported by existing transport infrastructures.
- **Income Disparities:** Lower average earnings for women can make transport costs a higher burden.

Income

- **Affordability Barriers:** Low-income individuals often struggle with the high cost of transport, limiting their ability to access jobs, healthcare, and social services.
- **Geographic Disparities:** People in economically deprived areas may have fewer transport options, with inadequate investment in infrastructure and services.
- **Transport Choices:** Limited financial resources may force individuals to rely on inefficient or unsafe transport modes, increasing travel time and economic stress.

Employment Status

- **Job Accessibility:** Lack of affordable and reliable transport options can prevent unemployed individuals from seeking and maintaining employment.
- **Shift Work and Transport Availability:** Those working non-standard hours (e.g., night shifts) may face limited public transport availability, forcing dependence on costly or impractical alternatives.
- **Informal Employment:** Workers in informal or unstable jobs may struggle with unpredictable transport costs and availability.

Ethnicity

- **Discrimination and Social Exclusion:** Ethnic minorities may face direct or indirect discrimination in transport services, reducing their access to equitable mobility solutions.
- **Language and Information Barriers:** Non-native speakers may have difficulty understanding transport information, schedules, and ticketing systems.
- **Residential Segregation:** Ethnic minority communities often reside in peripheral urban areas with poor public transport connectivity, exacerbating transport poverty.

Both Hungary and Spain face important regional disparities driven by urban-rural divides, infrastructure concentration in major cities, geographical, economic, and policy-driven factors. While both countries have made substantial investments in transport infrastructure, accessibility and quality vary greatly between urban and rural regions, leading to socio-economic inequalities.

The capital city of Hungary, Budapest, has a developed public transport network, including metro, trams, and buses, making it one of the most accessible cities in Central Europe. However, smaller town and villages face inadequate public transport services, with infrequent buses and limited rail connectivity, forcing

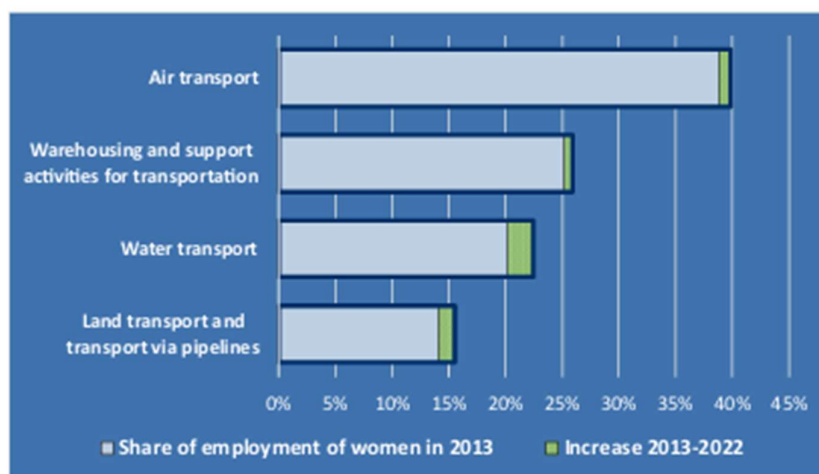
residents to rely on personal vehicles. Furthermore, the railway system is concentrated around Budapest, with major lines radiating outward. Eastern and southern regions experience underdeveloped railway connections. On the other hand, the western part of the country benefits from better road infrastructure, particularly near Austria, while the eastern and southeastern regions remain less connected, impacting trade and mobility.

Spain's high-speed rail network (AVE) is among the most advanced in Europe, connecting major cities like Madrid, Barcelona, Murcia and Seville, but many cities in the northern part of the country are not connected by AVE, and many rural areas lack direct access. Small towns and rural regions, particularly in Castilla-La Mancha and Extremadura, face very poor rail connectivity. Moreover, major metropolitan areas such as Madrid, Barcelona, Valencia or Málaga have extensive metro and bus networks, providing efficient urban mobility, but Spain's interior and less populated regions, such as Galicia and Andalusia, suffer from inadequate public transport, leading to greater car dependency.

3.2.3. ECONOMIC AND LABOUR MARKET EXCLUSION TRIGGERS

The twin transition process has a significant impact on the transport workforce due to job extinction or displacement in fossil-fuel dependent sectors, increasing skill gaps to adapt to new technical and digital requirements, or cuts in labour costs to shift investment in innovative technologies required by the twin transition process. Regions reliant on traditional automotive with combustion engines and fuel industries are most affected by these structural changes related to the twin transition. In parallel, the demand for new occupational profiles and skills has emerged in the sector to address sustainability and innovation challenges. At present, the sector faces significant labour shortages due to the changing working conditions, ageing population and new required skills in big data analytics, robotics, and artificial intelligence. Land transport is one of the most affected sub-categories, with heavy truck and lorry drivers in most demand with estimated 233.000 unfilled vacancies in the EU in 2023. Furthermore, women account for just 20% of the total transport labour force (European Commission: Directorate-General for Mobility and Transport, 2024).

Figure 12. Share of woman labour force in transport sectors, EU, 2022



Source: European Commission: Directorate-General for Mobility and Transport (2024)

The progressive decline in traditional transport sector jobs (e.g., fossil fuel vehicle mechanics, road transport drivers) and the shortage of technical skills in AI-driven logistics, smart mobility, and green energy

solutions in the mobility and transport sector may slow down the progress of the transition process. Furthermore, the increased use of AI, IoT, and connected vehicles raises risks related to data security and hacking, therefore cybersecurity and data privacy risks profiles will be needed in the sector.

3.2.4. COMMITMENT TO ENABLE A JUST AND FAIR TRANSITION PROCESS

In Hungary a new tariff policy of public transport modes allows free travel for people under the age of 14 and over 65, and half-price travel for people between 14 and 25. People with disabilities, long-term illnesses, and their companions, travel for free when using the railway infrastructure. Additional 50% discounts are provided for civil servants. became more affordable for a large number of people, including specific vulnerable groups (European Commission, 2024f).

In Spain, a core component of the National Integrated Plan of Energy and Climate 2023-2030 is a just transition, ensuring that negatively affected communities and industries receive adequate support. Car manufacturers and energy companies benefit from grants and tax incentives to drive innovation in low-carbon technologies. Consumers can access public subsidies and incentives for EV purchases and lower energy bills, especially aimed at protection of vulnerable groups. Regional disparities are also addressed, ensuring that rural areas receive investment in green infrastructure and energy-efficient transport.

The identified negative impacts in the National Integrated Plan of Energy and Climate 2023-2030 highlight transport poverty in the context of interurban mobility affecting people with lower income and residents in areas with low population density. The difficulty and vulnerability of people with lower income to purchase low-emission or electric vehicles.

The measures to address identified challenges of the twin transition process in the mobility and transport sector in Spain are focused on the enhancement of public transport modes and economic and fiscal incentives to deploy of electric mobility and supporting charging infrastructure:

- Installation of public access charging points or for fleets;
- Integration of electric vehicle fleets in public transport and companies;
- Incentives to replace conventional vehicles with electric ones;
- Creation of low-emission zones and restrictions on private vehicle use in densely populated areas;
- Promotion of sustainable urban logistics, load consolidation, the use of low-emission vehicles, and route optimization, while increasing the use of railway systems;
- Intermodal transport, with the establishment of infrastructure and services that facilitate smooth and convenient transitions between public transport and active mobility;
- Incentives to use non-motorized transport, with the creation of exclusive lanes, implementation of bike rental programs, the design of safe and attractive pedestrian infrastructure, and raising awareness of the benefits of active mobility.
- Awareness and education programs to promote behaviour change towards more sustainable mobility options.

Furthermore, Spain's Recovery, Transformation, and Resilience Plan (RTRP) channels significant investments into green infrastructure and mobility, aiming to revitalize the economy while supporting climate goals. Key strategic targets include the decarbonization of transportation, expansion of electric vehicle infrastructure, and promotion of sustainable urban logistics. By 2026, RTRP intends to have substantial progress in clean energy integration and smart mobility solutions. This plan prioritizes a just transition by allocating funds for reskilling of workers from carbon-intensive sectors and ensuring job opportunities in emerging green industries. Manufacturers in the automotive and logistics sectors are incentivized to adopt eco-friendly practices through grants and tax incentives. For consumers, RTRP focuses

on affordable access to sustainable transport options, reducing energy costs, and improving public transit systems, especially in underserved areas. The RTRP also emphasizes social inclusion, supporting rural and vulnerable communities through targeted investment. By fostering partnerships between local governments and private enterprises, the plan ensures that green infrastructure projects generate widespread economic benefits. The plan's strategic implementation timelines include quarterly assessments to adapt to societal needs and technological advancements (Gobierno de España, Plan de Recuperación, Transformación y Resiliencia, n.d.)

3.3. CHALLENGES AND OPPORTUNITIES OF THE JUST TWIN TRANSITION PROCESS

Both Hungary and Spain face structural and economic challenges in ensuring a just and inclusive transition in the mobility and transport sector. This transition must balance environmental objectives with social and economic equity, ensuring that no one is left behind. Achieving this requires strong government support, effective public-private partnerships, and a clear understanding of potential negative impacts on vulnerable social groups.

The just transition in the mobility and transport sector must be viewed through a multidimensional lens, considering key factors such as:

- Transport poverty: limited access to affordable, efficient, and sustainable transportation.
- The digital divide: unequal access to digital mobility solutions, such as smart ticketing and ride-sharing services.
- The urban-rural gap: disparities in transport infrastructure and services between cities and rural areas.
- Employment vulnerability: job losses in traditional transport sectors due to automation and electrification.
- The skills gap: the need for reskilling and upskilling in emerging green and digital transport technologies.

Addressing these factors is critical to achieving the EU's climate neutrality goals by 2050 while ensuring that the benefits of green and digital innovation are accessible to all.

While green and digital innovation can significantly improve quality of life, certain groups, due to factors such as age, disability, income, or employment status, may struggle to fully engage with or benefit from these changes. To ensure an equitable transition, they require targeted social and economic support to mitigate financial burdens, and adaptation measures to make new mobility solutions more accessible and user-friendly.

Timely policymaking and sustained public-private collaboration are essential to overcoming these challenges. Governments must ensure that digital and green innovations in transport are inclusive and widely accessible. Additionally, strategic workforce reskilling is crucial to help workers transition from phasing-out sectors to emerging technological and green jobs. Expanding vocational training programs in electric mobility, smart transport systems, and sustainable logistics will be key to ensuring long-term economic stability.

Finally, it is critical to ensure that the deployment of smart mobility solutions (e.g., AI-driven traffic management, digital ride-sharing platforms), electric mobility infrastructure (e.g., charging stations, electric vehicle fleets) and public transport services e.g., expanded routes, multimodal transport options) is equitable across national territories. Accessibility, availability, and affordability must be prioritized to ensure that all regions and social groups can benefit from the transition process.

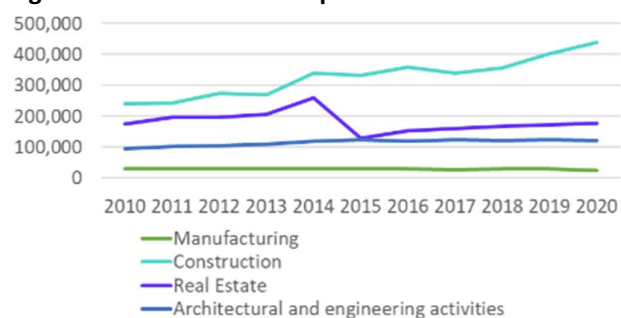
4. HOUSING AND BUILT ENVIRONMENT SECTOR

An overview of housing sector in Germany

From 2010 to 2020, Germany's GDP grew by 11.3%, reaching EUR 3.1 trillion in 2020, although it contracted by 4.6% in 2020 compared to 2019. The economy showed signs of recovery in 2021, with a projected growth of 2.8%, although there was a 0.7% contraction in the final quarter due to persistent supply shortages (European Commission, 2022a).

During the same period, the number of enterprises in the construction sector increased by 33.2%, reaching 715,198 in 2020, largely due to the growth of narrow construction (+71.7%) and architectural and engineering activities (+20.6%) (Figure 13). The real estate activities and manufacturing sub-sectors, however, saw declines of -6.4% and -4.9%, respectively (European Commission, 2022a).

Figure 13. Number of enterprises in the German broad construction sector between 2010 and 2020

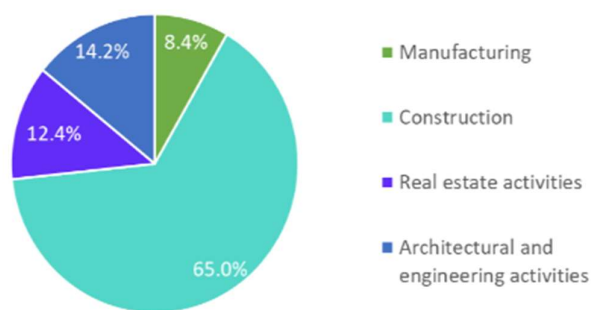


Source: European Commission (2022a)

According to the same source, the broad construction sector's production volume rose by 16.6% from 2015 to 2020, driven by civil engineering and building construction, which saw increases of 26.9% and 15.1%, respectively. Turnover in the sector grew substantially, from EUR 377.3 billion in 2010 to EUR 638.9 billion in 2020, marking an overall growth of 69.3%. All sub-sectors saw an increase in turnover, with narrow construction growing by 112.0%.

In terms of profitability, the gross operating rate of the sector was 20.1% in 2019, a slight decline from 22.6% in 2010 (European Commission, 2022a). In 2020, there were 4,662,042 persons employed in the German broad construction sector, 58.7% more than the 2010 level (2,938,001 persons). This was driven by a significant increase in the number of persons employed in the narrow construction sub-sector (84.8%), followed by the architectural and engineering activities (46.8%) and the real estate activities (31.5%) sub-sectors over the 2010-2020 period.

Contrarily, the manufacturing sub-sector registered a decline of 3.9% over the same period. In 2020, the narrow construction sub-sector employed 65.0% of the total workforce in the broad construction sector (i.e. 3,028,566 persons), followed by the architectural and engineering activities (14.2%, i.e. 663,293 persons), the real estate activities (12.4%, i.e. 579,659 persons) and the manufacturing (8.4%, i.e. 390,524 persons) sub-sectors (Figure 14).

Figure 14. Percentage of people employed per construction sub-sectors in Germany in 2020

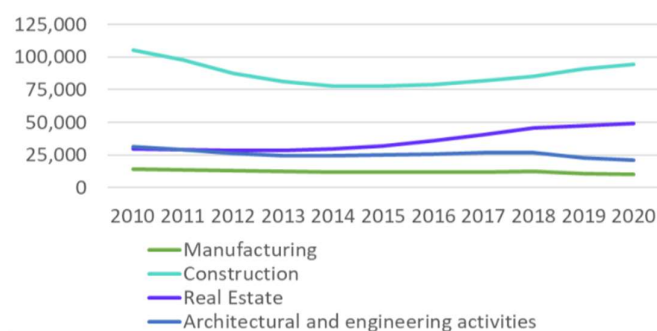
Source: European Commission (2022a)

To support housing development, the German government has allocated EUR 8 billion in subsidies for the housing market from 2018 to 2024 and introduced a tax incentive law for new rental apartments in 2020. Additionally, a rental price break was extended for five years, and incentives were increased for adopting climate-friendly heating systems.

As part of its 2021-2026 National Recovery and Resilience Plan (NRRP), Germany has allocated EUR 2.5 billion for energy-efficient building renovations, aiming to achieve at least 30% energy savings. The residential programme under the NRRP plans to renovate 40,000 dwellings by 2026 (European Commission, 2022a).

An overview of housing sector in Portugal

Portugal's GDP decreased by 1.4% from 2010 to 2020, reaching EUR 184.9 billion in 2020, which marked a 7.6% decline compared to 2019. The number of enterprises in the broad construction sector fell by 3.6% from 2010 to 2020, totaling 174,298 companies in 2020 (Figure 15). The most significant decline occurred in architectural and engineering activities, which dropped by 33.7%, followed by a decrease in manufacturing by 30.2%, and narrow construction by 10.4%. In contrast, the real estate activities sub-sector saw an increase of 65.5% (European Commission, 2021b).

Figure 15. Number of enterprises in the Portuguese broad construction sector between 2010 and 2020

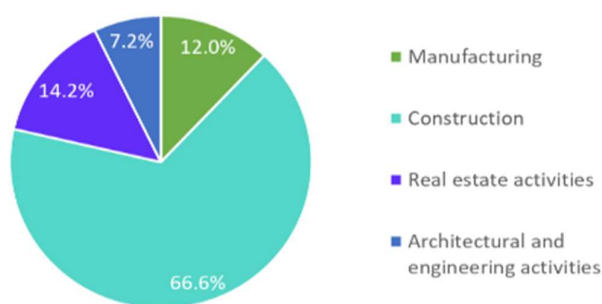
Source: European Commission (2021b)

The volume index of production in the construction sector grew slightly by 0.7% between 2015 and 2020. The total turnover of the broad construction sector amounted to EUR 40.3 billion in 2018, which was a 21.1% decrease compared to 2010. By 2020, the turnover rose slightly to EUR 41.0 billion but still showed a 19.7% decrease from 2010 (European Commission, 2021b). The main reasons for this decline were

reductions in narrow construction, architectural and engineering activities, and manufacturing, while real estate activities experienced a growth of 53.4% (European Commission, 2021b).

Employment in the construction sector in Portugal decreased by 15.6% from 2010 to 2020. The largest declines occurred in manufacturing, architectural, and engineering activities, both of which saw a reduction of 28.8%. On the other hand, real estate activities saw a significant increase in employment, rising by 42.7% (European Commission, 2021b). In 2020, 545,060 people were employed in the broad construction sector in Portugal. Despite a 15.6% drop from the 2010 employment level of 646,085, the sector has been growing since 2014, with a 19.1% increase. The narrow construction sub-sector employed 66.6% of the total workforce (363,030 people), followed by real estate activities (14.2% or 77,168 people), manufacturing (12.0% or 65,520 people), and architectural and engineering activities (7.2% or 39,342 people) (Figure 16).

Figure 16. Percentage of people employed per construction sub-sectors in Portugal in 2020



Source: European Commission (2021b)

To support affordable housing, the Portuguese government plans to create a pool of vacant properties that will be converted into affordable rental housing, with an investment of EUR 2.3 billion. Additionally, the National Investment Programme 2030 (PNI) has allocated EUR 22.0 billion for transport, energy, and environmental projects, contributing to a total estimated investment of EUR 43.0 billion (European Commission, 2021b).

According to the European Commission data, in terms of recovery and resilience, Portugal will receive EUR 13.9 billion in grants and EUR 2.7 billion in loans under the Recovery and Resilience Facility. The country's recovery plan includes measures to promote green and digital transitions, such as EUR 610 million for energy efficiency renovations and EUR 690 million for investments in infrastructure, road transport, and electric vehicle networks.

4.1. COMMON GOALS AND PRIORITY ACTION LINES OF THE TWIN TRANSITION PROCESS

The European Housing Strategy refers to the initiatives and policies adopted at the European Union (EU) level to address housing challenges in Europe. Although the EU does not have direct powers over housing policy, as this is mainly the responsibility of the Member States, the European strategy seeks to influence national policies and encourage cooperation and the exchange of good practices between countries. The strategy focuses on several key objectives, including accessibility and affordability, sustainability and energy efficiency, inclusive and quality housing and the development of social housing, among others. In short, the European housing strategy seeks a balance between the supply and demand of housing, the

well-being of citizens and the fulfilment of climate and sustainability objectives, promoting urban development that is inclusive, affordable and environmentally friendly.

Context in Germany

Germany's National Long-Term Housing Strategy is aligned with the European Union's objectives. Since 2021, there has been a new Federal Ministry of Housing, Urban Development, and Construction (BMWSB), led by the SPD (Social Democratic Party). Their work focuses on three main areas:

- The first area is housing and urban planning, with the aim of building 400,000 new homes per year, of which 100,000 will be social housing. This goal will be achieved through the affordable housing alliance. The BMWSB provides the appropriate framework and funding conditions to boost the housing market, including subsidies to federal states through the social housing subsidy program. It also includes social services such as housing allowances, social housing, action against homelessness, and municipal heat planning.
- The second area is construction, focusing on energy-efficient buildings and renovation, such as the energy renovation of cities, the Building-Energy-Act (GEG), and energy certification.
- The third area is spatial and regional development, including municipal heat planning, among other initiatives.

Due to Germany's federal structure, the responsibility for public construction law is divided between the central government and the federal states (Länder). Zoning law (building planning law) is under federal jurisdiction, determining the purpose for which a property may be used and whether a building project is suitable for its surroundings. The federal states are responsible for building regulations (Bauordnungsrecht), which govern how buildings should be designed and constructed to meet planning law requirements. The Model Building Code (MBO) 158, which serves as a guide for the federal states, also includes regulations on thermal and sound insulation (Section 15 MBO) and barrier-free construction (Section 50 MBO). Some state building regulations also include additional sustainability criteria, such as the solar obligation for new buildings specified in Section 32a of the Lower Saxony Building Code (NBauO) 159 (DENA, 2023, p. 38).

Context in Portugal

Portugal's housing and built environment strategies are aligned with the broader goals of the European Union to promote sustainable and energy-efficient construction, addressing both climate change and social inequalities. The country's long-term plans in this sector focus on reducing energy consumption, improving housing quality, and ensuring that vulnerable groups are not left behind in the green transition.

The housing and built environment sector in Portugal has been shaped by several long-term strategies, policies, and plans that align with the country's broader socio-economic objectives and environmental goals. One of the most significant frameworks for addressing housing in Portugal is the National Energy and Climate Plan 2021–2030 (NECP, 2030), which is part of Portugal's commitment to achieving the European Union's climate targets. This plan focuses on improving energy efficiency, reducing carbon emissions, and advancing the transition towards renewable energy sources in the housing sector (Portugal DGEG, 2019). The NECP 2030 specifically targets residential buildings, which are one of the largest contributors to the nation's greenhouse gas emissions and aims to increase energy efficiency through retrofitting and promoting sustainable construction practices, such as the passive house concept.

At the same time, the housing sector is shaped by the need to address pressing social issues like housing affordability and urban inequality. Social housing policies have been central to Portugal's housing strategy, especially following the economic crisis of the late 2000s. During this period, housing prices surged, and

housing insecurity among vulnerable groups increased, leading to a re-evaluation of the role of public housing (Pinto, 2017). To address these challenges, the government has implemented a combination of rent regulation policies, public housing investments, and urban regeneration initiatives designed to balance economic development with social inclusion (Pinto & Guerra, 2019).

Additionally, urban governance frameworks such as the Plan for the Recovery of the National Housing Stock aim to improve the overall quality of housing, especially for lower-income groups. This strategy includes significant investments in the renovation and rehabilitation of existing urban housing stock, with a focus on creating sustainable, energy-efficient homes in city centres, which are typically more accessible to the urban poor (Medeiros, 2023). These strategies are essential for meeting the sustainability targets outlined in the NECP 2030 and ensuring that housing remains accessible to all segments of society, particularly those most affected by market volatility.

Table 31. Main National Long-Term Strategies in Portugal

Policy/Plan	Objective	Key Targets	Progress	In-text Citation
National Energy and Climate Plan (NECP 2030)	Improve energy efficiency in buildings	- Reduce energy consumption in residential buildings by 35% (baseline: 2015). - 40% of energy from renewables by 2030.	As of 2023: 18% reduction in consumption, 32% renewables share in energy mix.	(Portugal DGEG, 2019)
Plan for Recovery of Housing Stock	Renovate degraded housing stock	- Renovate 1.3 million homes by 2030.	Renovate degraded housing stock.	(Medeiros, 2023)
Social Housing Programs	Address housing inequality and affordability	- Provide 30,000 new social housing units by 2030.	Social Housing Programs	(Ramos, Gonçalves, Lameira, & Rocha, 2021)

Source: Own elaboration

4.1.1. MAPPING OF KEY TARGETS

The housing sector in Europe plays a crucial role in addressing both climate change and social challenges, with a growing focus on improving energy efficiency, housing renovation, insulation, and the development of smart housing solutions. Energy consumption in buildings is one of the largest sources of greenhouse gas emissions in Europe, making energy efficiency improvements a central goal for achieving climate targets. Renovation and insulation of existing housing stock are key measures in reducing energy consumption and enhancing the sustainability of Europe's built environment. In addition, the integration of smart housing technologies is becoming increasingly important in creating homes that are energy-efficient, cost-effective, and adaptable to the needs of their occupants.

These efforts are supported by European policies and frameworks, such as the European Green Deal and the Energy Performance of Buildings Directive, which aim to drive the transformation of the housing sector. By focusing on energy efficiency, sustainable renovation, and smart housing, Europe seeks to create a more sustainable, resilient, and inclusive housing market, while also improving the quality of life for its residents.

In both Portugal and Germany, the housing sector plays a pivotal role in the broader European agenda to enhance energy efficiency, promote housing renovation and insulation, and support the development of smart housing. In Portugal, the National Energy and Climate Plan 2021–2030 (NECP 2030) prioritizes energy-efficient retrofitting of residential buildings, aiming to reduce carbon emissions and improve sustainability, especially in the wake of economic challenges. The country is also focused on urban regeneration and creating energy-efficient homes for lower-income groups, aligning with broader EU goals for environmental and social sustainability.

Similarly, Germany is advancing its housing sector with a strong emphasis on energy-efficient construction and renovation. The newly established Federal Ministry of Housing, Urban Development, and Construction (BMWSB) targets a significant reduction in the carbon footprint of buildings through energy-efficient renovation projects and sustainable construction practices. Both countries are also exploring smart housing solutions to improve the overall quality of life, reduce energy costs, and facilitate the green transition.

Through these efforts, Portugal and Germany contribute to the EU's broader vision of transforming the housing sector into a more energy-efficient, sustainable, and inclusive environment. Further details are described as follows:

Context in Germany

Germany has made significant efforts to improve the energy efficiency of its housing and building sector, which is central to the country's sustainability goals. One key initiative is the introduction of tax incentives for homeowners undertaking energy-saving renovations such as wall insulation and upgrading heating systems. These efforts are part of a broader plan to reduce carbon emissions and promote the energy-efficient use of buildings.

A major program to support this transition is the Bundesförderung für effiziente Gebäude (BEG) (Bundesamt für Wirtschaft und Ausfuhrkontrolle, n.d.), which provides financial support for energy-efficient building projects. This program covers a range of activities, including the construction, acquisition, and renovation of properties to meet energy-efficient standards. This initiative aligns with Germany's overall goal to reduce emissions and improve energy performance in both residential and non-residential buildings.

To further enhance energy efficiency and reduce carbon emissions, in 2019 Germany implemented the Fuel Emissions Trading Act (BEHG) (Bundesministerium der Justiz, 2019). This law establishes a national emissions trading scheme for the heating and transport sectors, which includes carbon pricing for fossil fuel emissions, such as natural gas and coal, used in buildings. This initiative supports Germany's commitment to decarbonizing the building sector.

Germany's National Recovery and Resilience Plan (NRRP) (Federal Ministry of Finance, 2020) includes investments to improve the energy efficiency of buildings, with a focus on deep renovation. By 2026, the renovation of 40,000 dwellings is expected to improve energy performance significantly, making these buildings more energy-efficient. The plan also introduces municipal living laboratories, which foster the development of innovative energy solutions and encourage the integration of renewable energy technologies in urban neighbourhoods.

In terms of smart housing, Germany is fostering technological innovation through the Building Energy Act (GEG) (Bundesministerium der Justiz, 2020) and its amendments. These regulations include the mandatory use of renewable energy for new heating systems, a measure that can support the transition to smart energy solutions in homes. The act also incorporates the European Union's nearly zero-energy building requirements, promoting the development of more energy-efficient homes that can incorporate smart energy management systems.

Additionally, grants are available for renovating heating systems, such as upgrading to smart, energy-efficient systems like gas-fired boilers. The country also aims to use digital tools like the Building Information Modelling (BIM) (BIM Deutschland, 2025) platform to improve construction planning and enhance the efficiency of building projects, which can be a key enabler of smart housing development. These efforts demonstrate Germany's comprehensive approach to integrating energy efficiency, renovation, insulation, and smart housing solutions in its housing policies. These measures contribute significantly to the country's climate targets while ensuring that its housing sector becomes more sustainable, affordable, and energy-efficient.

Context in Portugal

The strategic goals of Portugal’s housing policies are diverse, with a focus on both environmental sustainability and social equity. Central to these objectives is a commitment to reducing the housing sector’s carbon footprint. As part of a broader sustainability agenda, a key goal is ensuring that all new housing developments adhere to stringent energy efficiency standards, such as those outlined by the passive house movement, which reduces energy consumption while enhancing comfort (Figueiredo et al., 2020). The implementation of energy-efficient standards for both new and existing buildings is expected to reduce national emissions and contribute to long-term housing affordability by lowering residents’ utility costs.

Another crucial aim is addressing housing affordability, which continues to be a major challenge for many low-income families. The Portuguese housing market has seen rising property prices in urban areas, particularly in Lisbon and Porto, making homeownership increasingly unattainable for many. To address this issue, policies targeting rent control and the development of affordable housing have been introduced (Alves et al., 2023). These policies include rent regulation and subsidies for low-income tenants, as well as investments in social housing construction to ensure that vulnerable populations have access to stable housing (Pinto & Guerra, 2019).

Urban regeneration also plays a significant role in Portugal’s housing strategy. Revitalizing inner-city areas that have faced disinvestment is a core aspect of the national strategy, aimed at promoting social integration and providing affordable housing in locations with better access to services and employment opportunities. These regeneration projects are designed to balance environmental, economic, and social goals, ensuring that the benefits of urban renewal reach all residents, particularly marginalized groups (Carvalho, 2024).

Table 32. Environmental Sustainability

Target Area	Baseline Year	2030 Goal
GHG Emissions (Residential)	16 Mt CO ₂ e (2015)	12 Mt CO ₂ e (-25%)
Renewable Energy Share (Homes)	14% (2015)	32%
Energy Efficiency (Buildings)	180 kWh/m ² /year	130 kWh/m ² /year (-28%)

Source: National Energy Plan and Climate 2021-2030 (Necp 2030)

The focus on sustainability and energy efficiency in housing is driving significant policy initiatives aimed at improving both new and existing buildings. Efforts include ensuring that new homes meet high energy efficiency standards, retrofitting existing properties to reduce emissions, and promoting deep renovations for long-term sustainability. In parallel, urban regeneration projects are revitalizing marginalized areas,

while digitalization and smart technologies are being integrated into the construction sector to foster resource optimization and sustainability.

These initiatives are complemented by policies aimed at increasing social housing availability, addressing homelessness, and developing green and digital skills to support the transition to more sustainable construction practices. Additionally, efforts to tackle regional challenges and reduce construction waste through circular economy principles further contribute to the broader goal of building a more sustainable and energy-efficient housing sector.

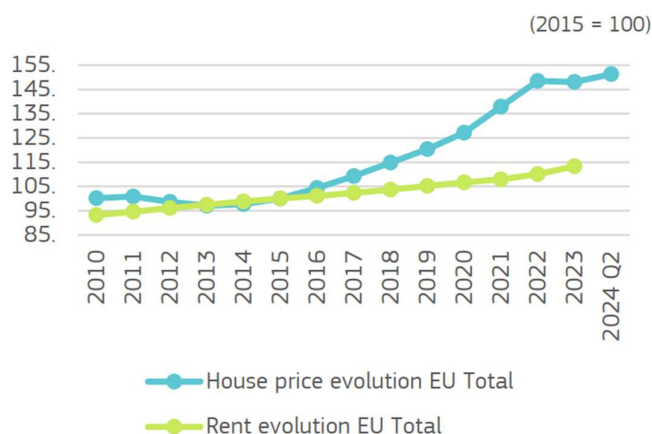
Table 33. Summarizing common goals and priority action lines of the twin transition process in Germany and Portugal

Goal	Portugal	Germany
Energy efficiency	New homes with passive energy efficiency standards.	Deep energy renovation and ecological construction.
Renovation and insulation	Urban regeneration and improvement of existing infrastructure.	Spatial planning, digitalization and municipal heating.
Smart homes	Implementation of implicit smart technologies.	Digitization of the construction sector and smart homes.
Accessibility and social housing	Increase in social housing and subsidies for tenants.	Increase in social housing and rent control policies.
Development of green skills	Training in sustainable technologies implicit in the renovation.	Training in green and digital skills in construction.
Spatial and regional challenges	Urban regeneration in marginalized areas.	Protection against heat and rural-urban approach.
Circular economy	Implicit approach to green construction.	Reduction of waste in construction through circular economy laws.

Source: Own elaboration

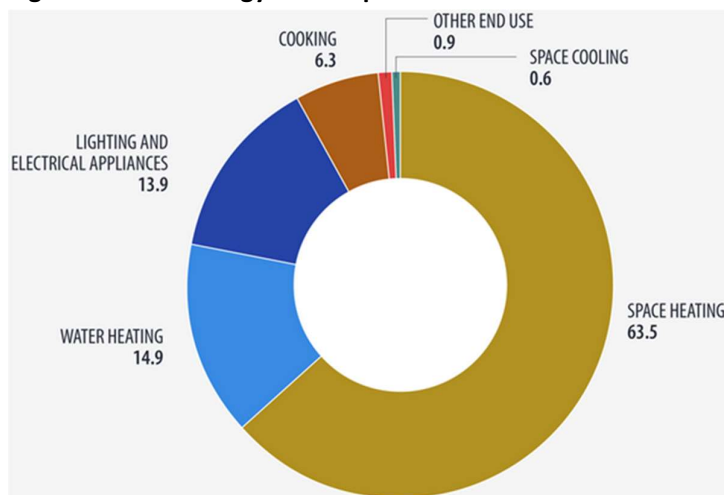
4.2. IMPACT OF THE TWIN TRANSITION PROCESS ON STRUCTURALLY VULNERABLE COHORTS

Housing affordability and availability in the EU are under increasing pressure due to several interlinked trends. Rising construction costs, the limited supply of new housing, and the acceleration of demand – particularly in urban areas – have driven both rent and house prices up substantially over the past decade (Figure 17). House prices have surged by more than 50% since 2015, and rents have risen by more than 13% across the EU (European Commission, 2025a).

Figure 17. Rent and house price evolution in the EU

Source: European Commission (2025a)

The housing shortage in the European Union exacerbates energy poverty, particularly affecting low-income households that often reside in older, energy-inefficient buildings. These properties require higher energy consumption for heating and cooling, leading to increased utility bills, which places additional strain on already limited financial resources. In 2022, most of the EU final energy consumption in the residential sector was covered by natural gas (30.9 %) and electricity (25.1 %). Renewables accounted for 22.6 %, followed by petroleum products (10.9 %) and derived heat (8.2 %). A small proportion was still covered by coal products (solid fossil fuels) (2.3 %) (Figure 18).

Figure 18. Final energy consumption in the residential sector by fuel, EU, 2022

Source: Eurostat (2024c)

Improving energy efficiency in housing presents both opportunities and challenges in meeting the EU's social and climate goals. Upgrades such as improved insulation and the installation of renewable heating systems can significantly reduce utility costs, lower emissions, and improve overall living conditions. These energy improvements align with the EU's climate targets and simultaneously help to alleviate energy poverty by making homes more affordable to heat and cool. This, in turn, enhances the comfort and financial stability of vulnerable populations (Figure 19).

Figure 19. (%) Share of final energy consumption in the residential sector by type of end-use, 2022

	Households energy use: space heating	Households - energy use: space cooling
European Union (27 countries from 2020)	63,50%	0,64%
Euro area (20 countries from 2023)	63,70%	0,78%
Germany	66,10%	0,20%
Ireland	57,90%	0,00%
Spain	39,50%	1,03%
Italy	67,30%	2,07%
Hungary	71,90%	0,47%
Portugal	32,20%	0,89%

Source: Eurostat (2024c)

Addressing housing challenges through energy efficiency also supports the EU's goals for a just transition, contributing to the reduction of emissions from the built environment and decreasing energy consumption across Member States (MSs). Policies that integrate housing affordability, energy efficiency, and climate resilience are vital for supporting vulnerable populations, fighting energy poverty, and ensuring a fair and inclusive energy transition in Europe (European Commission, 2025a).

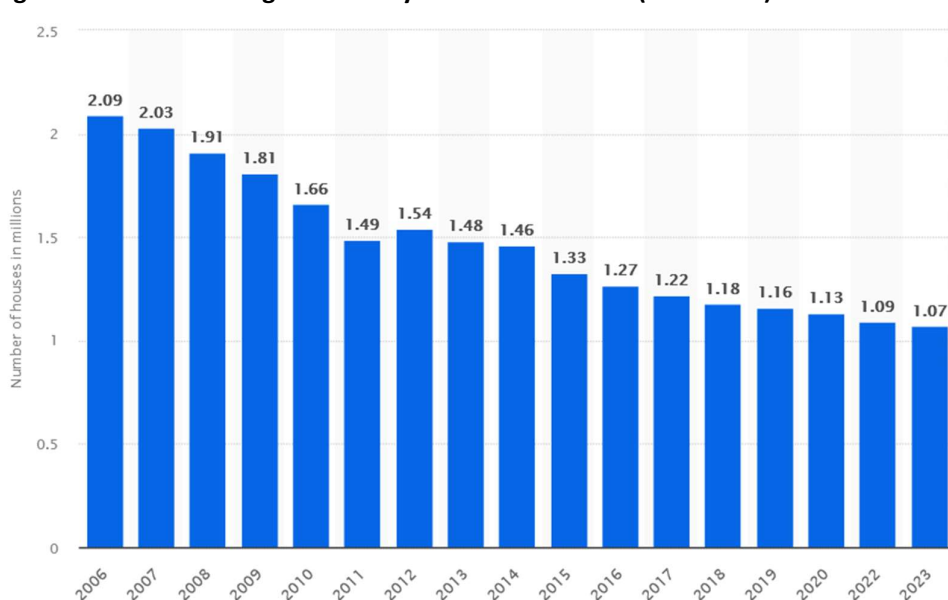
In Portugal, the transition to a more sustainable and equitable housing system has been driven by an acute awareness of its impact on vulnerable groups. Social housing policies have been specifically designed to address the needs of low-income and marginalized populations, including those at risk of homelessness, refugees, and elderly residents (Bomfim, 2024). A key component of these efforts is rent regulation policies, which play a crucial role in mitigating the financial burdens faced by tenants, particularly in urban areas where rent inflation has significantly outpaced income growth. Rent controls aim to ensure that low-income tenants are not displaced by rising housing costs and have access to secure, long-term housing options (Alves et al., 2023).

Moreover, a focus on urban regeneration ensures that marginalized communities benefit from improvements in housing quality and infrastructure. These projects are designed with anti-gentrification policies to prevent existing residents from being displaced as property values increase (Carvalho, 2024). By prioritizing these vulnerable populations, Portugal aims to ensure that its shift toward a more sustainable and modern housing sector does not exclude those who are already the most disadvantaged.

4.2.1. HOUSING POVERTY: KEY INDICATORS AND DETERMINANTS

Germany

In the field of social housing, since 2006 the responsibility for implementing measures has been with the federal states as a result of the first stage of the German federal reform (Föderalismusreform I). As a result, the focus and scope of policy measures vary considerably from one federal state to another. In 2019, the German government introduced Article 104d of the Basic Law (Gesetz zur Änderung des Grundgesetzes), which allows the federal government to provide direct, earmarked financial support to the federal states for the construction of social housing from 2020 onwards. There were about 1.07 million units of social housing in Germany in 2023. This was actually the lowest figure on the timeline (Figure 20).

Figure 20. Social housing in Germany from 2006 to 2023 (in millions)

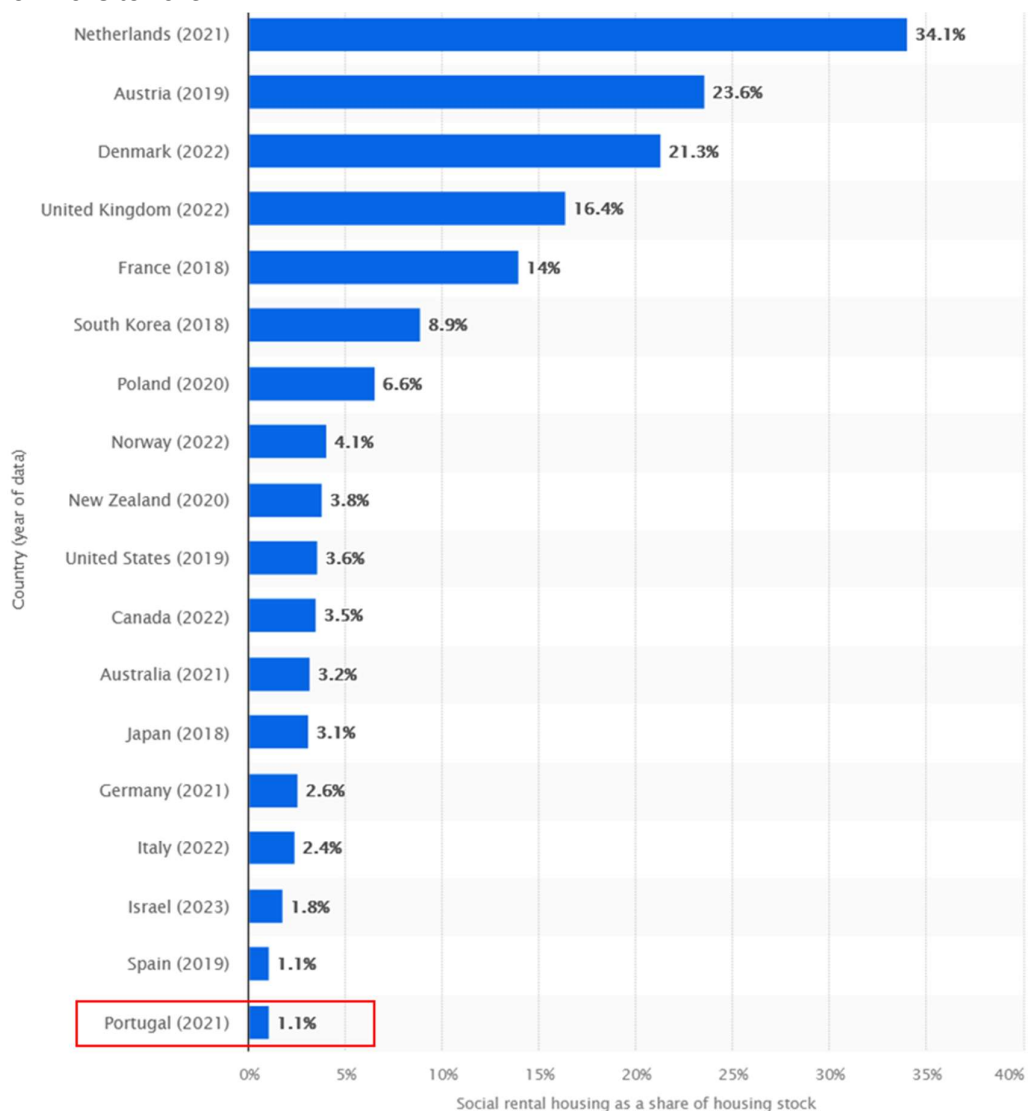
Source: Statista (2025)

Housing cooperatives play an important role in the provision of social housing and affordable rent in Germany. Such cooperatives are eligible for funding from the EUR 1.5 billion Federal Compensation Fund for Social Housing. In 2018, the federal government gathered stakeholders for a Housing Summit (Wohngipfel).

The government announced a new package of measures to strengthen housing affordability, aimed at supporting the construction of 1.5 million new housing units. The package includes measures to secure land for building, reduce construction costs and address skills shortages in the construction sector. The key measures have been grouped into three broad categories: stimulating investment in housebuilding; affordable housing; and reducing costs and tackling skills shortages. Social housing is a key component of this programme, with a total budget of €5.0 billion for the period 2018-2021 (EC 2022, pp. 26-27).

Portugal

A survey was conducted in Portugal last year on public opinion and forecasts regarding the housing situation and government measures (Statista, 2024a). The results revealed that 54 percent of respondents viewed the government measure of providing full financing for the purchase of a property positively. This data may have some relation to the low percentage of social rental housing in that country in relation to the total housing stock (Figure 21).

Figure 21. Social rental housing as a share of the total housing stock in selected countries worldwide from 2018 to 2023

Source: OECD (2024)

To effectively implement a just transition in the housing and built environment sector, Portugal has devised several national action plans that align with the objectives of FITTER-EU. These plans focus on integrating sustainability with social inclusion, and they can provide valuable insights into how to ensure that both producers and consumers benefit from the transition to a more sustainable housing sector (Table 33).

Table 33. Housing: National Action Plans Aligned with FITTER-EU Objectives

Action Plan/Policy	Strategic Targets	Impact Assessment	Objectives Alignment with FITTER-EU
Social Housing Policies	Increase affordable housing supply, improve living	Impact on low-income groups and social integration,	Aligns with FITTER-EU's focus on inclusivity by ensuring housing is affordable and accessible to vulnerable

	standards for vulnerable groups.	housing affordability.	groups, promoting a just transition for low-income households.
Housing Policies, Market and Home Ownership	Regulate housing market, promote homeownership access.	Focus on reducing housing inequalities, particularly in urban areas.	Supports FITTER-EU's goal of reducing housing inequalities and promoting equitable access to homeownership, ensuring both producers (developers) and consumers benefit.
Territorialised Housing Policy	Address territorial housing disparities, improve regional housing quality.	Focus on regional imbalances in housing provision, especially in rural and suburban areas.	Aligns with FITTER-EU by ensuring housing equity across different regions, reducing rural-urban housing disparities, and supporting local development.
Urban Policies in the 2030 Agenda	Implement sustainable urban development, improve housing conditions.	Assess impact on urban regeneration and sustainable housing development in cities.	Supports FITTER-EU's sustainability objectives by promoting green, resilient cities, benefiting both urban producers and consumers with eco-friendly solutions.
State-Subsidised Housing	Provide state-subsidised housing, improve architectural integration.	Impact on low-income and social housing provision, especially in urban regeneration.	Aligns with FITTER-EU's goal of promoting equitable housing and ensuring that low-income households have access to quality housing.
Rent Regulation	Regulate rent prices to ensure affordability and prevent displacement.	Focus on protecting renters, reducing housing cost burdens for low-income populations.	Supports FITTER-EU's consumer protection goals by ensuring fair and affordable rent prices, reducing displacement risks, and promoting a just transition for renters.
National Energy and Climate Plan (NECP 2030)	Improve energy efficiency in housing, reduce carbon emissions in buildings.	Significant environmental benefits through energy-efficient housing and renewable energy use.	Directly aligns with FITTER-EU's sustainability goals by promoting energy-efficient housing, benefiting both producers and consumers through reduced environmental impact.

Source: Medeiros (2022); Marques et al., (2014); Santos y Teles (2015); Pinto (2017); Alves (2019); Pinto & Guerra (2019); Portugal DGEG, (2019); Figueiredo et al., (2020); Laranja et al., (2020); Oliveira (2020); Ramos et al., (2021); Lameira et al., (2022); Alves et al., (2023); Bomfim (2024); Carbone (2024); Carvalho, (2024); Chamusca (2024).

Considering the previous information, both countries are working on:

- **Availability:** Increasing the social housing stock and public investment in areas of greatest need seek to improve the availability of affordable housing. Social housing is housing that is owned by the government or local authority, it is rented out to people who receive benefits at a discounted rate compared with other housing in the area.
- **Accessibility:** Promoting affordable housing through cost-restricted construction and supports accessibility improvements so that older people can remain in their homes as they age. Urban regeneration is also important, providing affordable housing in areas with better access to services and employment opportunities.
- **Affordability:** Several financial support mechanisms aimed at addressing inequalities in housing affordability, as well as rent control policies and the construction of affordable housing are seeking to mitigate the financial pressures faced by tenants.

The determinants of housing poverty are multifaceted and stem from various factors. Economic conditions play a significant role, as low wages, unemployment, and economic downturns can severely limit individuals' ability to afford housing. Additionally, imbalances in supply and demand contribute to housing poverty. A limited supply of affordable housing units, paired with high demand, intensifies the issue.

Government policies and regulations are also crucial. Zoning laws and housing subsidies directly affect both the availability and affordability of housing. When governments fail to invest adequately in social housing, poverty rates are more likely to rise, exacerbating housing challenges for vulnerable populations. Finally, social factors, such as discrimination based on race, ethnicity, or socioeconomic status, can further limit access to housing, reinforcing cycles of poverty. This can result in marginalized communities facing greater barriers to securing adequate and affordable housing, contributing to the persistence of housing poverty.

4.2.2. HOUSING POVERTY: ASSOCIATED VULNERABILITIES

In Europe, housing vulnerabilities are closely linked to various demographic factors, such as age, gender, income, and other social determinants. These factors influence the risk of experiencing housing poverty, insecurity, and exclusion. Below are some key demographic vulnerabilities related to housing:

Age:

- **Children and young people:** Families with children, particularly single-parent households, are at higher risk of living in poor-quality or overcrowded housing. Housing affordability issues affect young people as well, with many struggling to access housing due to low wages and high rental costs. In Germany, the Federal Ministry of Housing, Urban Development, and Construction (BMWSB) has special programmes that focus on supporting housing for young people and students.
- **Elderly people:** Older adults, particularly those on fixed pensions, often face challenges in accessing affordable housing. Many are at risk of living in inadequate housing conditions, especially in cities where rents and housing prices have increased significantly. Elderly people may also face difficulties with accessibility (e.g., lack of elevators, poor maintenance of properties) and may experience social isolation because of housing instability. In Germany, the "Altersgerecht Umbauen" programme helps seniors modify their homes to meet their needs as they age.

Gender:

- **Women:** Gender disparities in income and employment often lead to women being more vulnerable to housing poverty. Women are more likely to live in poverty, particularly single mothers or women over 65. Women are also more likely to face discrimination in the housing market, especially when dealing with issues related to child custody or domestic violence. Single mothers are in particular at risk of housing insecurity due to the challenges of balancing low wages with childcare responsibilities.
- **Gender-based violence victims:** Women who have experienced domestic violence are highly vulnerable to housing insecurity. They may need to leave abusive situations but struggle to find affordable housing due to financial constraints and a lack of specialized housing services for victims.

Germany's housing and built environment transition emphasises addressing the specific challenges faced by structurally vulnerable individuals and families, including those in precarious economic situations, older and younger age groups, and those experiencing gender-specific issues (notably single parents and individuals at risk of violence). Measures have been implemented to secure housing options for women and children affected by violence

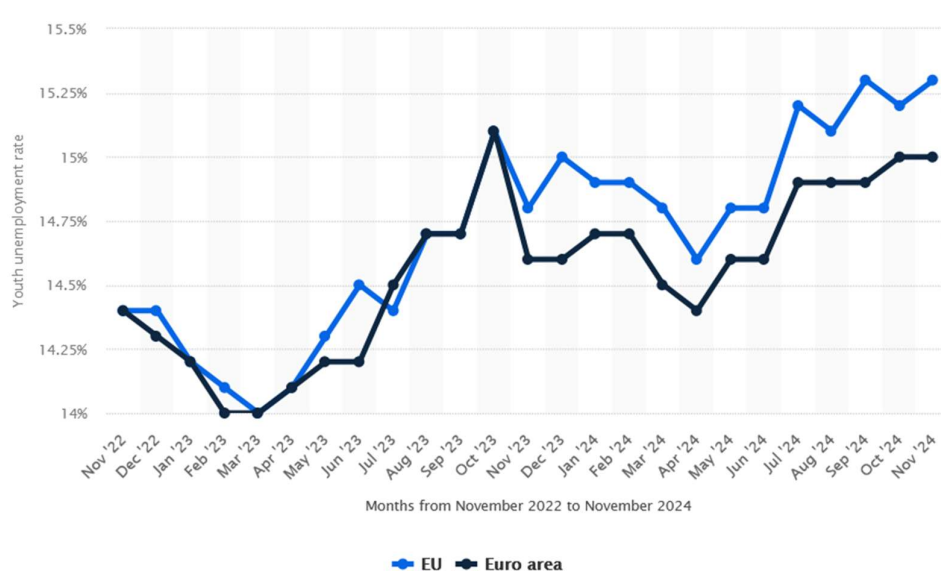
Income:

- **Low-income households:** Low-income households are the most vulnerable to housing poverty. These families often spend a large proportion of their income on housing costs, leading to financial strain.

In Germany, the government has committed to a carbon dividend (Klimageld) to especially relieve low-income households, ensuring that the social costs of the energy transition are equitably distributed (but has not initiated it yet). These measures are expected to be expanded with national funding beyond 2026, illustrating Germany's long-term commitment to decarbonising the building sector and transitioning toward a sustainable, energy-efficient future. Additionally, the "National Action Plan against Homelessness" aims to introduce the "Housing First" approach more widely into German housing and construction policies.

- **Youth and the unemployed:** Young people, especially those entering the workforce with low-paying or insecure jobs, face challenges in securing affordable housing. Similarly, individuals experiencing long-term unemployment are at high risk of housing insecurity, particularly as housing costs outpace income growth. In the European Union and the Euro area, the seasonally adjusted youth unemployment rate (under 25 years) has trended upward from November 2022 to November 2024. Over the past year, the EU's youth unemployment rate increased from 14.8 percent in November 2023 to 15.3 percent in November 2024 (Figure 22).

Figure 22. Youth unemployment rate (under 25 years) in the EU and the Euro area (November 2022 to November 2024)



Source: Statista (2024b)

Disability and Health Status:

- **People with disabilities:** Disabled individuals often face additional barriers when it comes to accessing affordable and suitable housing. Inaccessible housing (e.g., lack of ramps, inadequate wheelchair access) and high costs make it particularly difficult for people with disabilities to secure adequate housing. Furthermore, disabled individuals often have higher living costs due to the need for specialized equipment or care.
- **Chronic health conditions:** Those with chronic health conditions, particularly when combined with low income or age, are more likely to live in substandard housing conditions that exacerbate their health problems. Poor-quality housing, such as inadequate heating or dampness, can have significant negative effects on individuals with respiratory issues or other chronic conditions.

Ethnic minorities and migrants:

- **Immigrants and refugees:** Migrants, particularly refugees or undocumented individuals, face heightened vulnerability in the housing market. Language barriers, discrimination, and legal status can hinder their access to safe and affordable housing. Additionally, many migrants live in overcrowded conditions or are forced to live in informal housing arrangements due to limited access to the formal housing market.
- **Ethnic minorities:** People from ethnic minority backgrounds often face discrimination in the housing market, which can limit their access to quality housing. In some cases, these individuals may be subject to housing segregation or live in poorer neighborhoods with higher levels of poverty and insecurity.

Geographic location:

- **Urban vs. Rural:** People living in cities are often more vulnerable to housing poverty due to high rents and limited affordable housing. However, people living in rural areas may also face challenges, including a lack of affordable housing options and limited access to public services and employment opportunities. These factors create a unique set of vulnerabilities depending on the geographic location of individuals and families.

Overall, these demographic factors create intersecting vulnerabilities, meaning individuals in multiple categories (e.g., low-income single mothers, elderly people with disabilities) may face compounded risks related to housing insecurity and poverty. Addressing these challenges requires targeted policies and interventions that consider the diverse needs of vulnerable populations across Europe.

Regional disparities in housing and urban development are prominent due to differences in population density, economic conditions, and historical development patterns between urban and rural areas. These disparities affect housing affordability, access to social housing, and overall living conditions across regions.

Germany

Regional disparities in housing across Germany are complex and involve a range of factors, including economic, social, and environmental dimensions. While urban centers face issues of affordability and overcrowding, rural regions struggle with population decline and underdeveloped housing markets. The federal government, in collaboration with state authorities, is working towards bridging these gaps through targeted investments in social housing, regional development, and climate adaptation strategies. These efforts aim to ensure that all citizens, regardless of their region, can access quality housing and enjoy an improved quality of life.

- Urban vs. Rural divide.** The rural-urban divide is a central issue in Germany's housing landscape. Urban areas, especially large cities like Berlin, Munich, and Hamburg, have seen skyrocketing housing demand, leading to inflated housing prices and significant affordability challenges. Young families, single-parent households, and low-income groups often find it difficult to secure adequate housing in these cities. Urban housing markets are characterized by high rents, limited availability of social housing, and gentrification in certain neighbourhoods. In contrast, rural areas face challenges such as depopulation, especially in structurally weak regions, where housing demand is low, and the construction sector is often underdeveloped. Consequently, while there is a pressing need for housing in cities, many rural areas struggle with disused or empty homes due to migration to urban centers.
- Federal and state disparities.** Germany's federal structure plays a crucial role in addressing housing disparities. Housing policies and their implementation differ significantly between the federal states (Länder). Since 2006, the responsibility for social housing and public construction has been decentralized, with each state having the authority to design and implement policies tailored to its regional needs. However, some states have been more proactive in addressing housing poverty than others, creating a patchwork of housing opportunities across the country. For instance, Bavaria and North Rhine-Westphalia have been more successful in securing funding for social housing, while states in the former East Germany (e.g., Saxony and Brandenburg) often face challenges related to population decline and aging infrastructure.
- Government actions and regional development.** The Federal Ministry for Housing, Urban Development, and Construction (BMWSB) has recognized the need for equitable living conditions across all regions and has launched programs to address the disparities. The BMWSB's "Plan for Germany - Equivalent Living Conditions Everywhere" seeks to balance the development of urban and rural areas by fostering sustainable development and spatial planning. The program focuses on revitalizing structurally weak rural areas, providing funding for local development projects and promoting innovative approaches to spatial planning through initiatives like the Model Projects for Spatial Planning (MORO).

Additionally, measures to address housing inequality between regions include targeted financial support for social housing construction, which is currently supported by a joint federal-state

initiative. For example, in 2022-2027, Germany has proposed €18.15 billion in funding for social housing construction, with each federal state co-financing the efforts. This aims to create more affordable housing in regions where supply is limited, particularly in rural and economically disadvantaged areas.

- **Climate and heat protection strategies.** The impact of climate change is another critical factor in regional disparities. Extreme heat events disproportionately affect urban areas due to the urban heat island effect, which results in higher temperatures compared to surrounding rural areas. Vulnerable populations, such as the elderly, children, and those with chronic illnesses, are more susceptible to heat stress. The BMWSB has responded by launching the “Heat Protection Strategy” to address the impacts of extreme weather. This strategy includes measures such as creating more green spaces, improving urban cooling infrastructure, and protecting vulnerable groups (e.g., the homeless and outdoor workers) in urban areas.

On the other hand, rural areas may face challenges related to energy efficiency and decarbonization. Many rural homes and buildings lack the infrastructure for energy-efficient renovations, and the decentralization of heating systems makes the decarbonization of heating networks a more complex issue. To address these challenges, Germany has introduced the Wärmeplanungsgesetz (Heat Planning and Decarbonisation of Heating Networks Act), which mandates heating plans in cities with populations above 100,000 by 2026, and in smaller municipalities by 2028.

Portugal

In the context of “regional disparities” in Portugal, the information reveals a series of dynamics that vary significantly across the country's territory. According to Marqués et al., (2014) here are some key points that highlight these disparities:

- **Evolution of the housing stock.** Although there has been significant expansion in the housing stock in Portugal over the past few decades, the increases have not been uniform at the regional level. For example, the Algarve and the Autonomous Region of Madeira experienced the highest increases in the number of housing units, largely due to the importance of tourism, which drove the construction of second homes and hotels. In contrast, the North and Alentejo saw lower increases (10% compared to 24% in the Algarve).
- **State of housing conservation.** Newer housing units are concentrated in urban areas such as Lisbon and Porto, while more rural areas, particularly in the interior and certain parts of Alentejo and the Azores, face greater challenges in terms of aging infrastructure. Older and more deteriorated housing units are found in the historic centres of large cities like Lisbon and Porto, as well as in certain urban centres in the North and Central regions of the country.
- **Living conditions and comfort.** Living conditions have generally improved, but regional disparities in housing comfort remain evident. Rural areas in the North and South of Portugal face more difficulties, especially in terms of bathroom facilities and thermal comfort.
- **Overcrowding in housing.** Housing overcrowding, a form of congestion, is a particularly acute problem in densely populated urban areas, especially in the North and metropolitan areas like Lisbon and Porto. This is a clear manifestation of regional disparities, as the most affected areas include urban agglomerations where populations with lower incomes and less access to education predominantly reside.
- **Impact of the economic crisis and geographical distribution of the crisis.** The economic crisis has exacerbated regional disparities, affecting vulnerable groups such as young people, immigrants,

and temporary or low-wage workers more severely. Furthermore, the crisis has unevenly affected municipalities' ability to finance public housing projects, creating more inequality in housing conditions between different regions.

The impact of the crisis has also been territorially unequal. More urbanized areas, such as the metropolitan areas of Lisbon and Porto, have experienced greater fluctuations in the real estate market, contributing to increased regional inequalities in terms of access to quality housing. In contrast, some rural areas in the interior have faced less pressure in terms of housing prices but have had fewer resources and less investment in housing renovation.

Regional disparities in Portugal, especially in the context of housing, are evident in terms of quality, access, and housing conditions in different regions of the country. Urban and metropolitan areas, such as Lisbon and Porto, face challenges related to overcrowding and aging infrastructure, while rural areas, especially in the interior, struggle with a lack of basic services and adequate housing. The economic crisis has exacerbated these inequalities, requiring more attention in public policies to address regional disparities and design solutions better suited to the diverse realities of the territory (Guerra, 2011; Ferrão, 2014).

In conclusion, both Germany and Portugal face significant regional disparities in housing, each with unique challenges linked to their respective economic, social, and environmental contexts. In Germany, the urban-rural divide is marked by overcrowded cities with skyrocketing housing prices and rural areas experiencing depopulation and underdeveloped housing markets. Federal and state-level policies, such as targeted investments in social housing and regional development, aim to bridge these gaps and improve living conditions across regions. Additionally, climate change considerations, such as the urban heat island effect, further complicate housing disparities, necessitating comprehensive strategies for both urban cooling and rural energy efficiency.

Similarly, Portugal's housing challenges are characterized by uneven regional development, with urban areas like Lisbon and Porto experiencing overcrowding and deteriorating infrastructure, while rural regions, particularly in the interior, struggle with insufficient investment in housing renovation and basic services. The economic crisis has deepened these disparities, leading to more pronounced inequalities in housing quality and access. Public policies must focus on addressing these regional imbalances, ensuring that all citizens, regardless of location, can access safe, affordable, and comfortable housing. Both countries need continued efforts in tailored policy interventions to reduce these regional disparities and foster more equitable housing opportunities.

4.2.3. ECONOMIC AND LABOUR MARKET EXCLUSION TRIGGERS

Economic and labour market exclusion are significant challenges in the context of Europe's "twin transition" — the simultaneous shift towards a green and digital economy. This transition, while necessary for long-term sustainability and competitiveness, can inadvertently exacerbate social inequalities and create barriers for certain groups in the labour market. The shift toward cleaner technologies, digitalization, and decarbonisation often requires new skills, which can leave those without access to education or retraining opportunities at a disadvantage. At this stage, several groups are particularly vulnerable to labour market exclusion during the twin transition:

- **Older workers:** Older employees who have spent their careers in traditional industries may find it difficult to transition to new, greener or digital sectors. They are often less likely to engage in lifelong learning or digital upskilling, putting them at risk of unemployment or underemployment.
- **Low-income individuals:** Those with lower levels of education or living in economically disadvantaged regions may struggle to access the necessary training or qualifications for emerging

green and digital jobs. This lack of access to education and job opportunities can further entrench poverty and inequality.

- **Youth:** Although young people may be more digitally literate, they may still face barriers to entry in high-skill industries due to a lack of work experience or access to mentorship and networks in emerging sectors.

In the green transition, sectors such as renewable energy, energy efficiency, and low-carbon technologies offer new job opportunities, but they also require a workforce with specialized knowledge. Similarly, the digital transition demands highly skilled workers in areas such as information technology, data science, and artificial intelligence. Without targeted policies and investments in education, training, and social support, many individuals could face economic and labour market exclusion, hindering the overall progress towards a fair and inclusive transition.

The challenge lies in ensuring that the twin transition does not deepen existing disparities but instead offers equitable opportunities for all Europeans. Policy interventions, such as reskilling and upskilling programs, social protection measures, and inclusive labour market strategies, are essential to mitigate these exclusion risks. Addressing these issues will not only enhance social cohesion but also ensure that the twin transition leads to a more resilient, sustainable, and inclusive European economy.

Germany

Germany, with 84.3 million people, is the largest economy in the EU and the fourth largest globally. It is home to major global companies in automotive, chemicals, and electronics. Around 55.1% of its workforce is employed by small and medium-sized enterprises (SMEs). In March 2023, Germany had 45.72 million employed people, with an unemployment rate of 5.7%. Eastern Germany faces higher unemployment (7.2%) than the West (5.3%), though the gap has narrowed. The influx of Ukrainian refugees has likely raised the unemployment rate by 0.4% (EURES, 2023a).

Germany has 773,000 job vacancies as of April 2023, a 9% decrease from the previous year. The country also has a significant number of cross-border commuters. There is strong demand for skilled workers in healthcare, engineering, and STEM fields, as well as other roles like childcare workers, drivers, and agricultural workers (EURES, 2023a).

The Skilled Labour Strategy was adopted by the German government in autumn 2022 to address the challenges of having enough qualified technical workers, especially in building services engineering, energy, and construction sectors. This strategy for the 20th electoral term established a supportive framework, highlights areas for action and aims to enhance skilled worker recruitment and retention. It assesses current challenges, refines existing approaches, and sets new priorities. In this context, the Federal Government prioritized the following areas through legislative and sub-legislative measures as part of the Skilled Labor Strategy:

- Strengthening vocational training
- Promoting lifelong learning
- Increasing labour potential through higher workforce participation
- Improving work quality and transforming work culture
- Implementing modern immigration laws.

These do not have a specific timeline because the strategy is a comprehensive approach. Rather, they will be gradually achieved within this approach through increasing investment, legislation, and further administrative support (Federal Ministry of Labour and Social Affairs, 2022).

Portugal

In the first quarter of 2023, Portugal had a population of 10.3 million, with an activity rate of 60.8%. The unemployment rate was 7.2%, affecting women (7.5%) and youth (19.6%) the most. Part-time work is uncommon, making up only 7.3% of employment, especially among women. Remote work, boosted by the pandemic, still affects 19% of the workforce, mainly in sectors like IT and finance. Most registered unemployed individuals are women and foreigners, primarily from Portuguese-speaking countries (EURES, 2023b).

In terms of sectors, 72.1% work in services, with notable areas like vehicle repair, health, and education. The manufacturing and construction industries have shown signs of recovery, and traditional sectors like footwear and fashion are modernizing and expanding internationally (EURES, 2023b).

In Portugal, the twin presents both opportunities and challenges, particularly in terms of economic and labour market exclusion. While these transitions have the potential to drive economic growth, innovation, and sustainability, they also risk exacerbating inequalities, especially for vulnerable groups who may not have the skills or resources to adapt to the changes:

- **Green transition and economic exclusion:** Portugal's commitment to decarbonisation and renewable energy is evident in various national strategies, such as the National Energy and Climate Plan (NECP 2030). As the country shifts towards a low-carbon economy, certain sectors, such as coal mining, oil and gas, and traditional manufacturing, are expected to face significant downsizing. Workers in these industries, particularly those in rural and economically disadvantaged areas, are at risk of economic exclusion due to the closure of polluting industries without immediate alternatives in green sectors. While renewable energy and energy efficiency projects offer new job opportunities, these require specific technical skills, which many current workers do not possess.
- **Digital transition and labor market exclusion:** Digitalization is another key component of the twin transition in Portugal, with sectors like information technology, data analytics, and automation expanding rapidly. However, there is a significant divide in the country between those who have access to digital education and training and those who do not. Rural areas and marginalized communities often face challenges in terms of access to technology, digital skills training, and fast internet connections. This digital divide exacerbates inequalities in the labor market, as individuals without the requisite skills may find themselves excluded from the growing number of digital jobs.

To address these challenges, the Portuguese government has introduced several initiatives aimed at ensuring a just transition for all sectors of society. The National Energy and Climate Plan includes provisions for upskilling and reskilling workers in industries affected by the green transition. Additionally, Portugal is investing in educational programs to equip citizens with the digital skills necessary for the future workforce. The Portugal 2020 program, which aligns with European Union funding frameworks, focuses on supporting skills development, improving labour market inclusion, and promoting social innovation, which can help mitigate exclusion risks.

Moreover, Portugal is placing emphasis on social protection policies, including support for workers who are displaced by the transition. Social housing and rent regulation initiatives also aim to ensure that vulnerable groups do not face economic hardships as urban regeneration projects and the digital economy reshape Portugal's cities.

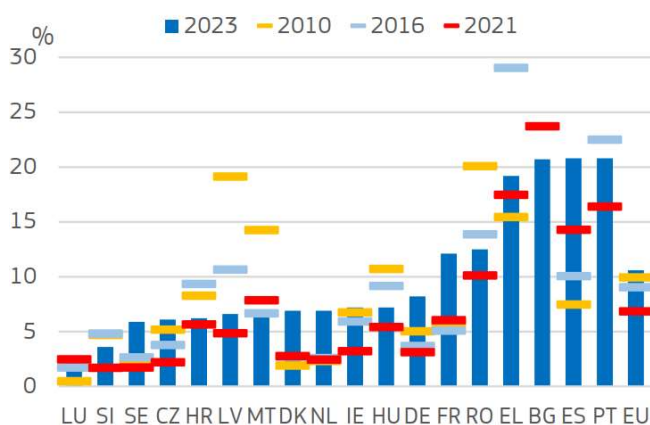
In conclusion, while Portugal is making significant strides towards a greener and more digital economy, the twin transition presents challenges in terms of economic and labour market exclusion. Addressing the skills gap, ensuring access to education and training, and promoting social inclusion through supportive policies will be key to ensuring that the benefits of the twin transition are shared equitably across all sectors of society.

4.2.4. VULNERABILITY TO CLIMATE-CHANGE RELATED DISASTERS

Energy poverty in the EU describes the struggle of households to afford essential energy services, such as heating, cooling and electricity. In 2023, approximately 10.6% of the EU population faced challenges in adequately heating their homes due to affordability, with notable variations between MS. This marks a 54% increase in this energy poverty indicator since 2021, reversing a decade-long decline (Figure 23).

We observe similar timelines in other energy poverty indicators such as the share of people living in households with arrears on utility bills. This stands at a lower level than in 2010 and followed a falling trend between 2014 and 2020. However, it subsequently increased slightly every year, reaching 9.3% of EU households declaring arrears in 2023 (European Commission, 2025a).

Figure 23. Individuals unable to keep the home adequately warm, 2010-2023 (%)



Source: European Commission (2025a)

In 2023, 18.5% of households in the EU lacked adequate insulation or heating systems to maintain a comfortable temperature during the winter, with those facing high housing costs being the most affected. This not only leads to discomfort and financial strain but also results in increased emissions and resource consumption, causing unnecessary environmental harm. Improving energy efficiency can help reduce these negative effects, enhance living conditions, lower energy bills, and contribute to decarbonisation goals (European Commission, 2025a).

Germany

As said, Germany's housing transition is keenly aware of the impact on structurally vulnerable cohorts such as low-income families, children, elderly individuals, single parents, and those affected by rural-urban disparities. These groups often face housing insecurity due to structural challenges in the housing market, such as rising housing prices, low homeownership rates, and limited rental options that match their incomes.

Between 2015 and 2020, housing prices in Germany increased significantly—38.7% overall, with new dwellings rising by 30.4% and existing ones by 40.2% (European Commission, 2022a). This has exacerbated the housing crisis, particularly for vulnerable groups who face disproportionate barriers to accessing affordable housing.

Germany has implemented several initiatives to address these disparities, including:

- The “Baukindergeld” program, which provided grants to families and single parents to encourage homeownership, benefiting those with incomes below EUR 90,000. This initiative aimed to reduce the gap between low and high-income households in terms of homeownership.
- The “Jung kauft Alt” and “Klimafreundlicher Neubau im Niedrigpreissegment” programs, which focus on promoting affordable housing through cost-restricted construction, particularly for young people and those at risk of exclusion.
- The “Altersgerecht Umbauen” (Age-Appropriate Renovation) program, which supports older adults by funding accessibility improvements, ensures that older individuals can remain in their homes longer. In 2024, funding for this program was doubled to EUR 150 million, allowing more elderly individuals to benefit from accessible living spaces.
- Programs aimed at women and children affected by violence, ensuring that housing is secured for these vulnerable groups.

Germany is also committed to a fair and just transition in the housing sector, with an emphasis on regional development and housing affordability for all. Key initiatives include:

- The “Alliance for Affordable Housing”, launched in 2022, which brings together all relevant stakeholders to develop a comprehensive set of policies aimed at stabilizing the housing market, particularly in regions with high demand or economic instability.
- The 14-point package of measures introduced in 2023 to promote investment in housing, including accelerating planning and approval processes through a pact signed by federal and state governments.
- The Growth Initiative and other government policies, such as the “National Action Plan against Homelessness”, which introduces the “Housing First” approach to ensure that everyone, including vulnerable populations, has access to secure housing.

By integrating climate adaptation into the German Building Code, these policies also seek to address the increasing risks posed by climate change, particularly for vulnerable populations affected by extreme heat, poor living conditions, and high energy costs. Through these comprehensive strategies, Germany is aiming to reduce regional disparities and ensure that its housing transition is inclusive and equitable for all citizens, especially those who are most vulnerable.

Portugal

As it was said, energy poverty, defined as the inability to afford adequate heating or cooling, affects vulnerable populations and exacerbates their social and economic hardships. Portugal’s long-term strategies aim to tackle this by improving the energy efficiency of housing, reducing greenhouse gas emissions, and ensuring that low-income groups are not left behind in the green transition. The National Energy and Climate Plan (NECP 2030) emphasizes retrofitting residential buildings, enhancing energy efficiency, and promoting sustainable construction practices. This plan seeks to reduce energy consumption and increase the use of renewable energy in homes, particularly for vulnerable populations who often experience high energy costs.

Moreover, social housing policies and urban regeneration projects play an essential role in tackling housing inequality and ensuring that marginalized communities benefit from the improvements in housing quality and access to energy-efficient homes. The Plan for Recovery of Housing Stock aims to renovate degraded housing stock, while social housing programs focus on providing affordable housing units for low-income groups.

In addition to addressing energy poverty, these policies are designed to ensure a fair and just transition for all, particularly structurally vulnerable cohorts such as low-income families, the elderly, and marginalized

groups. Rent regulation policies help to prevent displacement due to rising housing costs, while urban regeneration projects are intended to avoid gentrification and ensure that existing residents are not pushed out as property values increase.

Portugal's commitment to a just transition is evident in its emphasis on inclusive policies that promote affordable housing, energy efficiency, and social equity, ensuring that the benefits of the green transition are equitably shared across society. The alignment with European frameworks like the European Green Deal further strengthens this commitment, integrating sustainability and social justice into housing policy. In summary, both Germany and Portugal are tackling significant challenges related to energy poverty and housing inequality, with a focus on ensuring an inclusive, green transition in the housing sector. Rising energy costs and inadequate housing have worsened energy poverty, especially for vulnerable groups. Germany addresses these issues through policies promoting affordable housing, energy-efficient construction, and support for the elderly. Portugal focuses on energy-efficient housing and retrofitting as part of its National Energy and Climate Plan, alongside social housing programs for low-income groups. Both countries aim to ensure that the benefits of the green transition are shared equitably, fostering social justice, sustainability, and regional development.

4.3. CHALLENGES AND OPPORTUNITIES OF JUST TWIN TRANSITION PROCESS

In terms of common opportunities and challenges:

- **Job creation:** The transition to a sustainable construction sector can create new jobs in areas such as energy retrofitting, renewable energy installation, home insulation and green building. In the process, special attention must be paid to:
- **Skills gap:** The transition to a green economy in Portugal and Germany demands highly skilled workers in areas like renewable energy technologies, energy efficiency, and sustainable agriculture. However, a large portion of the workforce in sectors facing decline (such as fossil fuels and traditional industries) may not have the necessary qualifications for these emerging fields. The lack of access to reskilling and upskilling opportunities further increases the risk of job displacement and economic exclusion, especially for older workers or those with lower levels of formal education.
- **Job polarization:** The digital transition also leads to job polarization, where high-skill, high-wage jobs in technology and data science are expanding, while low-skill, low-wage jobs in industries such as retail and traditional manufacturing are being replaced by automation. Workers in low-skill jobs, especially those who are not digitally literate or cannot afford retraining programs, risk being excluded from the labour market altogether.
- **Reduced housing costs** through improved energy efficiency.
- **Improved quality and availability** of public housing.
- **Improved health and well-being:** Energy-efficient and climate-adapted housing can provide a healthier and more comfortable indoor environment.

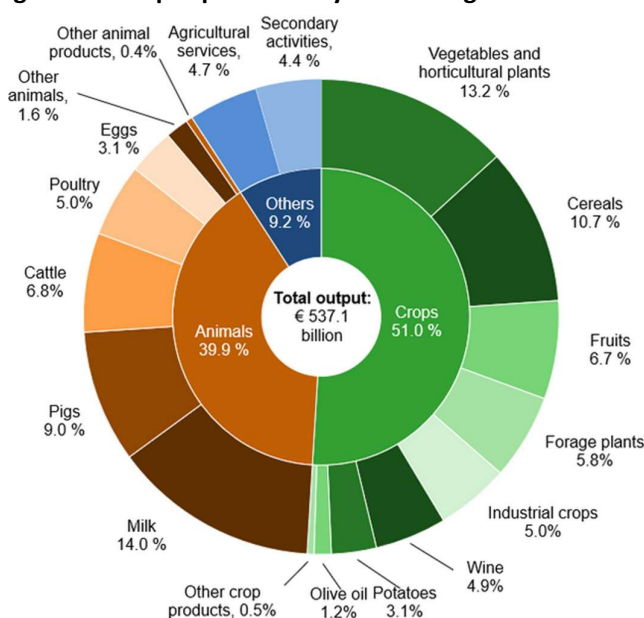
The successful implementation of the twin transition in Portugal requires continued attention to the needs of vulnerable groups and a comprehensive approach that combines environmental sustainability with social equity.

A similar idea is outlined in the case of Germany, as the plans and programmes described aim to address vulnerabilities related to housing affordability, energy poverty and the impact of climate change. By increasing the availability of affordable housing, promoting energy efficiency and creating a more climate-resilient built environment, Germany seeks to ensure a fair transition that benefits all members of society.

5. AGRICULTURE AND FOOD SECTOR

As a first approach to the sector, this introduction includes an overview of the situation of the agri-food industry in two senses: 1) in Europe, including economic data and challenges that affect the sector and; 2) main data to contextualize the sector in Italy and Ireland, the two countries on which the analysis focuses. The European agri-food ecosystem is a fundamental part of the European Union (EU) economy, contributing 1.3% to the EU's GDP, employing around 7.3 million people (CEDEFOP, 2023) and, in 2023, creating gross value added of €223.9 billion. According to Eurostat (2024d) "the value of everything that the EU's agricultural industry produced in 2023 was €537.1 billion; this includes the value of crops, of animals, of agricultural services, as well as some goods and services that were not strictly agricultural, but which could not be separately measured" (Figure 24).

Figure 24. Output produced by the EU's agricultural industry (% of total output, 2023)



Source: Eurostat (2024d)

In terms of composition, most of the companies in the sector are small and medium-sized enterprises (SMEs). However, it is the large companies that have the greatest employment capacity, since they cover 40% of the workforce and generate more than half of the sector's turnover (European Commission, n.d.-a).

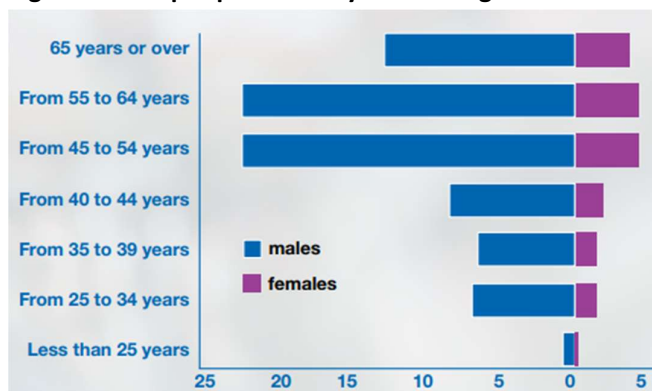
Despite these good figures, the sector faces at least three challenges today (European Commission, n.d.-a; CEDEFOP, 2023). All of them are deeply treated in the present document:

- Environmental impact.** Agriculture contributes to climate change mainly by emitting three greenhouse gases (GHGs). First, there is methane (CH₄) from livestock digestion processes (enteric fermentation), manure management and rice cultivation. Secondly, nitrous oxide (N₂O) from agricultural soils with organic and mineral nitrogen fertilization and manure management. Thirdly, carbon dioxide (CO₂) from soil because of agricultural land management such as ploughing or conversion of land use, e.g. from grassland to cropland. These practices can lead to mineralization of the soil organic carbon to CO₂ emitted into the atmosphere (European Commission, 2023f). Globally, the agri-food sector generates around a third of all greenhouse gas (GHG) emissions and

other air pollutants in the EU. Between 1990 and 2019, its share in global emissions fell by 10 percentage points, but in Europe the share increased (from 24% to 31%) (Tubiello et al., 2022).

- Innovation and digitalisation.** To meet future food demands and at the same time address environmental concerns, increasing innovation and efficiency is necessary (CEDEFOP, 2023). In this sense, advanced technologies are essential (both increasing efficiency and limiting environmental impact) due they can increase farm performance by enhancing sustainability, productivity, and resilience, especially through Internet of Things (IoT) technologies, sensors, data analytics (e.g. based on Artificial Intelligence) (European Commission, 2024c). Through Horizon 2020, “more than €200 million for Research and Innovation (R&I) were allocated to the deployment of digital technologies for the agricultural sector” (European Commission, 2023d).
- Labour productivity.** There is a long-established downward trend in the number of people working in the EU's agricultural sector (Eurostat, 2024d): agricultural income per annual work unit has declined for the EU in 2023 (-7.6%) although to a level that remained 34.3% higher than the index level in 2015. To this data, we must add the difficult generational change that the sector faces, because most young people are not attracted to living in rural areas and to low income jobs with strenuous working conditions. In 2020 only one in three farm managers was 40 years old or younger (CEDEFOP, 2023, p. 7) (Figure 25).

Figure 25. Output produced by the EU's agricultural industry (% of total output, 2023)



Source: CEDEFOP (2023, p. 7)

Overview of the agri-food sector in Italy

All these elements are not foreign to the two countries under study. Italy's agricultural sector and agri-food system "contribute greatly to the country's economy and represent around 2% and 15% of GDP, respectively" (European Commission, 2023f). The country has around 1.1 million agricultural holdings, covering around 12.6 million hectares of the country's agricultural area. Furthermore, more than 50% of the total area dedicated to agricultural use is classified as mountainous or with natural limitations and 53% of the Italian population lives in rural or intermediate areas, with the agricultural and forestry sectors being important economic drivers (European Commission, 2023f).

Thanks to the different climates, soils and topography that mix in the country, Italian agriculture has one of the most diversified agricultural productions in the EU (Tables 34 and 35).

Table 34. Agriculture output per sector in Italy. Crop products (million EUR)

Crop Product	Crop output % in MS	MS Crop output as % of EU
Totals	66.5%	8.5%
cereals (including seeds)	6.8%	0.9%
industrial crops	1.6%	0.2%
forage plants	3.3%	0.4%
vegetables and horticultural products	22.3%	2.9%
potatoes (including seeds)	2.1%	0.3%
fruits	10.5%	1.3%
wine	16.1%	2.1%
olive oil	3.8%	0.5%

Source: European Commission (2023h)

Table 35. Agriculture output per sector in Italy. Animal products (million EUR)

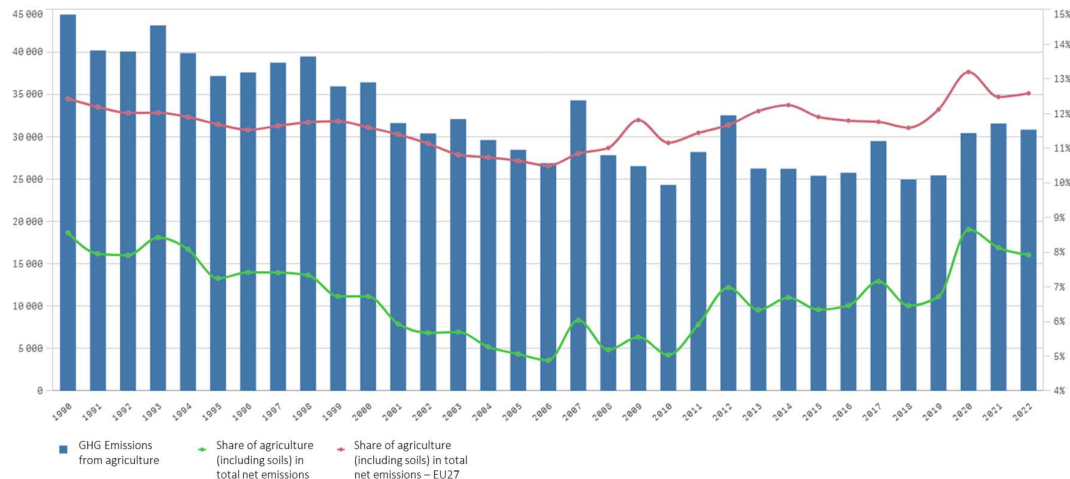
Animal Product	Animal output % in MS	MS Animal output as % of EU
Totals	33.5%	4.3%
cattle	6.5%	0.8%
pigs	6.7%	0.9%
sheep and goats	0.4%	0.0%
poultry	4.7%	0.6%
milk	12.1%	1.5%
eggs	3.1%	0.4%

Source: European Commission (2023h)

By geographical area, cereals, soy, meat and dairy products are produced in the north of the country, while the southern regions specialize in fruits, vegetables, olive oil, wine and durum wheat. "This makes the country one of the largest agricultural producers and food processors in the EU" (European Commission, 2023f), also having the largest number of agri-food products with Protected Designations of Origin and Protected Geographical Indications recognized by the EU. It is currently the main wine producer, by volume, in the world (European Commission, 2023f).

Regarding the pollution of the sector the indicator presented below (Figure 26) represents a sum of mainly non-CO₂ emissions of enteric fermentation, manure management, rice cultivation and soil management and of the CO₂ emissions and removals from LULUCF (Land Use, Land Use change and Forestry) from grassland and cropland. These are reported by Member States under the 'Agriculture' and 'LULUCF' sectors as defined by the International Panel on Climate Change (IPCC), in the national greenhouse gas inventory submitted to the United Nations Framework Convention on Climate Change.

Figure 26. GHG emissions from agriculture (including cropland and grassland) (1.000t of Co2 equivalent) in Italy



Source: European Commission (2023h)

EU GHG emissions from agriculture (including cropland and grassland) have fallen by more than 20% since 1990, but they have stagnated since 2010. In 2021, they accounted for 13% of total EU GHG emissions. This indicator does not include emissions of CO₂ from the energy use of agricultural machinery, buildings and farm operations or emissions from production of inputs, such as inorganic fertilizer.

Overview of the agri-food sector in Ireland

According to data from the European Commission (2023e), 99% of Ireland is covered by predominantly rural and intermediate regions, making the country one of the Member States with the highest percentage of the population living in rural areas compared to the EU average. There are around 135.000 farms in Ireland, with an average farm size of 33.4 hectares (Ireland has an estimated 6.9 million hectares of land, of which about 64% are suitable for agriculture). The total number of active farmers is 127.000. However, despite high levels of productivity, employees in this sector have low-income levels compared to other sectors of the economy. The agri-food sector accounts for 4.3% of the country's economy (total GVA). Additionally, Irish exports of agri-food products account for around 9% of total merchandising exports (European Commission, 2023e).

Productive diversification is also notable in Ireland, although its strong points are milk, cattle, forage plants and pigs (Tables 36 and 37).

Table 36. Agriculture output per sector in Ireland. Crop products (million EUR)

Crop Product	Crop output % in MS	MS Crop output as % of EU
Totals	24.3%	0.6%
cereals (including seeds)	4.0%	0.1%
industrial crops	0.3%	0.0%
forage plants	13.4%	0.3%
vegetables and horticultural products	3.6%	0.1%
potatoes (including seeds)	2.5%	0.1%
fruits	0.6%	0.0%
wine	0.0%	0.0%
olive oil	0.0%	0.0%

Source: European Commission (2023g)

Table 37. Agriculture output per sector in Ireland. Animal products (million EUR)

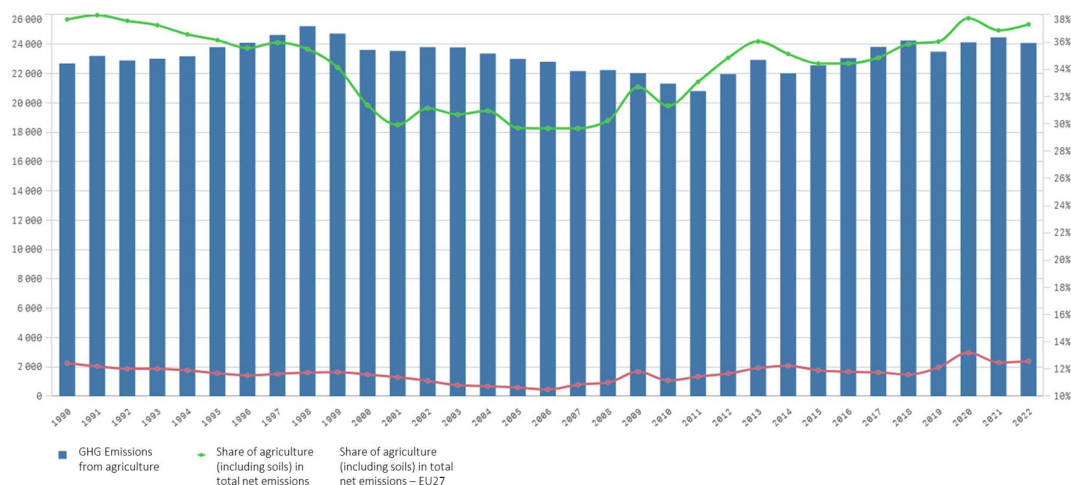
Animal Product	Animal output % in MS	MS Animal output as % of EU
Totals	75.7%	1.7%
cattle	28.8%	0.7%
pigs	6.7%	0.2%
sheep and goats	4.0%	0.1%
poultry	2.1%	0.0%
milk	33.1%	0.8%
eggs	1.0%	0.0%

Source: European Commission (2023g)

Among the measures that Ireland supports for the development of the agricultural sector, there is support for farmers who grow protein crops (such as peas, beans, lupins and soya) and for the fruit, vegetable and beekeeping sectors (European Commission, 2025b).

Regarding the GHG emissions from agriculture in Ireland, the following Figure (27) represents a sum of mainly non-CO₂ emissions of enteric fermentation, manure management, rice cultivation and soil management and of the CO₂ emissions and removals from LULUCF (Land Use, Land Use change and Forestry) from grassland and cropland in Ireland.

Figure 27. GHG emissions from agriculture (including cropland and grassland) (1.000t of Co2 equivalent) in Ireland



Source: European Commission (2023g)

The following sections address all of these elements and describe their impact on vulnerable groups in both countries. It should be noted that the approach adopted from now on involves an analysis in terms of employability and investments made to activate the double transition, all within the framework of European legislation. In this sense, consumer trends will be taken into account when they affect these areas, but an analysis of consumer preferences will not be carried out.

5.1. COMMON GOALS AND PRIORITY ACTION LINES OF THE TWIN TRANSITION PROCESS

The Common Agricultural Policy (CAP) of the European Union (EU) is a set of rules and measures that regulate the agricultural sector in member countries, with the aim of ensuring the sustainability of agricultural production, food security and supporting rural development. In this context, both Ireland and Italy have designed national strategies aligned with the principles of the CAP, each with specific approaches that respond to their national challenges and objectives (Table 38).

Table 38. Key common points

Goal	Approach
Ensuring an adequate and stable supply of food	The CAP aims to ensure a stable food supply at reasonable prices, guarantee farmers' income, promote sustainable practices, and increase competitiveness by modernizing farms. In Ireland, actions under Food Vision 2030 focus on reducing emissions, increasing carbon capture, and promoting biodiversity. Italy's CAP Strategic Plan 2023-2027 emphasizes ecological transition, focusing on agroecological practices, organic production, and animal welfare, while addressing depopulation through entrepreneurship and employment support in rural areas.

Sustainability and environmental protection	CAP supports environmental practices through direct payments to farmers linked to sustainability efforts, such as reducing greenhouse gas emissions and improving soil fertility. Ireland promotes precision grazing and methane reduction, while Italy supports organic farming and agroecological practices. Both countries use eco-schemes to incentivize voluntary environmental commitments like reducing pesticide use and promoting biodiversity. CAP also manages agricultural crises through intervention funds to stabilize prices during extreme weather events.
Rural development and digitalisation	The CAP supports rural development by improving infrastructure, job creation, and economic diversification. Ireland integrates digital technologies to manage resources and combat climate change. Italy fosters rural development through sustainable forestry and the circular bioeconomy while attracting young farmers. Both countries emphasize digitalisation and innovation to improve agricultural efficiency and sustainability.
Promoting competitiveness	The 2021 CAP reform focuses on sustainability, climate change adaptation, and digitalisation. Both Ireland and Italy align with the European Green Deal, working to reduce agriculture's carbon footprint and improve biodiversity. Italy's CAP strategy allocates resources for green measures, such as organic farming, while Ireland adapts agricultural practices to climate change, promoting long-term sustainability.

Source: Own elaboration

Summarising, the Italian CAP Plan aims at enhancing the competitiveness and sustainability of the country's diversified agriculture and rural areas. Its strategy addresses a wide scope of needs from the different territories. This includes an adequate income for farmers, protecting them from adverse climatic events, reducing the impact of agriculture on the environment, tackling labour exploitation, and improving the quality of life in rural areas (European Commission, 2023f). The Irish strategy has been designed "to promote the sustainable development of the farming and food sector by supporting viable farm incomes and enhancing competitiveness. The Irish CAP Plan also contributes to the achievement of environmental and climate objectives at national and EU levels and aims to improve socio-economic conditions in rural areas" (European Commission, 2023e).

5.1.1. MAPPING OF KEY TARGETS

Italy

Italy's CAP strategic plan aims to address the transition towards more sustainable and resilient agriculture while tackling key issues related to the environment, climate, and social aspects. These objectives are directly aligned with the reduction of greenhouse gas emissions, enhancing carbon sequestration, promoting sustainable energy, reversing biodiversity loss, and fostering efficient management of natural resources.

The plan incorporates national action plans such as the "Piano d'Azione Nazionale (PAN) per l'uso sostenibile dei prodotti fitosanitari" (National Action Plan for the Sustainable Use of Plant Protection Products), the "Piano d'azione per il miglioramento della qualità dell'aria" (Action Plan for Air Quality Improvement), and the "Piano d'azione nazionale per la produzione biologica e i prodotti biologici" (National Action Plan for Organic Production and Products). These action plans provide more specific measures and refer to legal sources like laws, regulations, and decrees, further detailing the CAP strategy's measures.

The strategy's specific objectives directly address several critical areas. These objectives are as follows:

1. Objective No. 2 (SO. No.2): Improving market orientation and increasing farm competitiveness both in the short and long term, with an increased focus on research, technology, and digitization. To achieve this, the strategy includes measures for the wine, fruit and vegetable, potato, oil, and beekeeping sectors to promote sustainable production. In the wine sector, competitiveness will be strengthened by enhancing quality and adapting production structures for full sustainability. In the fruit and vegetable sectors, the focus will be on strengthening producer organizations, implementing sustainable production methods, and contributing to climate change mitigation. For beekeeping, resilience will be increased by promoting training, technical knowledge, and research to combat declining bee productivity and mortality.
2. Objective No. 3 (SO. No.3): Improving the position of farmers in the value chain, which will be achieved by strengthening and harmonizing quality systems and promoting certification schemes that recognize sustainable production practices. Transparency will also be ensured through enhanced traceability systems for food products and forest raw materials. Moreover, local supply chains, including forestry, will be strengthened and diversified to create new marketing opportunities.
3. Objective No. 4 (SO. No.4): Contributing to climate change mitigation and adaptation, with a focus on reducing greenhouse gas emissions, enhancing carbon sequestration, and promoting sustainable energy. The strategy will support renewable energy production from agricultural, livestock, and forestry by-products, emphasizing energy-neutral balances and sustainable forest management.
4. Objective No. 5 (SO. No.5): Promoting sustainable development and the efficient management of natural resources. Specific actions include supporting organic farming through eco-schemes, promoting agro-ecological practices, and fostering renewable energy in agriculture. A budget of €9 billion is allocated for these actions over the period 2023-2029.
5. Objective No. 6 (SO. No.6): Contributing to halting and reversing biodiversity loss. This includes supporting agro-ecological crops, conserving and valorizing plant and animal genetic resources, and preserving habitats and natural species linked to agricultural and forestry activities. The strategy promotes collective approaches to natural resource management and targets regions affected by natural handicaps, mountain areas, and agro-climatic-environmental fragility.
6. Objective No. 7 (SO. No.7): Attracting and supporting young and new farmers, particularly women and the long-term unemployed, to foster sustainable entrepreneurial development in rural areas. This objective includes promoting the green and circular economy and encouraging digitalization and innovation in farming businesses. The planned budget for these actions is €1.1 billion over the 2023-2029 period.
7. Objective No. 8 (SO. No.8): Promoting employment, growth, and gender equality in rural areas by increasing sustainable employment opportunities, especially for women and young people. Initiatives will focus on sustainable tourism, bioeconomy, and social agriculture while promoting land management and recovery of abandoned areas. Furthermore, infrastructure gaps will be addressed, particularly in digital access and services, to improve the viability of rural areas.

These specific objectives are reinforced by national plans such as the National Action Plan for Organic Production and Products 2024-2026, which aims to increase the consumption of organic products in public and private catering, promote research, and foster innovations in the organic sector. The Organic School Canteen Fund, established by Decree-Law No. 50 of 24 April 2017 (converted with amendments by Law No. 96 of 21 June 2017), also aligns with these objectives by promoting the consumption of organic

products in schools and supporting vulnerable groups, particularly those living in poverty, women, and disadvantaged children.

To support the development of cooperative enterprises and network contracts (Action 6.2), measures will focus on innovation and digitalization at the network level, supporting small farms and promoting sustainable farming methods. Moreover, Action 7 will focus on improving environmental sustainability through research, innovation, and technical assistance within the context of Agricultural Knowledge and Innovation Systems (AKIS).

In conclusion, Italy's CAP Strategic Plan and the accompanying national action plans provide an integrated approach to achieving sustainability and resilience in agriculture, addressing climate change, biodiversity loss, natural resource management, and socio-economic development in rural areas. These plans, with a combined budget of over €50 billion for the 2023-2029 period, will foster a transition towards a more sustainable, competitive, and inclusive agricultural sector.

Ireland

Ireland's key objectives for the period 2023-2027, as part of its Common Agricultural Policy (CAP) Strategic Plan, are designed to address the country's environmental and agricultural challenges. These targets aim to reduce greenhouse gas emissions, enhance carbon sequestration, promote sustainable energy practices, reverse biodiversity loss, and foster efficient management of natural resources. The overarching goals include protecting farm family incomes, recognizing their hard work in food production, and contributing meaningfully to the country's climate ambitions (Department of Agriculture, Food and the Marine [DAFM], 2024).

The CAP Strategic Plan incorporates both Pillar I (Direct Payments and Sectoral Interventions) and Pillar II (Rural Development), with a performance-based approach that evaluates Member States on how effectively their plans contribute to EU-level objectives. Ireland's CSP, however, has faced criticism in certain areas, such as the limited environmental impact of its eco-schemes and insufficient measures to address the environmental effects of the dairy sector. Specifically, Ireland has been noted for having few interventions targeting methane emissions, and for focusing on output measures (like intervention uptake) rather than actual environmental outcomes (Meehan, 2024).

The Irish Government's Ag Climatise Roadmap, developed based on Teagasc's Marginal Abatement Cost Curves (MACCs), outlines key tasks for meeting climate and environmental objectives, particularly in the agri-food sector. These tasks include reducing greenhouse gas emissions from the sector, increasing carbon sequestration in land use, addressing nutrient loss to the environment, reducing ammonia emissions, and promoting sustainable food production systems. To ensure progress, Ireland has emphasized transparent communication of advancements, especially through the Origin Green Programme, which is instrumental in tracking sustainability efforts (Teagasc, 2023).

Another strategic initiative, Food Vision 2030, seeks to ensure that Ireland's agri-food sector aligns with sustainability goals, focusing on areas such as carbon reduction, innovation in agriculture, and improving environmental practices (DAFM, 2024). Furthermore, Our Rural Future is a key national initiative that supports rural communities through various measures, including promoting sustainable agriculture, enhancing broadband access, and investing in green economy initiatives such as renewable energy and afforestation (DAFM, 2021).

Ireland is also actively addressing the issue of gender equality in agriculture, recognizing that achieving gender balance is critical for the long-term sustainability of the farming sector. This includes efforts to promote women's participation in agricultural leadership, improve financial inclusivity, and support policies that encourage female succession in farming (DAFM, 2024).

Lastly, the Food Poverty Action Plan 2024 aims to tackle food poverty in Ireland through a cross-governmental approach, focusing on vulnerable groups and tackling the root causes of food insecurity. The plan emphasizes the importance of integrating food security into public policy, monitoring the effectiveness of initiatives, and providing emergency food provision when necessary (Department of Social Protection [DSP], 2024b).

In summary, Italy's national plans reflect a comprehensive and multifaceted approach to addressing climate change and sustainability challenges, aligning with FITTER-EU objectives. These plans tackle GHG emissions, carbon sequestration, sustainable energy, biodiversity, and efficient management of natural resources through the implementation of innovative and sustainable practices, with a focus on ecosystem protection and promoting a just transition for producers and consumers.

Overall, Ireland's approach to environmental and agricultural sustainability involves a combination of clear, ambitious targets, strategic plans for land use and farming practices, and policies that integrate social, economic, and environmental factors for the benefit of both rural communities and the broader national climate goals. These plans seek to balance the need for productive agriculture with the urgent need for environmental protection and climate action.

5.2. IMPACT OF THE TWIN TRANSITION PROCESS ON STRUCTURALLY VULNERABLE COHORTS

Both countries, Italy and Ireland, share a common approach to green and digital transition, but their strategies are tailored to their national contexts. Italy puts emphasis on the inclusion of small farmers, cooperatives and the promotion of organic farming, while Ireland combines its goal of climate neutrality with a strong focus on gender equality and support for farming families through concrete measures in social and climate policies. Both countries are committed to improving living conditions and competitiveness in rural areas, promoting employment and sustainable economic development.

5.2.1. COST BURDEN OF THE TWIN TRANSITION

Context in Italy

Italy faces significant costs associated with the twin transition to more ecological and digital agriculture. The country is committed to ambitious goals within the Common Agricultural Policy (CAP), focused on improving farm competitiveness, reducing greenhouse gas emissions, and promoting sustainability. Italy has allocated substantial budgets for the modernization of the agricultural sector, including investments in renewable energy, digitalization, and innovative technologies. These investments require considerable public support to help farmers implement sustainable practices, which incurs additional costs related to training, new technology adoption, and the implementation of greener agricultural practices.

This country also promotes a just transition approach, with support measures for vulnerable groups such as young farmers, women, and small farms. These initiatives aim to reduce the economic barriers these groups may face in adapting to greener and more digital agriculture. Additionally, Italy fosters the inclusion of sustainable practices in education and consumption, such as supporting organic farming and promoting organic products.

Italy takes a more detailed approach to specific sectors, such as organic agriculture. The country has developed a National Action Plan promoting the consumption of organic products in school canteens and other collective structures, which involves additional costs in terms of infrastructure and training. Italy has also allocated large budgets, such as the 16.5 billion euros for Objective No. 2 (CAP strategy), aimed at

improving agricultural competitiveness. Furthermore, Italy has specific measures for the digitalization of the agricultural sector, which presents an economic challenge in terms of technological innovation.

Context in Ireland

Ireland also faces significant costs related to the twin transition towards more ecological and digital agriculture. Similar to Italy, Ireland is committed to CAP goals, with a focus on farm competitiveness, reducing greenhouse gas emissions, and sustainability. The country is investing in the modernization of its agricultural sector through digitalization and innovative technologies, which requires considerable public funding to support farmers in adopting sustainable practices. This includes costs related to training, adopting new technologies, and implementing greener agricultural practices.

It also focuses on a just transition, offering support for vulnerable groups such as young farmers, women, and small farms. These measures aim to reduce the economic barriers these groups may encounter in the process of adapting to greener and more digital agriculture. Like Italy, Ireland promotes the inclusion of sustainable practices in education and consumption, supporting organic farming and promoting organic products.

In Ireland, one of the most prominent challenges is the criticism from the European Court of Auditors (ECA), which points out that certain practices within the CAP Strategic Plan (CSP) lack adequate environmental value, especially in the dairy sector. The ECA highlights the lack of interventions to reduce methane emissions, which implies a significant cost in terms of adapting agricultural practices. In response, Ireland has launched initiatives such as the Just Transition Fund, aimed at supporting large-scale projects and financing the transition to a carbon-neutral agricultural economy, such as the peatland restoration project. This initiative represents a considerable effort to cover the economic costs of the transition for rural communities.

Both Italy and Ireland are facing considerable financial challenges related to the twin transition towards greener and more digital agriculture. However, each country has adopted distinct approaches. Ireland emphasizes large-scale projects like the Just Transition Fund and peatland restoration to mitigate transition costs, while Italy focuses on supporting specific sectors such as organic farming and the digitalization of agriculture, with significant budgets allocated to implement these transformations.

5.2.2. ASSOCIATED VULNERABILITIES

The agri-food sector in Europe faces several demographic vulnerabilities related to age, gender, and income. A significant challenge is the aging workforce, with the average age of farmers increasing (across Europe, the number of farmers over 65 is notably high). This creates concerns about generational gaps in the sector and the need to attract younger individuals to farming. Indeed, only 11% of EU farm holdings are managed by individuals under 40 (Niranjan, 2024a, 2024b).

Gender disparities are another issue, with women underrepresented in certain agricultural roles. For example, rural women in Europe are less likely to own a mobile phone compared to rural men, indicating significant technology access disparities. Moreover, women tend to experience higher rates of food insecurity, even when controlling for household income levels, reflecting deeper gender inequalities in the sector (Niranjan, 2024a, 2024b).

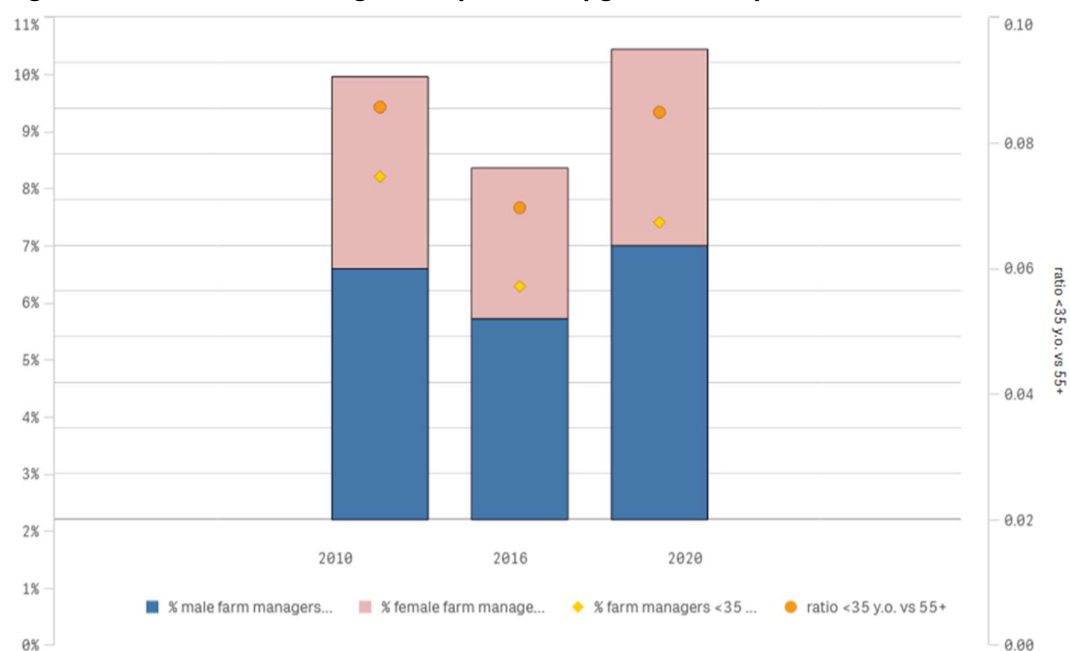
The income gap between large and small farms has been growing, with large farms benefiting from economies of scale while small farms struggle to compete (Niranjan, 2024a, 2024b). Over the past 15 years, the income disparity has doubled, putting immense financial pressure on smaller farms. Many small farms

are closing due to minimal profit margins, and farmers are increasingly seeking alternative income sources to stay afloat.

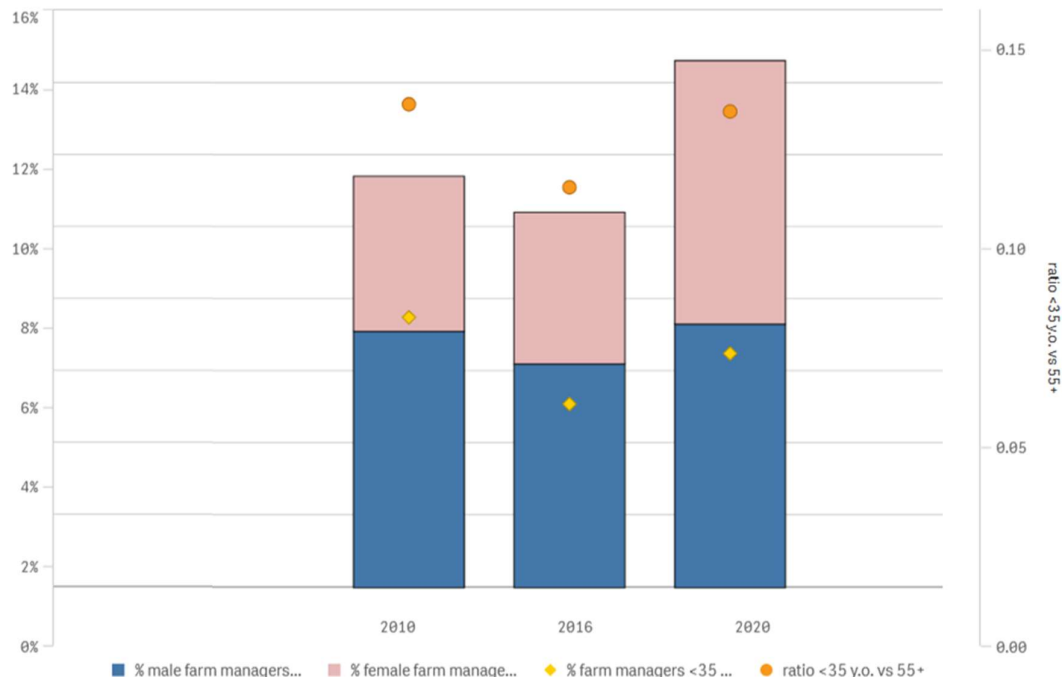
To address these challenges, it is important to support small farmers and create policies that can attract younger and more diverse participants to the sector, ensuring its resilience and sustainability in the future. In the case of Ireland and Italy, both countries are implementing measures in that way to mitigate the impact of the transition to sustainable and digitalised agriculture, with a focus on inclusion and support for vulnerable groups. Key common issues include:

- **Focus on vulnerable groups:** Both countries identify young farmers, women and small farms as the groups most affected by the transition to more sustainable and digitalised practices. In the period 2005-2020, the proportion of young farmers in the overall farming population declined in the EU. The trend differs among Member States. On average the share of female young farmers is especially low (Figures 28 and 29).

Figure 28. Share of farm managers <35 years old by gender in Italy



Source: European Commission (2023h)

Figure 29. Share of farm managers <35 years old by gender in Ireland

Source: European Commission (2023g)

Additionally, CAP strategy in Italy has this objective related to this topic: **promoting employment, growth and gender equality, including female entrepreneurship in agriculture, social inclusion and local development in rural areas, including the circular bio-economy and sustainable forestry.** To achieve Objective 8, the strategy will support through regional rural development interventions:

- Initiatives aimed at increasing and diversifying sustainable employment opportunities (e.g. sustainable tourism, bioeconomy, green jobs, social agriculture), strengthening the multifunctionality of agriculture and forestry, promoting the creation of new business, especially for young people and women, in related activities and in all those activities which can keep rural areas economically and socially viable;
- Initiatives that contribute to the sustainable management of land and landscapes by intervening in collective and public goods, promoting the recovery of abandoned, degraded and damaged areas and targeting them towards the needs of the community for tourism, recreation or the creation of innovative businesses capable of creating value from the valorisation of local resources.
- Initiatives aimed at bridging the infrastructure gap, with particular attention to the digital divide, and improving the availability/accessibility of services for the population and businesses, through cultural enhancement, securing and renovation of housing structures, residential centers and rural hamlets; recovery and re-use of rural structures and collective assets; energy efficiency and seismic adaptation of rural housing.
- Initiatives that contribute to the sustainable management of the countryside and landscape by intervening in collective and public assets, promoting the recovery of abandoned or degraded areas and adapting them to the needs of the community for tourism and recreation or for the

creation of innovative businesses capable of generating value from the valorisation of territorial resources.

- Access to basic services for workers, especially seasonal workers, guarantee them greater autonomy and security, also with a view to combating illegal employment.

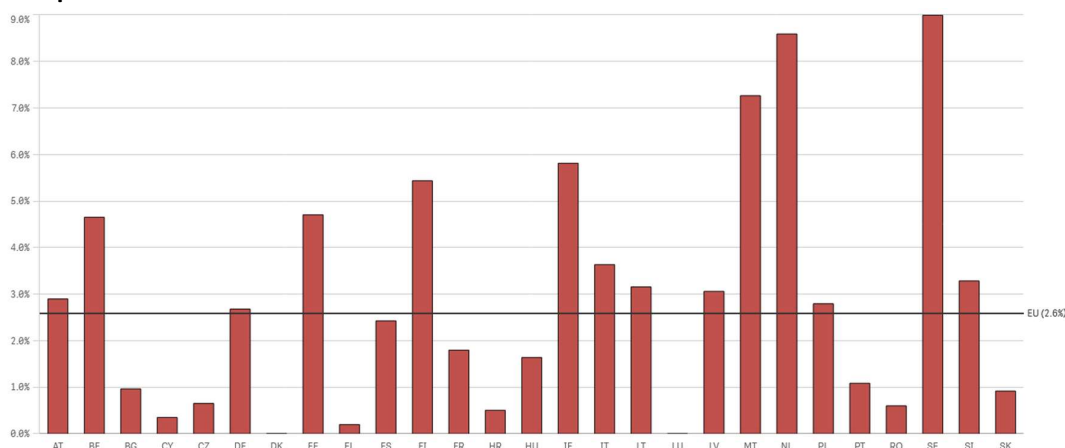
The available budget for pursuing this objective is 4 billion. The implementation period is 2023-2029.

- **Training and capacity building:** The lack of skills and resources necessary to use new technologies, as well as digital divides (urban vs rural) hinder digitalization in the agriculture sector. Furthermore, the cost of implementing certain digital technologies could be greater than the potential benefits, especially for smallholder farmers. In this regard, emphasis is placed on equipping farmers with the skills needed to adapt to changes, by promoting continuing education and training, including peer-to-peer training and teaching in the agricultural field.
- **Fostering cooperation and participation:** Both countries support the creation of cooperatives and producer organisations to improve competitiveness, transparency and sustainability in agricultural markets.
- **Financial measures and subsidies:** Funds and subsidies are implemented to support farmers in the transition, including funds to promote sustainability and digitalisation in agriculture, as well as cost reduction for vulnerable farmers.

Support for knowledge exchange, training, advice and innovation is key for securing smart and sustainable agriculture, forestry and rural areas. It includes investment in knowledge transfer and information actions, advisory services, farm management and farm relief services, and for cooperation / European Innovation Partnership. The level of support varies among EU Member States, reflecting their specific national context and priorities.

Figure 30 shows the Rural Development (RD) money spent for knowledge and innovation as a share of the total expenditures already realised for the 2014-2020 multiannual budget (including national contributions).

Figure 30. % RD realised expenditures for cooperation-EID, knowledge transfer and advisory services in Europe



Source: European Commission (2023g)

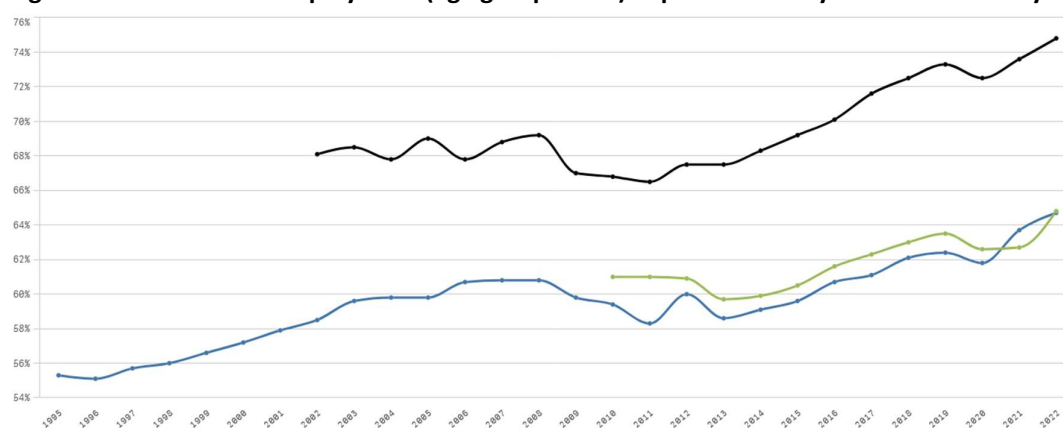
Regarding the uniqueness of each country

Ireland focuses especially on farmers dependent on the peat industry, with measures such as the Just Transition Fund, aimed at financing projects that support rural communities in their adaptation to the carbon-neutral economy. The country also stands out for its emphasis on supporting young people and women in agriculture, providing resources for training, access to land and credit. Furthermore, although some initiatives to support food security in the context of food poverty are mentioned, the country does not significantly address food insecurity in the strategic transition plans.

Italy, for its part, places particular emphasis on supporting organic agriculture, as demonstrated by the creation of the Organic School Canteens Fund, which seeks to promote the consumption of organic products, especially in low-income families. In addition, Italy is focusing on generational renewal and policies to support women in agriculture, with measures to facilitate their inclusion in the transition to more sustainable agriculture. Subsidies and technical assistance are also provided to small farms to encourage sustainability and digitalisation, helping to reduce the costs of adapting to new practices.

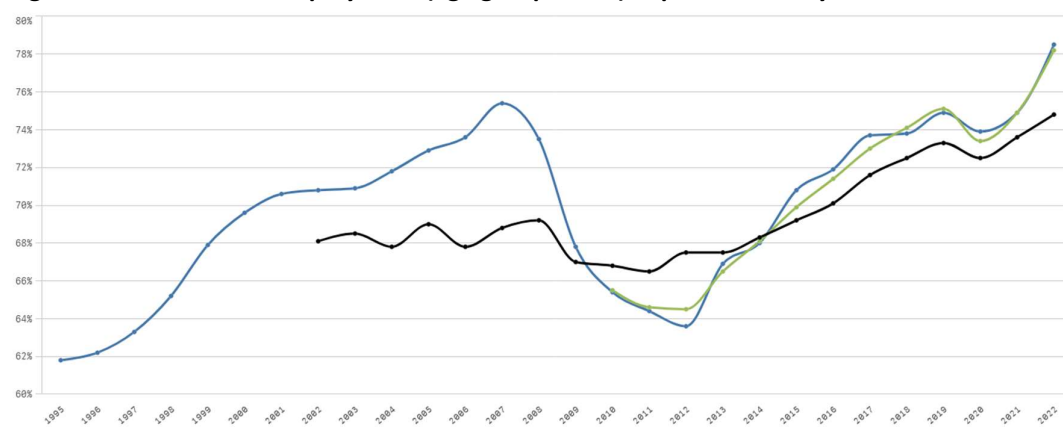
As the European Commission (2023g, 2023h) confirms, promoting rural employment is crucial to improve quality of life, including access to services, and to halt depopulation of rural areas. These graphs (Figures 31 and 32) compare rural and total (meaning rural and other areas) employment rates.

Figure 31. Evolution of employment (age group 20-64) in predominantly rural areas in Italy and EU



Source: European Commission (2023h)

Figure 32. Evolution of employment (age group 20-64) in predominantly rural areas in Ireland and EU



Source: European Commission (2023g)

Taking this into account, both Italy and Ireland recognise that there are vulnerable regions (such as rural, mountainous or areas with difficult agro-ecological conditions) that require special attention to mitigate regional disparities. Thus, both countries share the following communal points regarding regional disparities associated with the dual transition towards more sustainable and digital agriculture: **focus on vulnerable areas, promoting sustainability, rural development and employment, innovation and digitalisation and cooperation and strengthening networks.**

Italy

Regarding the topic of this section, CAP strategic plan in early December 2022 in Italy addresses the following general and specific objectives:

- Strengthening the resilience and vitality of rural territories, generating opportunities for new entrepreneurship based on the consolidation of natural and social heritage, creating the conditions for improving the attractiveness and inclusiveness of marginal areas. This general objective includes the following specific objectives:
 - Attracting and supporting youth and new farmers and facilitating sustainable entrepreneurial development in rural areas,
 - Promoting employment, growth, gender equality, including women's participation in agriculture, social inclusion and local development in rural areas, including the circular bio-economy and sustainable forestry
 - Improving EU agriculture's response to society's food and health needs, including high quality, healthy and nutritious food produced in a sustainable manner, reducing food waste as well as improving animal welfare and combat antimicrobial resistance
- Promoting and sharing knowledge, innovation and digitalization in agriculture and rural areas and encouraging their adoption (AKIS).

The Plan recognises, as key priority, the importance of organic farming, as a privileged production technique for contributing to the achievement of all the environmental objectives envisaged; to this end, the sector is around 2.0 billion euro over the five-year period under rural development.

The national CAP strategy is also aimed at fostering the quality of life in rural areas by improving social inclusion processes, the quality and accessibility of infrastructure and services, including digital services, for the population and businesses, halting depopulation and supporting entrepreneurship, also by strengthening the social fabric. More than 11.7 billion euros have been earmarked for this objective, representing 11% of the total resources, 11% of the rural development resources (EAFRD + national resources) and 13.4% of the regional programming resources, with an increase in resources compared to the average annual availability of the 2014-2022 programming.

This same program also describes action 6.2, which provides support for Cooperative enterprises and network contracts. The strategy envisages the following activities:

- Promotion of the characteristics and advantages of the two instruments (dissemination material, events, etc.) targeted at potential "leader" farms.
- The introduction of reward criteria for organic farmers (small farms) who join or associate.
- Encouraging innovation at network level by promoting digitalisation, integrated use of information, automation; studies within organic farms to identify the specific factors that hinder the formation of associations and those that can encourage it.

Additionally, the planned objectives for the three-year period 2024-2026 are:

- To increase the number of organic farmers;

- Organization of dissemination events and analysis documents promoted by the National Rural Network on good practices in agri-food cooperation and business networks.

Finally, concerning the specific topic of Agriculture 4.0, even if the research reveals the lack of information and insights, we can provide some anticipations provided by SVIMEZ and Centro Studi Tagliacarne on their dedicated analysis on a sample of 2,000 industrial enterprises with between 5 and 499 employees. In the South, 59.8% have invested or will invest in 4.0 technologies between 2017 and 2024, (against 56.3% in Centre North). While 50% have adopted an 'open innovation' model (i.e. open to collaborations with universities, customers and suppliers for a structured growth of the territory and the strengthening of production chains (against 46.1%)).

Ireland

In the Irish case, there are two main policies that focus on vulnerable areas, promotion of sustainability, rural development and employment, innovation and digitalisation and cooperation and strengthening networks.

On the one hand, the Food Vision 2030 program sets out key actions to increase diversification. Specifically, it proposes to develop alternative land uses and rural businesses that offer viable economic returns and have lower emissions than traditional ruminant based production systems (ruminant based means relating to cattle, sheep, goats, deer, antelope). Immediate actions include increased activity in organic farming, horticulture and forestry as well as anaerobic digestion for bio-methane production.

It also recognises that the family farm model is key to social sustainability in rural Ireland. Issues that impact the viability and well-being of primary producers and the broader rural community are identified. These include the challenge of generational renewal in agriculture; attracting new entrants, ideas and innovation; improving gender balance and LGBTI+ inclusivity; education and training; health and safety; mental health and well-being and rural development (DAFM, 2021: 119).

Additionally, its fourth mission (an Innovative, Competitive & Resilient Agri-Food Sector, Driven by Technology and Talent) includes the following objectives: Policy Coherence and Synergies in Sustainable Food Systems between Ireland's Domestic Policy and its Development Cooperation and Foreign Policy.

On the other hand, the "Our Rural Future 2021-2025" focuses on the following thematic objectives:

- Optimising the opportunities for rural communities from high speed broadband.
- Supporting improved quality employment and career opportunities in rural areas.
- Assisting the regeneration, repopulation and development of rural towns and villages.
- Enhancing the participation, leadership and resilience of rural communities.
- Enhancing public services in rural areas.
- Supporting a Just Transition to a climate neutral economy.
- Supporting the sustainability of Agriculture, the Marine and Forestry.
- Supporting the sustainability of our island and coastal communities.
- Nurturing our culture and heritage.

In this strategy, key deliverables relevant to farm families include:

- Implement Ag Climatise, a roadmap towards climate neutrality for the agri-food sector.
- Support research and development in areas such as agri-food, biobased systems, smart agriculture and precision agriculture to promote and encourage innovation and diversification.
- Develop Ireland's Common Agricultural Policy Strategic Plan for the period 2023- 2027, which addresses existing and emerging challenges including climate action, environmental protection, generational renewal, viable farm incomes, and to sustain vibrant rural areas.

- Support generational renewal, including young farmers and women in agriculture, through measures including the CAP, taxation measures and access to finance initiatives.
- Encourage and support Local Authorities to expand the number of farmers' markets, farm shops and community-owned markets in all towns, to showcase produce from local farmers, growers, and food producers.
- Develop a new integrated marine sustainable development plan, as a successor to Harnessing Our Ocean Wealth, focusing on all aspects of the marine sector.
- Deliver an ambitious afforestation plan to achieve an afforestation target of 8,000 ha/ year.
- Enact legislation implementing revised Nursing Home Support Scheme (Fair Deal Scheme) provisions in respect of assets which are family-owned and operated farms and businesses
- Through the Just Transition Fund, deliver flagship projects of scale, such as the €108 million Bord na Móna Peatlands Restoration Project, to assist communities in the transition to a carbon neutral economy.
- Maximise our resources and strengths in the Green Economy to support employment opportunities for rural communities in areas such as renewable energy, sustainable tourism, energy retrofitting, the Bioeconomy and the Circular Economy.
- Invest in rehabilitating our peatlands to contribute to reduced carbon emissions, carbon sequestration and enhanced biodiversity.

Regarding digitalization, Ireland has a well-structured and interactive approach to fostering knowledge, innovation and digitalisation in agriculture and rural areas. This involves clear coordination and collaboration within the Agricultural Knowledge and Innovation System (AKIS) (European Commission, 2025b).

As regards the uniqueness of each country, Italy places significant emphasis on sectoral diversification within its regions, crafting policies tailored to specific sectors such as wine, olive oil, and fruits and vegetables. The aim is to increase the competitiveness of these industries by adapting them to regional conditions and promoting sustainability, reflecting a strategy that is sector-specific and tailored to the needs of each region.

Additionally, Italy highlights the importance of mountain areas and regions with agro-ecological constraints, offering measures such as organic farming and the revitalization of struggling rural areas. The available budget for pursuing this objective is 6,7 billion. The implementation period is 2023-2029.

Furthermore, Italy's National Strategic Plan is more clearly focused on territorial cohesion and integrated regional development. This approach emphasizes that regions should address agroecological crises and challenges through national-level measures that are specifically tailored to the individual territories.

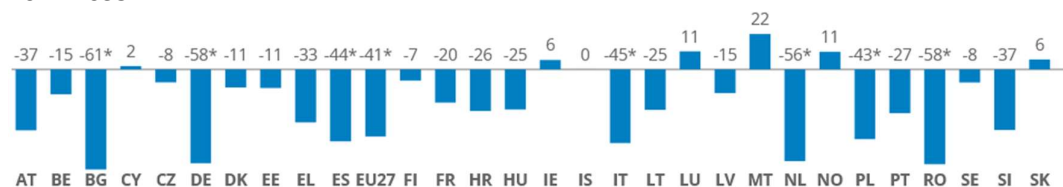
In Ireland, the approach to a just transition towards a low-carbon economy is particularly significant. The country emphasizes the involvement of farmers, scientists, environmentalists, and social groups in this process, with the establishment of a Dialogue on the Future of Agriculture being a key measure. However, this transition is not as explicitly linked to regional disparities as in Italy.

Ireland also focuses on gender inclusion within agriculture, with a comprehensive action plan aimed at promoting female leadership, succession in agricultural enterprises, and addressing barriers to women's participation in the sector, particularly in peripheral rural areas. Additionally, the country has developed a national strategy to combat food poverty, which includes a specific action plan to address disparities in food access between urban and rural areas.

5.2.3. ECONOMIC AND LABOUR MARKET EXCLUSION TRIGGERS

In 2023, approximately 7.2 million people were employed in agriculture across the European Union (EU), and the employment share of high-tech occupations was, in that year, 5.9% of the total. The top three occupations are Farmworkers and gardeners, Agricultural labourers and Drivers and vehicle operators. According to data from CEDEFOP (2020) future employment growth average over the period 2022-2035 is estimated at -21.8 in the agriculture sector (Figure 33).

Figure 33. Future employment growth (in %) in Agriculture, forestry & fishing sector across countries in 2022-2035



Source: CEDEFOP (2020)

Italy and Ireland share several commonalities in their approaches to addressing economic and labour exclusion, especially in rural areas and in specific sectors of agriculture. As it was said, both countries focus on the inclusion of vulnerable groups, such as young people, women and small farms. In Italy, this is achieved through measures that facilitate access to land, credit and digitalisation for young and new farmers. In Ireland, the emphasis is on generational renewal, offering support to young people and women in the agricultural sector through access to land, finance and training.

Furthermore, both Italy and Ireland promote economic diversification in rural areas as a means of reducing labour exclusion. In Italy, sectors such as the bio-economy and ecotourism are highlighted, while in Ireland the development of sustainable jobs in similar areas, such as the bio-economy, tourism and renewable energy, is encouraged. Both strategies also seek to improve the sustainability and competitiveness of agriculture, especially of small farms. Italy promotes sustainability in specific sectors such as wine and olive oil, while Ireland is betting on a fair transition and more sustainable management of agriculture, with the aim of making farming families more resilient, both economically and environmentally.

In terms of improving the position of farmers in the value chain, both countries implement actions that favour small producers. Italy focuses on quality certification and traceability of agricultural products, facilitating access to wider markets for small producers. In Ireland, measures are also adopted that seek to ensure that agricultural practices contribute to sustainability, which, in turn, improves farmers' access to more competitive markets.

Despite these similarities, each country has unique approaches to addressing economic and labour exclusion. Italy, for example, has specific schemes such as the Piano d'Azione Nazionale, aimed at the sustainable use of phytosanitary products and improving air quality, which complement its CAP strategy and address key environmental issues that can contribute to economic exclusion in rural areas. Furthermore, Ireland focuses on key agricultural sectors such as wine, olive oil and beekeeping, with a particular focus on improving the competitiveness and sustainability of these specific sectors.

Furthermore, Ireland developed a Dialogue on the Future of Agriculture (which involves diverse actors, such as farmers, scientists, environmentalists and social groups) and the National Dialogue on Women in Agriculture Report to find collaborative solutions to problems of economic and labour exclusion, with a particular focus on the transition to a low-carbon economy. Furthermore, Ireland has launched a Food Poverty Action Plan that comprehensively addresses the underlying causes of poverty, not only from the

perspective of food aid, but also considering aspects such as employment, housing and energy costs. This comprehensive approach reflects the effort to reduce economic exclusion in rural areas in the long term.

5.2.4. VULNERABILITY TO CLIMATE-CHANGE RELATED DISASTERS

Both Ireland and Italy recognise the vulnerability of agriculture and rural communities to climate change-related disasters and have integrated measures within their CAP strategies to mitigate these risks. Both countries consider climate change as a major threat that must be addressed in a comprehensive manner to ensure the long-term sustainability and competitiveness of agriculture.

Specifically, in Ireland, vulnerability to climate change is addressed through several measures and strategies within its CAP Strategic Plan 2023-2027. The country has identified that climate change presents a significant threat to agriculture and rural communities, especially due to the dependence of the agricultural sectors on climatic conditions.

To address this vulnerability, Ireland is working on promoting sustainable agricultural practices that contribute to climate change mitigation, such as reducing greenhouse gas emissions, particularly methane and nitrous oxide. In this regard, it focuses on improving the carbon absorption capacity of its agricultural land and reducing the loss of nutrients to the environment, which also contributes to the improvement of biodiversity and water quality. Ireland's strategy highlights the importance of the resilience of agricultural production systems to climate disasters through the use of adaptive agricultural practices, investment in research and the development of new technologies and systems that can withstand the impacts of climate change.

Italy, for its part, also addresses vulnerability to climate change-related disasters in its CAP strategy, although with a focus on the diversification and sustainability of its agricultural systems. Italy has recognised that natural disasters, such as floods, droughts and other extreme weather events, represent a growing challenge for agriculture, especially for small farms that lack the resources to adapt to these changes.

The country has promoted policies that foster resilience by supporting the transition towards more sustainable agricultural practices, such as conservation agriculture, rational use of water resources and the implementation of technologies to improve energy efficiency. In particular, Italy's CAP Strategic Plan focuses on the adaptation of its key sectors, such as viticulture, olive oil production and beekeeping, to the impacts of climate change, including the use of agricultural practices that can reduce the risk of climate disasters and improve the long-term sustainability of these sectors.

5.3. CHALLENGES AND OPPORTUNITIES OF JUST TWIN TRANSITION PROCESS

Broadly speaking, the double transition is a process of transformation towards a sustainable, low-carbon economy, ensuring that benefits are distributed equitably and that no one is left behind. As mentioned, this approach seeks to balance the need for climate action with the protection of labor rights and the well-being of affected communities.

One of the challenges of the just transition is the impact of these policies on employment (some sectors may experience job losses due to the closure of polluting industries). In addition, regional inequalities may appear, as well as the need for new training for workers.

These elements are not foreign to the situation in Italy and Ireland. Specifically, the following common challenges have been found along with the opportunities that may arise from them:

- **Generational and inclusion:** Both Italy and Ireland face challenges related to generational renewal in agriculture and the inclusion of women and other vulnerable groups, such as young farmers. Both nations recognise the need to support these groups through training policies, access to resources such as land and financing, as well as promoting economic diversification in rural areas.
- **Training and capacity building:** Digitisation and the use of advanced technologies in agriculture (which can be associated with **agriculture 4.0**) are perceived as a significant opportunity to improve the sustainability and competitiveness of the agricultural sector. Both countries are taking steps to train farmers in these new technologies, although they face economic barriers and the need for resistance to change on smaller farms, which represents a major challenge in the transition process.

According to the last available data (for the year 2023), 72% of Italian farms use agriculture 4.0 solutions. The Agricoltura 4.0 market in Italy reached in 2023 a value of EUR 2.5 billion, increasing 19% the value of the previous year. Despite lower growth compared to 2021-2022 (+31%), the sector continues to show a lively interest in digital solutions, with an increase in innovative solutions available on the market (+10%) and technology providers (+13%). Connected machinery and vehicle monitoring and control systems still account for about half of the market, but various factors - including the gradual reduction of government incentives - have led to a decline in these two items (-7% and - 10% respectively) in favour of software that allows the hardware to be interconnected, and the data collected to be analysed. 11% of expenditure is on management software and FMIS (Farm Management Information Systems), 8% on data integration platforms, 8% on crop and land mapping systems, 5% on DSS (Decision Support Software).

Companies that invest mostly in these solutions are those that have already embarked on digitisation in previous years. However, if the percentage of cultivated area is taken into account, the growth of 9% in 2023 is modest compared to the previous year 8% (Osservatorio digital innovation, 2024; di Giorgio dell'Orefice, 2023).

- **Transition to sustainability:** Countries are focusing on implementing policies to achieve a green and sustainable transition, which involves both the adoption of organic farming practices and the implementation of digital technologies to optimize production processes. This shift to greener agriculture presents both opportunities to reduce environmental impact and challenges for producers who must adapt to new demands and standards.
- **Fostering innovation:** Innovation in the agricultural sector is presented as a key to addressing the challenges of climate change and sustainability, with a strong emphasis on precision agriculture and the use of advanced technologies as tools to improve resource use efficiency.

The transition towards sustainable and digitalised agriculture presents both challenges and opportunities for farmers in Ireland and Italy. Advanced technologies and agriculture 4.0 have the potential to improve efficiency and sustainability in the agricultural sector, but digitalisation faces economic barriers, especially for small farmers. Young farmers and women are the key groups in this transition, with both countries implementing policies to support their inclusion in sustainable and digitalised agriculture.

6.EXPERT CONSULTATION

The twin transition process is a complex process that requires comprehensive research and informed decision-making to identify its potential negative impacts on structurally vulnerable cohorts. The external expert consultation process is a vital tool for ensuring that sectoral research on the twin transition is comprehensive. Engaging external experts in the task of vulnerability mapping under the baseline scenario in the four sectors, has been essential to deepening the understanding of sectoral critical challenges and emerging opportunities of just and fair transition.

The engagement of multidisciplinary sectoral experts took place in two online workshops (held on September 19th, 2024, and February 25th, 2025) and through an online consultation process conducted in February 2025. A total of 43 experts from Hungary, Germany, Ireland, Italy, Spain and Portugal were involved (25 experts attended two online workshops and 18 experts participated in the online consultation). External experts' contribution has provided FITTER-EU with holistic perspectives on the impact of the twin transitions in the four sectors of analysis in six European countries, and helped bridge gaps between theoretical and empirical considerations. Furthermore, expert input has enhanced the quality and accuracy of sectoral research with specific industry insights, fostering collaboration between FITTER-EU consortium partners, academia, industry, and civil society organisations.

The structure, key topics and the outcomes of the two workshops and the online consultation process are outlined below.

6.1. FIRST ONLINE WORKSHOP “TWIN TRANSITIONS AND ASSOCIATED STRUCTURAL VULNERABILITIES”

Date: 19 September 2024

Participants:

- 19 guest experts from Germany, Hungary, Ireland, Italy, Portugal and Spain. Expertise in policy making, academic research, social justice, green and digital transitions.
- 17 FITTER-EU consortium partners as moderators and observers.

Workshop structure:

- One plenary session
- Two break-up rooms (green and digital)
- Personal validation process.

Total time: 1h 45 min (03:00 pm - 04:45 pm CET)

Key discussion topics:

- The impact of the twin transition on generating and/or exacerbating vulnerabilities in the EU countries.
- Identification of the most vulnerable social groups in the context of the twin transition process.
- Critical challenges and emerging opportunities of just and fair twin transition process.

Workshop dynamic

The session began with a welcome and introduction, where the specific goals of the workshop were presented. A brief overview of FITTER and its project partners was provided, followed by a round of introductions from the experts in attendance, where they shared their names, companies/organizations, and positions.

After the introductions, there was a presentation of the key concepts of the FITTER-EU project. This included an explanation of the twin transition (green and digital), the disadvantaged groups associated with vulnerabilities and inequalities, eco-social justice, and the four sectors of analysis within the project. The first plenary discussion was focused on problematizing vulnerabilities in the context of the twin transition. Participants discussed which social groups in their countries are considered vulnerable to the impacts of the twin transition and why. Existing structures and mechanisms in the twin transition that generate or exacerbate these vulnerabilities were also addressed. Additionally, the group reflected on whether the twin transition has a multiplying effect on vulnerabilities in their countries, and if so, how. In the second discussion, participants were divided into two breakout thematic rooms: one focused on the challenges of the green transition and the other on the challenges of the digital transition. The discussions in these rooms centered on the following questions:

How do current norms and structures in each transition process advantage some population cohorts and disadvantage others?

Which groups are absent or not fully included in the decision-making process of the green/digital transition?

What are the most critical challenges that need to be addressed to ensure a fair green / digital transition for all?

What are the current mechanisms of representation and recognition of specific vulnerable groups in the context of green/digital transition?

Can vulnerabilities be mitigated through effective inclusive mechanisms of recognition and representation?

Is a fair green transition possible without financing alternatives for vulnerable audiences?

The session concluded with a validation of the initial concept of the just twin transition. The proposed definition was: "A just twin transition is a process that addresses intersectional inequalities, with fair burden-sharing, while controlling climate change and taking into account the dimensions of environmental, climate, energy, and digital justice". Participants reflected on whether this definition accurately captures the challenges of the twin transition in relation to the vulnerabilities and inequalities involved.

The workshop concluded with a brief wrap-up.

Main outcomes

Identifying critical challenges and vulnerabilities produced or exacerbated by the twin transition process in Europe from a social justice perspective was the dominant focus of the discussions. The need to consider structural vulnerabilities through a justice lens, ensuring that the transition does not leave certain groups behind was particularly highlighted. Some of the key ideas shared by the experts addressed the following themes:

Structural complexity of vulnerabilities

Existing vulnerabilities, such as poverty, low income, and a lack of education or work opportunities, are exacerbated by the twin transition. Vulnerabilities directly caused by the twin transition include insufficient digital training, limited access to resources, and policies that tend to favor wealthier individuals (e.g. grants for clean energy systems, subsidies for private electric vehicles). A common factor that amplifies vulnerabilities is low income and education, which play a significant role in most cases.

Territorial disparities and regional inequalities also present a major issue, particularly in rural and peripheral areas, where unprivileged social groups face compounded challenges. Moreover, climate

change disproportionately impacts certain regions, even within the same country, as seen in areas like inland Italy. Another key concern is that low-income workers are frequently employed in sectors that are highly vulnerable to digital disruption, adding another layer of difficulty to their situation.

Identification of Vulnerable Groups

Among the most vulnerable groups experts highlighted the need to focus specific attention on:

- Rural low income and education people, in mining, forest and agricultural areas;
- Rural women, migrant and travelling workers;
- People with disabilities;
- Elderly people;
- Ethnic minorities and migrants;
- Low-skilled workers
- Organizations with low resources/know-how to adapt to the transition process;
- New vulnerable groups caused by the transition policies and measures, and market forces;
- Unemployed people;
- Remote underprivileged communities;
- Young people from the peripheral and rural areas;
- People suffering from poverty and social exclusion, homeless people.

Critical challenges of the green transition process

The viability of the twin transition is closely linked to economic factors. As one participant mentioned: "The more money you have, the more possibilities you have". This point highlights how individuals with greater resources have more opportunities to adapt and benefit from the green and digital transitions, while those with fewer resources face significant challenges.

Citizen participation is crucial for the success of the green transition. One participant emphasized that "The green transition is supposed to emerge from a participatory political process, but in reality this process is not as open as expected". This comment underscores the importance of ensuring that the transition process is inclusive and reflects the voices of all sectors of society, which doesn't always happen effectively. Finally, it was commented that when the transition process is not inclusive, decisions made at the political level can negatively impact vulnerable groups. According to one participant, "policymakers are not giving us the opportunity to speak. This affects vulnerable groups (such as immigrants who are not visible in society)". This highlights how decision-makers do not always provide an adequate platform for these groups to voice their concerns, further exacerbating their vulnerability.

Critical challenges of the digital transition process

Discussing the critical challenges of the digital transition process, experts argued that skills, digital literacy and accessibility are most relevant drivers of social inclusion and access to the benefits of technological innovation in all economic sectors. Digital upskilling of the workforce is critical in traditionally low-skilled sectors (e.g. agriculture, transportation). Furthermore, digitisation of access to welfare services exacerbates existing complexities, as noted by a guest expert: *"In Ireland the access to those service is very complex because of the means testing rules. You need a college level degree to understand those rules. Digitising those services creates a layer of exclusion. People are less likely to apply and that hits the most vulnerable households"*.

Jobless households, people with disabilities, elderly people, low-skilled workers are most vulnerable within the context of digital transition. Elderly people and people with disabilities need special attention as

physical barriers which they face often in their daily life are compounded by digital barriers of technology costs, complex usability of devices, or lack of resources and skills: *“Accessibility barriers, affect the ability of people with disabilities to use certain types of energy and technologies, which can limit their ability to access a reliable and affordable supply of energy”*.

The workshop highlighted the urgency of integrating social justice into the twin transition process. Vulnerable groups, such as rural populations, low-income workers, and marginalized communities, face compounded challenges that must be addressed to ensure an inclusive transition. The importance of economic support to most vulnerable groups and participatory decision-making was also emphasized, as these factors significantly influence the fairness and success of the green and digital transitions. Policymakers must take proactive steps to involve vulnerable groups in the process, ensuring that no one is left behind as Europe navigates this complex transformation.

6.2. SECOND ONLINE WORKSHOP “ENERGY POVERTY AND ITS INTERSECTIONAL IMPACT ON VULNERABILITIES”

Date: 25 February 2025

Participants:

- 6 guest experts from Germany, Hungary, Ireland, Portugal and Spain. Expertise in policymaking, academic research, social justice, green and digital transitions.
- 10 FITTER-EU consortium partners as moderators and observers.

Workshop structure:

- One plenary session

Total time: 1h 30 min (04:00 pm - 05:30 pm CET)

Key discussion topics:

- Double energy vulnerability in the context of twin transition.
- The role of renewable energy in mitigating energy poverty.
- The impact of social innovation in addressing energy poverty.

Workshop dynamic

The online workshop began with a welcome and introduction, followed by a presentation on the scope and overview of the FITTER-EU project. The next segment addressed the impacts of the twin transition process on vulnerable groups, presenting a review of national baseline scenarios and associated vulnerabilities. It also examined the alignment with the common EU energy transition goals across the six countries studied, along with an explanation of the related vulnerabilities.

A discussion session followed, moderated by the Complutense University of Madrid team, focusing on the issue of energy poverty within the context of the twin transition process. The discussion revolved around the following key questions:

- How can the twin transition effectively reduce persistent energy poverty among the most vulnerable groups?
- Can renewable energy sources play a key role in ensuring fair and secure access to energy for all?
- And, what types of social innovations are needed to effectively combat energy poverty?

The workshop concluded with a summary of the findings, followed by a wrap-up and closing remarks.

Main outcomes

Digitalization and supporting vulnerable households

- **Potential of digitalization:** Digital services provide a powerful tool for better understanding energy consumption. They can help vulnerable households identify inefficiencies and adjust their energy usage, potentially leading to significant savings. Moreover, the digitalization of building information (such as performance certificates) could help solve the challenge of unavailable data, which often hinders the allocation of funding for energy-efficient renovations. By increasing access to these tools, vulnerable households can gain more control over their energy use and understand available support options.
- **Data gap and targeted support:** While governments hold data about energy consumption (e.g., through smart meters), there is a crucial gap in understanding people's specific circumstances, which often prevents targeted interventions. Bridging this gap with more digitalization can lead to better integration of social protection systems with the energy market.
- **Limitations of energy efficiency advice:** While advice on energy efficiency is valuable, it must be sensitive to the immediate needs and financial constraints of vulnerable groups. Energy efficiency measures can often be unaffordable for people already struggling with financial hardship. Before promoting energy-saving strategies, governments and organizations must address these groups' more pressing needs, such as their ability to pay for basic energy consumption.
- **Integrated support packages:** Providing an integrated approach is crucial, combining technological solutions (like smart meters) with information on social welfare supports.

Renewable energy and fair access

- **Community-driven energy solutions:** Citizen-driven energy initiatives, such as community cooperatives, are key for addressing energy poverty and promoting access to renewable energy. These bottom-up solutions empower communities to support each other in reducing energy costs, especially when many households, particularly renters, cannot afford to install renewable systems like solar panels. In these cases, energy communities can step in, pooling resources to invest in shared renewable energy infrastructure, such as community solar projects.
- **Barriers to accessing renewables:** Even though renewable energy sources, like solar, are cheaper in the long run than conventional energy, they remain inaccessible to many vulnerable groups. The barriers include financial limitations, lack of ownership (tenants cannot invest in renewable systems), and the availability of space for installations. This further highlights the importance of energy communities in providing equitable access to renewable energy solutions.

Policy integration and the role of governments

- **Need for policy integration:** There is a need for better integration of policies addressing energy poverty with those targeting housing inequality and the climate crisis. Often, the policies addressing these issues are siloed, resulting in missed opportunities for creating holistic solutions. Governments need to be proactive in identifying the barriers vulnerable households face and designing policies that integrate solutions across multiple sectors.
- **Challenges with technology assumptions:** It is crucial to acknowledge that not all vulnerable groups benefit equally from the rollout of technology. Often, policies assume a one-size-fits-all approach, but the needs of different groups—such as renters, low-income households, or people without digital access—can vary significantly. Social innovations that listen to the unique challenges of these communities are essential for creating inclusive solutions. Governments need

to focus on human-centered solutions, ensuring that technology serves people, not the other way around.

Best practices overcoming energy poverty

Learning from best practices around the world is essential. A noteworthy example is the initiative in Ireland, where a community surrounding a football club worked together to learn about energy efficiency, housing retrofitting, and solar energy. The project transformed the perception that retrofitting is only for the wealthy, empowering people to take action on energy efficiency and renewable energy. This grassroots approach fostered resilience within the community without relying solely on state resources.

In Latin America and Africa, community-driven cooperatives for self-consumption are helping members build their own renewable energy systems, such as solar panels. These initiatives could provide valuable insights for European contexts as well, showing how people can collaborate to reduce energy costs and promote sustainability.

6.3. ONLINE EXPERT CONSULTATION

Date: February 2025

Participants:

- 18 experts from Germany, Hungary, Ireland, Italy, Portugal and Spain. Expertise in policy making, academic research, social justice, green and digital transitions.

Consultation dynamic

The relevant experts were identified in each of the six countries by the consortium partners. Each expert received a personalized email introducing the objectives of the FITTER-EU project and explaining the purpose of the consultation. Based on their area of expertise, they were asked to provide written responses to four or five questions regarding the progress of the twin transition in their country within a specific economic sector.

Questions shared with experts regarding each of the sectors:

Energy sector experts

1. Based on your expertise, what three key trends do you observe in the energy sector related to the green and digital transition in your country?
2. From your perspective, what impact do the energy transition and the push for renewable energy sources have on the energy poverty situation in your country?
3. Which population groups are most negatively affected by the transition to renewable energy sources in your country?
4. Could you provide any examples of the most critical situations of energy poverty in your country?
5. What measures could alleviate the negative impact of the transition to renewable energy sources in your country?

Transport and mobility sector experts

1. Based on your expertise, what three key trends do you observe in the transport and mobility sector related to the green and digital transition in your country?
2. From your perspective, what impact do the energy transition and the push for electric mobility have on the transport poverty situation in your country?

3. Which population groups are most affected by the transition to electric mobility in your country in terms of transport accessibility, transport availability or employment opportunities in your country?
4. Could you provide any examples of the most critical situations of transport poverty in your country?
5. What measures could alleviate the negative impact of the transition to electric mobility on the most vulnerable groups?

Housing and built environment sector experts

1. Based on your expertise, what three key trends do you observe in the housing and built environment sector related to the green and digital transition in your country?
2. From your perspective, what impact could the climate change and the energy transition have on the housing and built environment sector in your country?
3. Which population groups are most affected by the climate change impact (urban heat and extreme weather events) on housing insulation and wellbeing?
4. Could you provide any examples of the most critical situations of housing vulnerability as a consequence of the climate change in your country?
5. What measures could alleviate the negative impact of climate change and energy transition on housing vulnerability?

Agriculture sector experts

1. Based on your expertise, what three key trends and impact do you observe in the agriculture sector related to the green and digital transition?
2. Which population groups are most negatively affected by the digital and green transition in the agriculture sector in your country in terms of economic and technical investments, lack of skills or employment opportunities?
3. Could you provide any examples of the most critical situations of the technical, legal and economic adaptation to the digital and green transition in the agriculture sector in your country?
4. What measures could alleviate the negative impact of the green and digital transition in the agriculture sector?

Main outcomes

Energy sector

1. Key trends of the green and digital transition in the energy sector

The twin transition plays a significant role in reshaping the energy sector, with a strong push toward renewable energy sources, digitalization, and energy efficiency. However, challenges remain, including economic disparities, infrastructure limitations, and regulatory uncertainties. A coordinated effort between governments, businesses, and society to ensure a just and inclusive transition is essential. Based on the contributions of experts from Germany, Spain, Hungary, and Portugal, the key developments highlight the following trends:

Strong Commitment to Renewable Energy and Decarbonization

Countries are increasingly shifting towards renewable energy sources such as solar, wind, and renewable gases. Large-scale renewable energy parks are gaining prominence as a more efficient alternative to individual installations. Governments are prioritizing energy storage technologies (e.g., virtual batteries) to

enhance energy reliability and grid stability. The increasing share of photovoltaic in the energy mix, the spread of green mobility, and the emergence of hydrogen technology are also remarkable. In Hungary and Portugal, replacing fossil fuel imports with domestic renewables is a key driver of the transition. However, the integration of large-scale renewable energy projects into existing landscapes is facing important challenges. Upgrading energy infrastructure requires significant investment, particularly in regions that have historically relied on fossil fuels. High costs and public acceptance issues hinder the rollout of new technologies like smart meters. The intermittent nature of renewable energy requires investments in grid stability, storage solutions. Energy Sharing is not yet implemented which could provide measures to face energy poverty. The challenge of balancing rapid technological advancements with the need for stable, long-term energy policies is also identified. Experts also argue that, from a geopolitical perspective, energy independence is crucial, and currently, only renewables can provide this solution. The electrification of home heating and cooling is feasible, though it requires investment. However, efficient climate control systems are becoming increasingly affordable and will continue to do so. Additionally, the industrial sector stands to benefit from access to cheap and clean energy.

Digitalization as a Catalyst for Energy Transformation

Smart grids and smart metering systems are being deployed to optimize electricity distribution and integrate distributed energy resources (e.g., self-consumption systems, battery storage). IT-based energy management, such as smart homes and smart meters, to support more conscious energy consumption is also emphasised.

Artificial Intelligence (AI) and digital platforms are facilitating real-time monitoring of energy systems, enabling better decision-making, predictive maintenance, energy management, and efficiency optimization. Digital solutions are recognized as enablers of sustainability, helping to optimize resource use and improve energy infrastructure efficiency. Furthermore, digital tools are enhancing sustainability reporting, making it easier to track environmental, social, and governance compliance.

Cybersecurity and data privacy concerns are rising as digitalization expands in the energy sector, particularly in countries like Germany and Spain.

Focus on Energy Efficiency and Consumer Empowerment

Energy efficiency is becoming a top priority across households, industries, and public administration, with growing government support for investing in insulation and modern heating systems. There is an emphasis on reducing overall energy demand through regulatory measures and efficiency-driven initiatives. In some countries, consumers are gaining near real-time access to their energy consumption data, improving their ability to manage energy use effectively.

2. The impact of the energy transition and the push for renewable energy sources on the energy poverty situation

The financial burden of energy transition is a growing concern from the lens of investment costs, affordability, and equitable access to renewables. The impact of the twin transitions in the energy sector on the low-income and vulnerable groups is acknowledged by experts. The high costs of renewable energy investments may lead to potential social inequality. Particularly in the short term, it has a negative, price-increasing effect. In the longer term, if the issue of grid balancing and buffer storage can be solved and renewable energy becomes cheaper, it can help improve the situation locally.

In Germany, the number of households spending over 10% of their income on energy rose sharply during the European energy crisis, from 26% in early 2022 to 43% by mid-2023. Low-income households are

particularly vulnerable due to poorly insulated rental properties and rising CO2 prices. In Hungary, energy poverty is above the EU average (someone is considered poor if they spend more than 10% of their income on purchasing acceptable heating and electricity services). Today, there is electricity everywhere in Hungary, and the majority of homes have natural gas or district heating. Energy-poor people use poor quality heating equipment, live in completely uninsulated homes (often with damp, mouldy walls) and often use coal, lignite or wood as fuel. Continuous electricity supply is also a problem. This segment is significantly burdened by the green transition and the associated additional costs. For them, the use of renewable energy sources only increases the gap between them and the non-poor, making it hopeless. Ensuring financial support mechanisms and addressing the energy costs of transitioning to renewable energy sources is also highlighted as essential in Portugal to avoid the exacerbation of energy poverty in the process. In Spain, experts argue that, despite years of green transition efforts and the development of large-scale renewable energy projects, energy poverty has continued to rise, suggesting that some of the underlying causes of energy poverty were not adequately anticipated or addressed in national strategies, such as the National Strategy Against Energy Poverty, or similar policy documents. Although regulatory frameworks include risk mitigation measures for vulnerable populations, these policies have not yet effectively reached those in need. Addressing these gaps requires a stronger, more targeted approach to ensure that the benefits of the energy transition are equitably distributed. Among the most critical challenges in Spain, experts highlight: the energy price volatility, which affects households' ability to plan and manage energy costs; physical constraints in the residential sector, where certain energy solutions cannot be applied due to the type of housing; financial barriers, as many households cannot afford large upfront investments required for the domestic energy system upgrades.

The focus is also placed on the role of energy self-consumption mechanisms such as citizen energy, energy sharing, citizen energy funds. Energy transition needs to balance sustainability with affordability and social equity. Citizen energy provides just and fair concepts for the energy transition. In Spain, Local Energy Offices and Local Energy Communities support vulnerable populations by allocating part of their energy production to families in situations of energy vulnerability. Some of these communities are already 100% solidarity-based, dedicating their entire production to supporting vulnerable households. A notable example is COENSOMA, located in the Malvarrosa neighborhood of Valencia.

Experts also note that policy adjustments are critical to precisely address the impact of the twin transition process on the most vulnerable groups. The role of energy efficiency measures is highlighted as continued investments in retrofitting buildings and expanding renewable capacity are critical. A shift toward investment-cost subsidies for renewables could further accelerate progress. Support mechanisms like climate bonuses or tax relief for low-income groups to offset rising CO2 prices and ensure equitable access to clean energy solutions are also deemed essential.

To conclude, experts argue that affordable energy is a fundamental element of well-being, the greater the share of renewable energy, the lower the overall energy bill. Lower energy costs also make it easier to access essential energy services, such as hot water and climate control. Additionally, lower energy prices directly lead to cheaper consumer goods, since energy costs have been one of the main drivers of inflation in recent years.

3. Population groups most negatively affected by the transition to renewable energy sources

The following vulnerable groups have been identified by experts:

- **Lower-income households** have fewer opportunities to access the benefits of renewable energy in their homes (e.g., installing solar panels) due to their inability to afford the required upfront investment. Additionally, public subsidies for renewable technology investments often end up

benefiting those who are not the most in need. Reduced electricity consumption from the traditional grid means that infrastructure costs—which are partly distributed based on energy consumption—are now spread among fewer consumers, many of whom have lower disposable income. In Hungary, Roma population in areas burdened by unemployment is heavily affected. Moreover, more than 500,000 **pensioners** living on pensions below HUF 200,000 (especially those living alone) are also heavily burdened. This group tries to manage the situation by reducing consumption, which leads to a significant decline in the quality of life. In Portugal, **low-income households, elderly people, and rural communities** are most affected. In Spain, households with elderly people, single-mother households, and households where the head of the family has low educational attainment and/or is unemployed are most affected.

- **People who have lost their jobs** due to the closure of fossil fuel-based power plants in coal phasing-out affected regions that disproportionately impacts already vulnerable populations and worsens poverty levels in these areas.
- **People with lower levels of education and fewer opportunities for reskilling** face greater challenges in adapting to new job markets created by the energy transition.
- **Households, self-employed workers and companies** that rely on traditional fuels if they lack the resources to switch to alternative energy sources (diesel for transport or heating oil in remote areas).
- **Tenants** and people living in social housing are more vulnerable than homeowners due to the lower access to energy efficiency measures, subsidies or energy self-consumption options. While the transition should not create disadvantages, not everyone has the means to benefit from direct renewable energy installations. Having available space for solar panels or wind turbines is not feasible for everyone, and the initial investment required is a significant barrier.

4. Evidence of critical situations of energy poverty

Experts noted that extreme weather events, such as heat waves or cold spells, make energy poverty situations more evident. Not being able to heat water, using the kitchen sparingly, limiting ironing—everyday activities consume energy, and lacking it or having to ration its use creates inequality.

In Spain, the most common form of energy poverty is the inability to afford energy costs. However, the most critical cases, when there is a complete lack of access to energy, are not solely an energy sector issue but rather a reflection of general poverty.

In Hungary, Roma settlements in Nógrád and Borsod, where they heat with stolen wood show extreme energy poverty. The technical condition of the houses is deteriorating, the windows and doors are in bad shape or broken, the heating equipment is an old, obsolete stove with poor heat output. Every member of the North Balaton Retired Club (95% lives alone) pays approximately HUF 6,000 for electricity and HUF 20,000 for gas, which would require a pension of at least HUF 260,000. However, the majority live on a pension of HUF 130-200,000 (including widow's pension). Another expert from Hungary noted that In 2023, 247 of their compatriots died as a result of hypothermia. The vast majority of those deaths were in their own homes.

Experts have shared examples of some local project and intervention initiatives to address persistent energy poverty in their countries. In Germany, Gender4Power project - [GENDER4POWER - Electra Energy](#) aims to address social inequality and the vulnerable situation of single parents households, women, elderly people.

In Portugal, poor infrastructure and inadequate insulation, such as in social housing neighbourhoods, are major issues. Portugal is the second worst country in the EU in terms of housing conditions, with infiltration,

rotting elements, and other structural problems. In 2023, 20.8% of the population in Portugal lived in households unable to keep their home warm, an increase of 3.3 percentage points compared to 2022. In 2022, Portugal was among the five EU-27 countries with the highest levels of this incapacity, affecting 17.5% of the population, nearly double than the European average of 9.3%.

5. Measures that could alleviate the negative impact of the transition to renewable energy sources

Experts identify the following measures to address energy poverty in their countries.

- Targeted programs and assistance for vulnerable groups: renovation of homes at risk of energy poverty, promotion of comprehensive energy-efficient refurbishment, fast-track renovations, and appliance replacement programs exclusively for vulnerable groups; A regulated, stable energy tariff exclusively for vulnerable consumers;
- Public support to energy self-consumption installations;
- Promoting the role of social energy advisors and establishing advisory offices for vulnerable individuals at municipal levels.
- Public support for covering monthly energy costs without discouraging energy efficiency;
- Promoting awareness of energy efficiency;
- Collaboration between different levels of government, businesses, and the third sector is crucial to ensuring that the energy transition benefits all segments of the population;
- Engaging citizens, including the most vulnerable groups, in the design of these measures (defining action plans, setting objectives, and allocating budgets).

Mobility and transport sector

1. Key trends of the green and digital transition in the mobility and transport sector

Spread of battery-powered electric vehicles, development of alternative fuel infrastructure, emergence of hydrogen-powered vehicles, development of multimode public transport hubs (bus + train + car + cycling facilities) to avoid unnecessary trips and encourage public transport use, and enhancement of rail transport both of passengers and freight are the key trends highlighted by experts in the mobility and transport sectors in the context of the twin transition process. Research and innovation are also acknowledged a top priority of car manufacturers to ensure an effective transition to electric mobility.

2. The impact of the energy transition and the push for electric mobility on the transport poverty energy situation

Experts argued that clean energy transport is more expensive than fossil fuels in terms of total cost, so in the short term it is more likely to worsen the situation than to improve it. Significant change can only be expected with a further increase in the price of petrol and a significant decrease in the price of green electricity and green hydrogen. Despite public incentives and subsidies to purchase electric vehicles, this option is affordable by a minority as, in addition to high price, the need for widely available charging stations is not yet addressed.

On the other hand, in Spain, the ageing car fleet has become a barrier to foster adoption of electric mobility. Sales of new cars are still below the volumes sold before the pandemics, therefore cars in Spain are more polluting and less safe than they could be. On the other hand, the deployment of the EV charging stations is still slow and hinders the interest in electric mobility of potential buyers. On the other hand, young people are using more public transport and shared mobility options and are not that interested in owning a car. Furthermore, high speed rail transport is more efficient, convenient, and less polluting, connecting

easily regional nodes. The entrance of new rail operators has decreased travel costs and has become more affordable.

3. Population groups most negatively affected by the transition to electric mobility in terms of transport accessibility, transport availability or employment opportunities

The following vulnerable groups have been identified by experts:

- **Low-income groups** living in social housing and without garages, those who cannot afford the purchase of electric vehicles and their charging.
- **People living in remote rural areas** as access to efficient public transport options is not widely available.
- **People who lost their jobs** due to the transition of the mobility and transport sector to clean energy. The transition disproportionately impacts already vulnerable populations and worsens poverty levels in the regions where factories have been laying off workers. The situation is critical for people aged over 50, whose skills do not fit the needs of clean energy transport, and whose upskilling or reskilling is not prioritized due to the increasing automatisisation and digitisation of the manufacturing process.
- **Elderly people**, over 50 and 60, who are not particularly enthusiastic about electric mobility as public incentives are not significant, the cost is high, and the charging infrastructure is underdeveloped.

4. Evidence of critical situations of transport poverty

Experts highlighted that people with reduced mobility (elderly people, people with disabilities) often do not have their own vehicle suitable for individual transport, and public transport might not be an option for them due to large distances and poor transport conditions. As a result, they are unable to go to work or meet their private needs (shopping, family, culture) at an adequate level due to mobility restrictions. Other experts highlighted the increasing costs of private vehicles in the last decade that have become a barrier for personal mobility; and the disappearance of regional and local rail connections, leaving many rural towns and villages disconnected from the main transport routes. While governments invest heavily in high-speed rail connections, benefitting large urban areas, rural remote areas are most negatively affected.

5. Measures that could alleviate the negative impact of the transition to electric mobility

Experts agreed that the targeted support schemes (e.g. vehicle subsidies) for those in need, special leasing schemes, and possibly special differentiated transport services could alleviate access to electric mobility. Moreover, the deployment of charging stations network should be significantly enhanced. Furthermore, measures to ensure employment, upskilling and reskilling options for people over 50-year-old laid off or at risk of losing their jobs in the process of restructuring and transition of the transport and mobility sectors should be emphasised. Developing technical training programs to foster the most demanded digital and advanced technological skills is essential.

Housing and built environment sector

1. Key trends of the green and digital transition in the housing and built environment sector

The key trends in the green and digital transition in the housing and built environment sector include the growing demand for affordable housing in cities, particularly middle-sized and larger cities, where there is a need for hundreds of thousands of flats to meet demand. At the same time, rural areas are experiencing underutilized housing stock, with millions of flats not in use due to insufficient support for rural development over the past few decades. This has led to increased urban migration and a lack of affordable housing in cities.

Digitalization in housing and construction is progressing, although at a slower pace. Notable trends in this area include the implementation of Building Information Modeling (BIM) systems in project execution and development, as well as the rise of smart buildings. Sustainable construction systems and materials are increasingly being used, with a focus on managing recycled materials and waste effectively.

Energy efficiency is a major focus, with thicker insulation layers and advanced window technologies, like triple glazing, being used to reduce energy expenditure. Home automation services, such as smart plugs, sensors, and systems for monitoring electrical and thermal consumption, are becoming more popular, contributing to energy savings.

Another significant trend is the shift towards self-consumption of electricity and thermal energy, particularly through the use of photovoltaic solar panels, geothermal, and aerothermal systems, leading to greater energy efficiency and reduced reliance on gas.

The housing market is also seeing an increase in international investment, driven by the digitalization of real estate management and transactions. Corporate landlords are utilizing digital platforms, algorithms, and surveillance technologies to optimize property management and increase rental profits. This digital capitalism contributes to the financialization of the housing market and can result in gentrification, which exacerbates shortages of affordable housing and increases housing prices.

Finally, there is a growing focus on energy-efficient buildings through renovation, the integration of renewable energy sources, and the continued digitalization of building management systems, which are enhancing energy monitoring and supporting the development of smart homes.

2. The impact of climate change and the energy transition on the housing and built environment sector

The impact of climate change and the energy transition on the housing and built environment sector is significant, with several key challenges and opportunities emerging:

- **Increased vulnerability to extreme weather:** As climate change intensifies, buildings are increasingly at risk of heat waves, extreme temperatures, and torrential rains. In regions like Spain, this is already causing issues where buildings, constructed for a different climate, are no longer suitable. The overuse of air conditioning to compensate for poor insulation is a direct consequence, and with rising temperatures, vulnerable households are experiencing discomfort and health risks, especially during "noches tórridas" (nights with temperatures over 25°C). Additionally, events like heavy rainfalls, as seen in Valencia in 2024, are overwhelming existing infrastructure, leading to flooding and damage to buildings that were not designed for such extreme weather.
- **Energy efficiency and refurbishment opportunities:** One of the main opportunities presented by climate change is the energy transition, particularly through the refurbishment of buildings to improve energy efficiency. As buildings become more energy efficient, they can better withstand both hot summers and cold winters. The energy transition also provides a chance to create more comfortable living spaces, reducing reliance on excessive heating and cooling systems, which can have significant energy-saving benefits. Energy-efficient buildings, such as those following the Passivhaus standard, offer substantial savings by improving the thermal envelope, resulting in energy savings of 70-90% compared to traditional homes.

- **Impact of extreme weather on building standards:** In response to changing weather patterns, there is a growing trend to rethink construction practices. For example, the intensity and frequency of heavy rainfall are forcing builders to reassess the size and capacity of rainwater evacuation systems, such as drainage pipes. Buildings need to be designed with greater resilience to such events to prevent damage and flooding, highlighting the need for updated construction methods and materials.
- **Vulnerable housing stock and retrofitting needs:** Older housing stock, especially in countries like Portugal, is vulnerable to both extreme weather and rising energy costs. The need to adapt and retrofit these older buildings is becoming more pressing to meet stricter sustainability regulations and climate resilience standards. Without significant investment in retrofitting, older homes will struggle to meet the demands of climate change, both in terms of energy efficiency and protection against extreme weather events.
- **Job creation and economic growth:** The energy transition in the housing sector offers potential for job creation, particularly in the construction and supply chain industries. The push to make buildings more energy-efficient and resilient can stimulate employment in retrofitting, sustainable construction, and renewable energy systems installation, providing economic growth while addressing environmental concerns.

Overall, the housing and built environment sector is undergoing significant adaptation to cope with the impacts of climate change. While there are challenges in terms of adapting old infrastructure and increasing energy demands, there are also significant opportunities for energy-efficient building practices, job creation, and climate resilience improvements.

3. Population groups most negatively affected by the climate change impact on housing insulation and wellbeing

- **Low-income households:**
 - They live in poor-quality housing with recycled materials that lack proper thermal insulation.
 - These homes are not adapted to current climatic conditions, so their occupants suffer from extreme temperatures, whether heat or cold.
 - They do not have the financial resources to make the necessary adaptations (insulation, window replacement, installation of air conditioning, etc.).
- **Middle-class families with economic difficulties:**
 - While not belonging to the lowest income groups, they face difficulties in affording renovations to improve insulation or cover the higher energy costs caused by poorly insulated homes.
 - They also lack the technical or financial means to install cooling systems like air conditioning.
- **Vulnerable communities:**
 - **Elderly, children,** and individuals with **pre-existing health conditions** are particularly vulnerable to extreme heat or cold due to poor thermal insulation in their homes.
 - The lack of proper insulation puts their health and well-being at risk, especially during climate phenomena such as “torrid nights” or extreme rainfall events like the “**Dana**”, for example in Spain.
- **Residents of homes in climate-risk areas:**

- The cheapest homes are usually located in areas most exposed to climate impacts, such as flood-prone or high-temperature zones. These locations are more accessible to low-income households, but the homes are not adapted to the new climate conditions.
- These groups cannot afford the necessary renovations (insulation, window replacements, etc.) to adapt to new climate realities, which exacerbates their vulnerability.

4. Evidence of critical situations of housing vulnerability because of the climate change

Some general evidences:

- **Climate impact on property values:**
 - In countries experiencing ecological degradation (e.g., loss of green areas, desertification), homes lose value, and depopulation of affected areas occurs.
 - The risks of extreme weather events increase insurance premiums and property devaluation, disproportionately affecting vulnerable households with fewer resources to adapt.
- **Vulnerable housing types:**
 - **Metal roofs:** Many families in vulnerable situations live in homes with poorly insulated metal roofs. These roofs cause extreme internal temperatures in summer (very hot) and winter (very cold), leading to significant discomfort and health risks.
 - **Inadequate thermal insulation:** Some homes have minimal or no insulation, making them cheap to build but uncomfortable and energy-inefficient. This situation worsens with extreme weather conditions.

Evidences by country:

- **Germany:**
 - Buildings that are not renovated are losing value in the real estate market due to high renovation costs.
 - Owners of such buildings may struggle to obtain loans for renovations because the buildings' value is insufficient to guarantee the loan, creating a vicious cycle.
 - The inability to renovate and adapt buildings increases the number of people living in substandard conditions.
- **Spain:**
 - **"Noches Tórridas" (Torrid Nights)** in cities like Barcelona exacerbate the discomfort and health risks for residents of poorly insulated homes, especially vulnerable groups who cannot afford to improve their living conditions.
 - **"Dana" events** (intense rainstorms) in Valencia and other parts of Southern Spain highlight the vulnerability of homes located in flood-prone areas. Such homes are often inadequately prepared to withstand extreme weather, leading to potential damage and displacement.
- **Portugal:**
 - **Poorly insulated houses** contribute to overheating in summer and cold conditions in winter, furthering energy poverty, especially in rural or low-income urban areas.
 - **Flooding risks** in coastal and riverfront regions put homes at significant risk of damage.
 - **Wildfire-prone communities** face threats to housing and infrastructure, especially in rural or remote areas.

- The **degraded social housing conditions** increase vulnerability for those living in these areas, as they lack the necessary infrastructure to withstand climate-related impacts.

5. Measures that could alleviate the negative impact of climate change and energy transition on housing vulnerability

To alleviate the negative impact of climate change and the energy transition on housing vulnerability, several measures should be implemented. Practical support for homeowners is crucial, such as offering consultation services and establishing “One-Stop-Shops” for renovations, along with long-term financial mechanisms for energy renovations. This is vital, as renovation rates need to increase significantly to meet climate goals by 2045.

Energy efficiency in existing buildings should be prioritized, with retrofitting for better insulation and the use of renewable energy sources like solar panels and geothermal systems. Additionally, electrifying industrial processes to avoid polluting energy sources is important. Governments must also promote smart buildings with sustainable and energy-saving systems.

Affordable housing can be supported through low or zero-interest loans for vulnerable populations, but care must be taken to ensure these homes are not given away for free to avoid neglect. Quality control should be improved with state inspectors to ensure homes meet necessary standards. Education is also key, as informed citizens are better equipped to avoid vulnerability.

Specific measures include improving insulation, implementing renewable energy, and enforcing bans on construction in flood-prone areas. More ambitious climate-resilient building regulations and expansion of energy-efficient housing programs are needed, with a focus on vulnerable populations. Integrated urban planning, decentralized renewable energy adoption, and stronger disaster preparedness will further enhance housing resilience to climate change.

Agriculture and food sector

1. Key trends of the green and digital transition in the agriculture sector

The agriculture sector is experiencing increased attention on the green and digital transitions from policy, media, and academia. Young farmers are adopting these changes at higher rates, while older farmers face challenges such as more paperwork and anxiety. Digital and data management tools are being implemented to optimize processes and reduce inputs, though restrictions on chemical products (like fertilizers and pesticides) may lead to lower yields and profit margins.

The green transition aims to mitigate greenhouse gas emissions and combat climate change by promoting sustainable practices. Key measures include making Europe climate neutral, reducing the use of harmful fertilizers and pesticides, and enhancing biodiversity. Technology plays a critical role in helping achieve these goals more efficiently.

There is also a growing need for farmer education, specialized advisors, and more time dedicated to managing digital tools and green traceability. Labour shortages remain a challenge, and public morals are increasingly aligned with sustainability, encouraging more eco-friendly farming practices.

2. Population groups most negatively affected by the transition to renewable energy sources

- **Older farmers:** The aging farming population, particularly those with limited digital skills, is struggling to adapt to new technologies. They often face high costs for technical assistance, investments in equipment, and training, which many cannot afford. This results in a slower

adoption of new practices, as older farmers typically lack familiarity with modern technologies and may find it difficult to keep up with the changing requirements of the green transition.

- **Smaller and less profitable farmers:** Small producers, especially those with fewer economic resources, are at a disadvantage when it comes to investing in new technologies and complying with green transition measures. These farmers often cannot afford the necessary investments in digital tools, specialized advisors, or equipment, which exacerbates existing inequalities in the sector.
- **Farmers in remote areas:** Producers located in rural or mountainous areas with poor internet and phone network coverage are especially impacted. The lack of reliable digital infrastructure makes it difficult for them to adopt new technologies, access information, or participate in digital platforms for the green transition.
- **Self-employed and low-income farmers:** Small-scale, self-employed farmers from lower social classes are the most negatively affected by the financial barriers to accessing the tools and advice needed to implement the green and digital transition. With limited financial resources, they cannot afford the necessary technology or the high costs of expert advice, further widening the gap between larger, more resourceful producers and small farmers.
- **Farm workers facing job insecurity:** The seasonal nature of agricultural work combined with the demands of adapting to new technologies and green transition measures leads to financial instability. Workers in agriculture, especially those from lower social classes, face the perception of their work being physically demanding and unstable compared to other sectors, which could limit their employment opportunities in the future. These groups face significant challenges due to the economic and technical investments required for the digital and green transition, compounded by a lack of skills, limited access to training, and financial resources.

3. Evidence of critical situations in terms of technical, legal and economic adaptation to the twin transition

- **Connectivity issues:** Many rural areas in Ireland lack fibre broadband, hindering farmers' ability to adopt digital technologies.
- **Skills gap:** Older farmers face significant challenges in adopting digital technologies due to skill gaps, which limits their ability to participate in the digital transition.
- **High costs:** The cost of digital and green technologies is a major barrier, particularly for small-scale farmers with limited financial resources.
- **Data collection burden:** Producers face significant workloads due to mandatory data collection, with little support from the distribution sector, resulting in poor communication of product origins and sustainability to consumers.
- **Consumer awareness:** There is a lack of consumer training and information at points of sale, hindering the promotion of sustainable products and farm practices that support rural populations and biodiversity conservation.
- **Commercial interests:** The focus on digital and green tools has often been driven by external commercial interests, leading to an emphasis on form over substance in their implementation.
- **Economic and social sustainability:** The transition may worsen issues like depopulation, loss of traditional production, and family farm closures if not properly aligned with economic and social sustainability.

- **Legislation complexity:** Farmers face the challenge of navigating a complex and ever-changing mix of European, national, and regional laws, such as the Common Agricultural Policy (CAP) and Green Deal, which increases administrative burdens and costs.
- **Misinformation and Lack of Guidance:** The lack of clear communication on legal updates and insufficient private advisory services creates additional costs and time for farmers to remain competitive.
- **Investment inequality:** Small farmers with low net equity struggle to make the necessary high investments in technology, which increases inequalities within the sector.
- **Higher costs for green products:** Green phytosanitary and fertilizer products are more expensive compared to those used in integrated production systems, increasing costs for farmers.
- **Slower technological adaptation:** The aging farming population leads to slower adaptation to new technologies, further hindering the digital transition.
- **Data recording challenges:** Many farmers rely on apps for data recording, but lack of technical knowledge and limited communication from instructors leads to poor data management. Key practices, like herbage measurement, are often left to younger or more tech-savvy staff, and without proper support, these practices are seen as compliance requirements rather than tools for improving farm management.

4. Measures that could alleviate the negative impact of the green and digital transition in the agriculture sector

To alleviate the negative impact of the green and digital transition in the agriculture sector, several key measures could be implemented. First, providing more training and support for both farmers and advisors would help improve skills and enable better adoption of digital and green technologies. Financial support through accessible grants would reduce the burden on smaller producers. Aligning regulations between domestic and imported products would create a level playing field, while better consumer information at points of sale would promote sustainable goods.

A thorough assessment of new measures by sector experts would ensure they are viable and beneficial. Additionally, recognizing and compensating rural ecosystem services, such as landscape maintenance and carbon fixation, is essential. Improving services in rural areas, like connectivity and healthcare, would also help farmers. Offering public education, simplifying processes, and reducing administrative complexity would ease the transition for farmers.

Incentives such as tax reductions for those adopting green practices would encourage early adoption. Fostering peer-to-peer learning among farmers and increasing communication support would build confidence, particularly for older generations. Finally, hiring communication experts would help bridge gaps and provide tailored guidance, ensuring a smoother, more inclusive transition for the agriculture sector.

7. CONCLUSIONS

The purpose of this deliverable was to contextualize and systematize the advancement of the twin transition process in four key economic sectors: energy, mobility and transport, housing and the built environment, and agriculture and food, across six EU countries (Germany, Hungary, Ireland, Italy, Portugal, and Spain). The analysis focused on a comparative review of national targets within the Fit for 55 EU framework and the identification of associated negative impacts on structurally vulnerable groups.

A qualitative research approach was employed to conduct a comprehensive policy review of national long-term strategies, sectoral action plans, and relevant policies in each of the six countries. Additional research methods included two online workshops with international experts in policymaking, academic research, social justice, and the green and digital transitions, as well as an online consultation with experts in the twin transition process across the four sectors. Further desk research of reports and statistical data provided additional insights into the progress of the twin transition and its related vulnerabilities.

Three key themes emerged from the analysis, outlining the current status of the twin transition process and its associated negative impacts:

1. **The twin transition as a driver of structural transformation.** The twin transition process has become a fundamental force reshaping national economies, fostering structural change, and driving innovation to mitigate environmental impact and combat climate change. Common overarching goals among EU countries include addressing global energy needs sustainably, reducing GHG emissions by at least 55% by 2030, and achieving climate neutrality by 2050. These goals are supported by national climate and energy plans that prioritise energy efficiency, security, and diversification. Reducing dependence on fossil fuels is also a central objective, aimed at alleviating energy costs and preventing energy poverty.
2. **Challenges in ensuring an inclusive and equitable transition.** While the transition towards clean energy, digitization, and energy efficiency is progressing rapidly, ensuring an equitable and human-centered approach remains a critical challenge. Highly skilled individuals and those in a favorable economic position are more likely to benefit from innovative RE solutions, clean-energy transport options, and digital business innovation. However, a significant share of the population, due to age, lack of skills, economic distress, or geographic location, is struggling to access these benefits. Indicators of energy poverty, transport poverty, and housing vulnerability have increased in recent years, exacerbating inequalities between those who can access and enjoy the benefits of the twin transition and those who are left behind. The intersection of multiple vulnerabilities has led to the phenomenon of double energy vulnerability, where individuals experience both domestic energy poverty and transport energy poverty (Robinson & Mattioli, 2020; Simcock et al., 2021). Additionally, the digital divide remains a critical issue, particularly in the context of metering devices for household energy efficiency, electric mobility and the skill gap between workers in fossil fuel-dependent industries and emerging opportunities in the RE sector. Geographic disparities, particularly between urban and rural areas, further complicate the transition. Unequal public investment in RE infrastructure, modernization of electric grids, and the integration of public transport modes risk leaving remote and rural areas behind. A balanced and just approach is essential to ensure that all regions can fully participate in and benefit from the twin transition.
3. **Emerging opportunities and pathways toward a just transition.** Consultations with experts have highlighted key priority areas for action to ensure a fair and inclusive twin transition. A holistic understanding of vulnerabilities is crucial to designing effective economic support mechanisms and policy tools to mitigate inequalities. Targeted research on overlapping vulnerabilities and the

identification of newly emerging at-risk populations within specific economic sectors are essential to inform evidence-based policymaking. Public awareness campaigns focused on energy efficiency and household insulation measures are needed to bridge knowledge gaps and enable informed decision-making. Furthermore, enhancing digital literacy and advanced digital skills is imperative to ensure that marginalized groups can access employment opportunities in emerging industries linked to the twin transition. Finally, fostering citizen engagement and grassroots initiatives can play a transformative role in driving community-led energy solutions. The promotion of renewable energy communities for self-consumption and the active involvement of vulnerable groups in decision-making processes are critical to achieving an inclusive transition. Strengthening public participation mechanisms will help ensure that the twin transition process not only accelerates decarbonization efforts but also contributes to social equity and long-term economic resilience.

The twin transition presents both immense opportunities and significant challenges. While it has the potential to drive economic transformation, innovation, and environmental sustainability, it also risks deepening social and regional inequalities if not managed equitably. Ensuring a just transition requires a strategic policy focus on mitigating vulnerabilities, fostering digital inclusion, addressing regional disparities, and promoting citizen participation. By implementing targeted measures and inclusive policies, EU countries can ensure that the twin transition process benefits all segments of society, fostering long-term resilience and sustainable growth across all sectors.

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